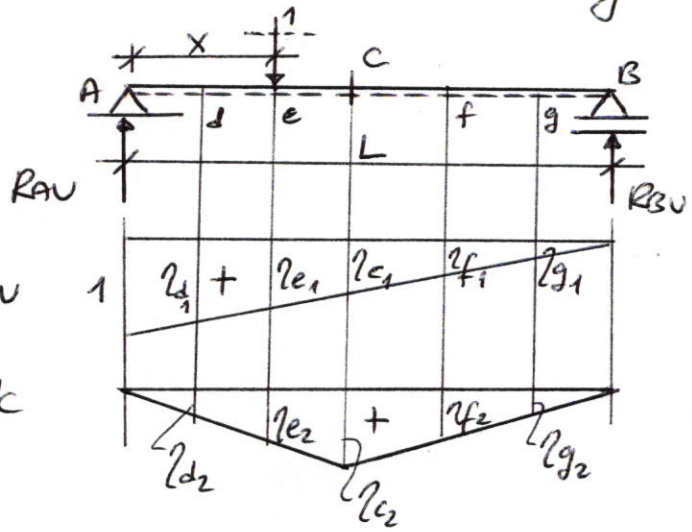
YAPI STATİĞİ I DERS NOTLARI

BÖLÜM II. TESİR ÇİZGİLERİ

VII - SİSTEMLERİN HAREKETLİ YÜKLERE GÖRE HESABI

VII-1- TESİR ÇİZGİLERİ

Bir sistem üzerinde belirli doğrultuda hareket eden bir birimlik kuvvetten, verilen bir kesitte meydana gelen kesit tesirlerinin (veya mesnet tepkilerinin) değerini bir birimlik yük hizasında ordinat olarak göstererek elde edilen diyagrama o kesite ait kesit tesiri (veya mesnet tepkisi) tesir çizgisi diyagramı denir. η ile gösterilir.



η_{d_1} = Bir birimlik yük d noktasında iken RAV'nin değeri

η_{e_1} = Bir birimlik yük e noktasında iken RAV'nin değeri

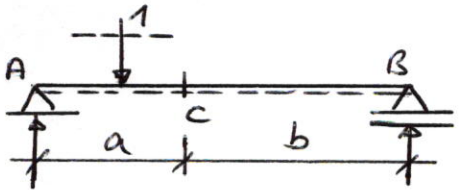
η_{d_2} = Bir birimlik yük d noktasında iken Mc'nin değeri

η_{e_2} = Bir birimlik yük e noktasında iken Mc'nin değeri

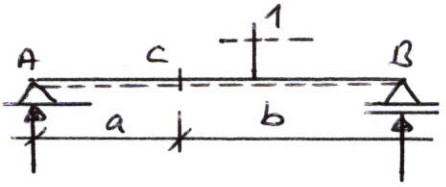
VII-2- TESİR ÇİZGİSİNİN TAYİNİ

Belirli bir kesitteki bir kesit tesirine veya mesnet tepkisine ait tesir çizgisini elde etmek için sistem üzerinde belirli bir doğrultuya paralel 1 birim değerindeki sabit bir kuvvet x'e bağlı olarak gezdirilir. Sabit yük durumunda kesit tesiri hesaplarında yapıldığı gibi, belirli kesitte bir birimlik yükten meydana gelen kesit tesiri ve mesnet tepkisi x'e bağlı olarak ifade edilir. Bunlara o kesit tesiri veya mesnet tepkisine ait tesir fonksiyonu denir. Diyagramına da tesir çizgisi diyagramı denir.

"1" Birimlik kuvvet kendisine paralel olarak hareket ettiği zaman x değişken olacağından x'e bağlı kesit tesiri veya mesnet tepkisi tesir çizgisinin fonksiyonunu verecektir. Bu fonksiyon çizilerek de tesir çizgisi elde edilir. Çoğu kez kesit tesiri veya mesnet tepkisi bütün sistem için bir tek fonksiyonla ifade edilemez. Bundan dolayı bu fonksiyonun yürürlükte olduğu her bölge için ifadeler ayrı ayrı yazılmalıdır.



Yük A-C arasında $f(R_{AV}, M_C, \ldots)$



Yük C-B arasında $f(R_{AV}, M_C, \ldots)$

Tesir çizgilerinin belirlenmesinde dikkat edilecek hususlar

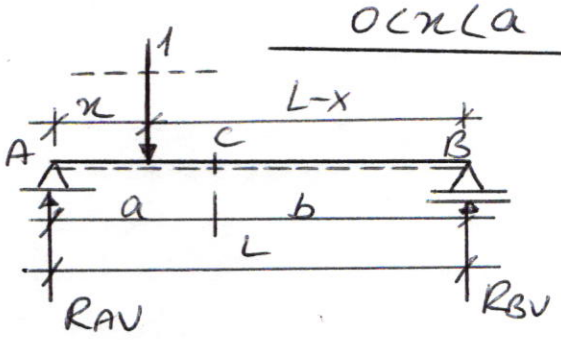
1- İspatlardan da görüleceği gibi izostatik sistemlerin tesir çizgileri lineer (doğrusal) fonksiyonlardır. Bu nedenle tesir çizgilerinin belirlenmesi için birkaç özel noktanın hesaplanması yeterlidir.

2- Verilmiş olan bir kesitte kesit tesirlerinin hesabı için önce tepkilerin bulunması gerekli ise bu tip tesir çizgilerinin hesabında da önce tepkilerin tesir çizgilerini belirlemekle başlamak kolaylık sağlar.

3- Belirli bir kesitte kesit tesirlerinin tesir çizgileri belirlenirken sistem söz konusu kesitte ikiye ayrılmış gibi düşünülür. Bu durumda;

a) Dış etki kesim yapılan kesitin solunda ise; denge sağda kalan kısım üzerinde düşünülür ve böylece sol parçanın tesir çizgisi ordinatları bulunur.

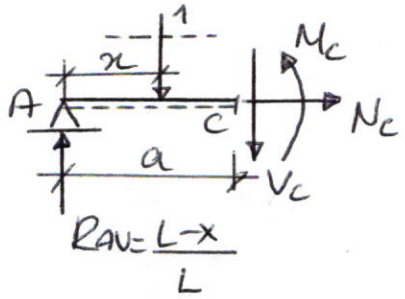
$0 < x < a$



Mesnet Tepkilerinin Hesabı

$\Sigma M_B = 0 \rightarrow R_{AV} = 1 \cdot \frac{(L-x)}{L} = \frac{L-x}{L}$

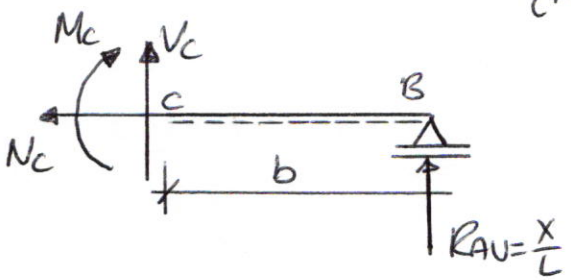
$\Sigma M_A = 0 \rightarrow R_{BV} = 1 \cdot \frac{x}{L} = \frac{x}{L}$



$\sum V_c = \frac{L-x}{L} - 1 = -\frac{x}{L}$

$\sum M_c = \frac{L-x}{L} \cdot a - 1(a-x) = a - \frac{x}{L} \cdot a - a + x$

$\sum M_c = x - \frac{x}{L} a = \frac{x(L-a)}{L} = \frac{x}{L} \cdot b$

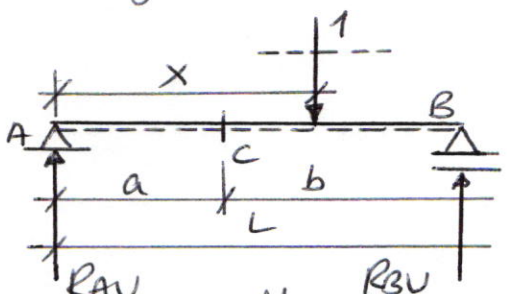


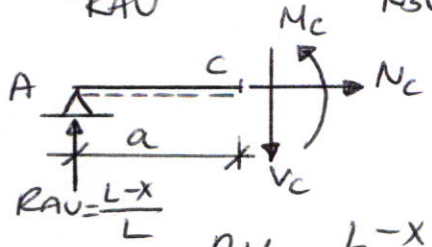
$\sum V_c = -\frac{x}{L}$; $\sum M_c = \frac{x}{L} \cdot b$

Hesaplardan da görüldüğü gibi yüksüz olan bölgede hesap yapılması işlemleri kısa tutmaktadır. Ancak hesap sonucu bulunan değerler birim yükün A-C arasında gezinmesi durumu içindir ve x değişkeni 0 – a arası değerler alacaktır.

b) Dış etki kesim yapılan kesitin sağında ise; denge solda kalan parça üzerinde düşünülür ve böylece sağ parçanın tesir çizgisi ordinatları bulunur.

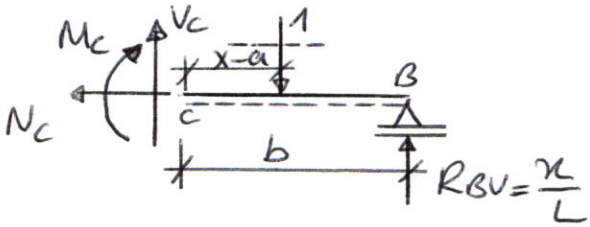
$a < x < L$





$\sum V_c = \frac{L-x}{L}$

$\sum M_c = \frac{L-x}{L} \cdot a$



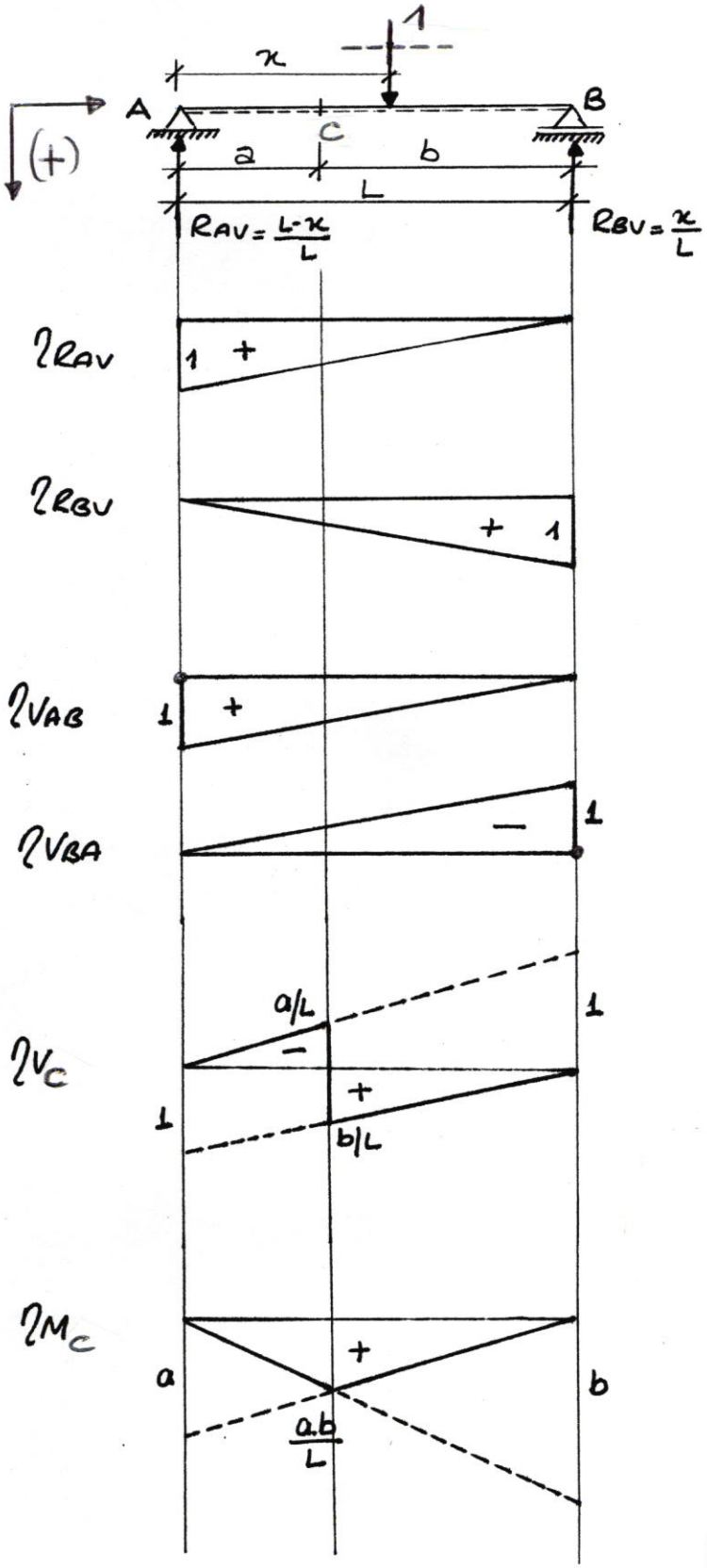
$\sum V_c = 1 - \frac{x}{L} = \frac{L-x}{L}$

$\sum M_c = \frac{x}{L} \cdot b - 1(x-a) = \frac{x}{L} b - x + a$

$\sum M_c = x\left(\frac{b-L}{L}\right) + a = -\frac{xa}{L} + \frac{aL}{L} = \frac{L-x}{L} \cdot a$

VII_3) BASIT KIRIŞ TESIR ÇIZGISI

a) Düşey Yük Etkisi



$\eta_{V_{AB}}$'nin A mesnedinde sıfır olması durumu $\eta_{V_{BA}}$'nın da B mesnedinde sıfır olması durumu ile aynıdır.

$$\Sigma M_B=0 \rightarrow R_{AV}=\frac{L-x}{L}.1=\frac{L-x}{L}$$

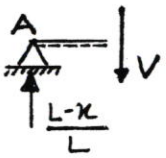
$$\Sigma M_A=0 \rightarrow R_{BV}=\frac{x}{L}.1=\frac{x}{L}$$

Mesnet Tepkisi Tesir Çizgisi

$$\eta_{R_{AV}}=\frac{L-x}{L}\begin{cases}x=0 \rightarrow \eta_{R_{AV}}=1\\ x=L \rightarrow \eta_{R_{AV}}=0\end{cases}$$

$$\eta_{R_{BV}}=\frac{x}{L}\begin{cases}x=0 \rightarrow \eta_{R_{BV}}=0\\ x=L \rightarrow \eta_{R_{BV}}=1\end{cases}$$

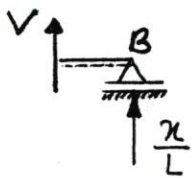
Kesit Tesiri Tesir Çizgileri



$$V_{AB}=\frac{L-x}{L}$$

$x=0 \rightarrow \eta_{V_{AB}}=1$

$x=L \rightarrow \eta_{V_{AB}}=0$

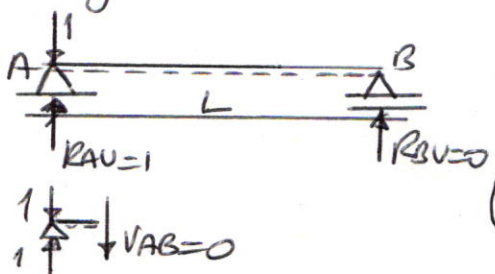


$$V_{BA}=-\frac{x}{L}$$

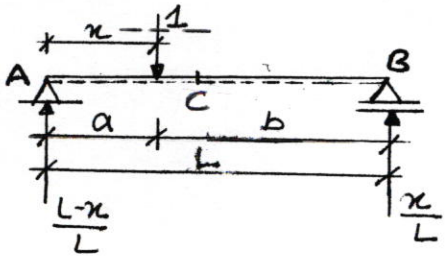
$x=0 \rightarrow \eta_{V_{BA}}=0$

$x=L \rightarrow \eta_{V_{BA}}=-1$

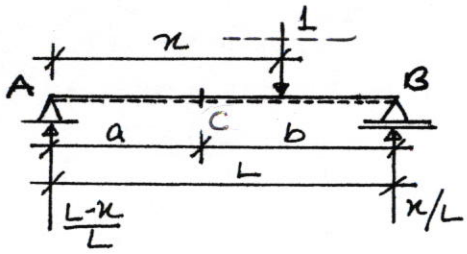
$\eta_{V_{AB}}$ tesir çizgisi çizilirken birim yük A noktası üzerinde iken $\eta_{V_{AB}}=0$ olacağı dikkate alınmalıdır.



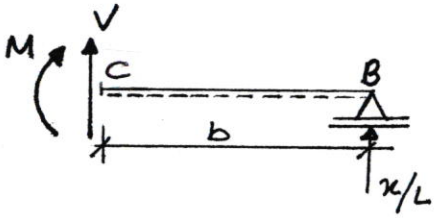
"C" noktası tesir çizgileri:



Not: Herhangi bir "C" noktasındaki kesit tesiri tesir çizgilerini bulmak için; tıpkı kesit tesirlerinde olduğu gibi sistem bu "C" noktasından ikiye ayrılır. Sol veya sağ parça üzerinde denge denklemleri yazılarak tesir çizgisi değeri x'e bağlı olarak bulunur. Kolaylık olması bakımından yükün olmadığı kısım üzerinde denge denklemini yazmak daha pratiktir.



$0 < x < a$ (yük A – C arasında)



$$?V_C = -x/L$$

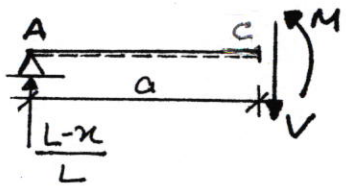
$$?M_C = \frac{x}{L}.b$$

$$x=0 \begin{cases} ?V_C = 0 \\ ?M_C = 0 \end{cases}$$

$$x=a \begin{cases} ?V_C = -a/L \\ ?M_C = \frac{a.b}{L} \end{cases}$$

$$x=L \begin{cases} ?V_C = -1 \\ ?M_C = b \end{cases}$$

$a < x < L$ (yük C – B arasında)



$$?V_C = \frac{L-x}{L}$$

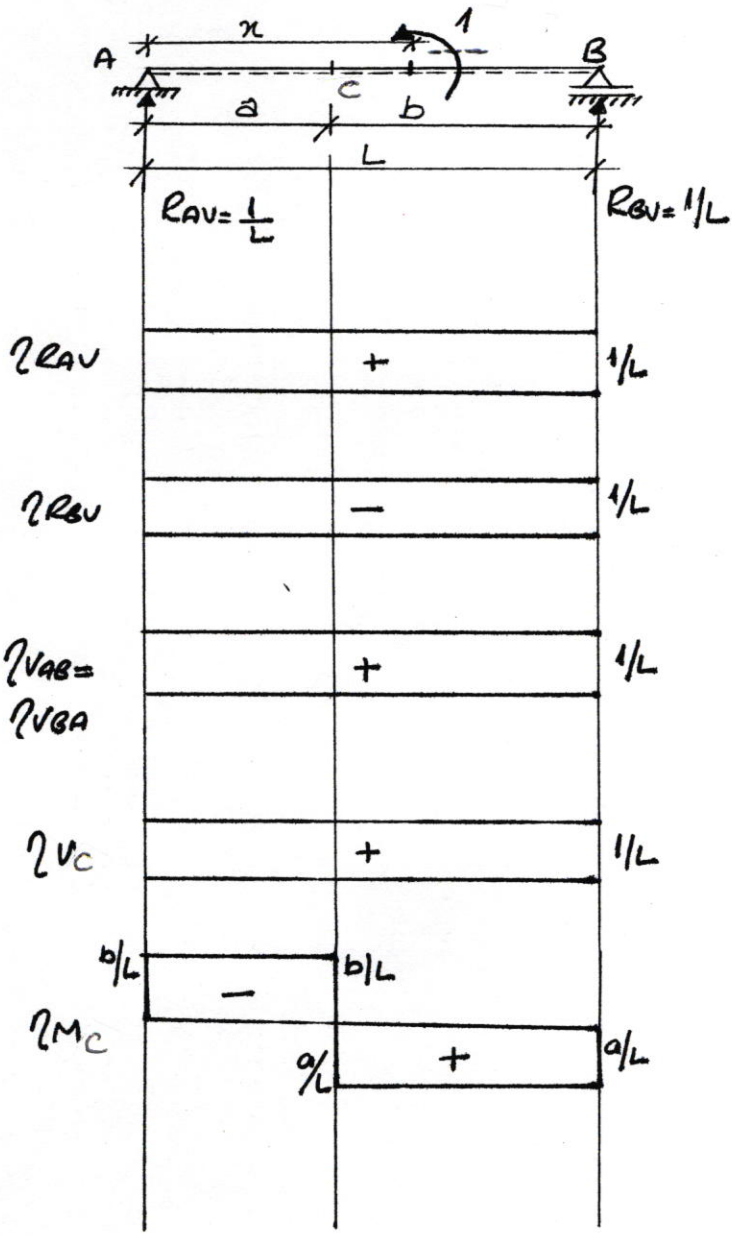
$$?M_C = \frac{L-x}{L}.a$$

$$x=a \begin{cases} ?V_C = b/L \\ ?M_C = \frac{a.b}{L} \end{cases}$$

$$x=L \begin{cases} ?V_C = 0 \\ ?M_C = 0 \end{cases}$$

$$x=0 \begin{cases} ?V_C = 1 \\ ?M_C = a \end{cases}$$

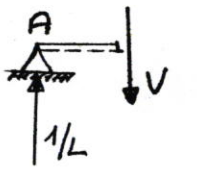
b) Birim Moment Etkisi



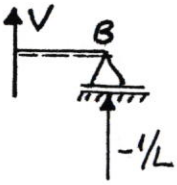
$\Sigma M_B = 0 \rightarrow R_{AV} = 1/L$

$\Sigma M_A = 0 \rightarrow R_{BV} = -1/L$

Kesit Tesiri Tesir Cizgileri

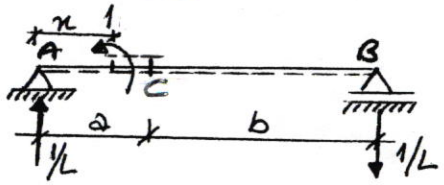


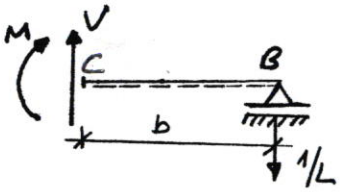
$\eta V_{AB} = 1/L$



$\eta V_{BA} = 1/L$

$0 < x < a$

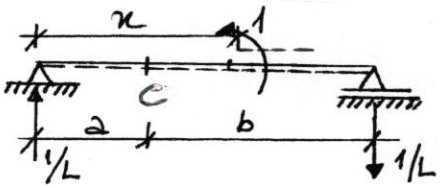


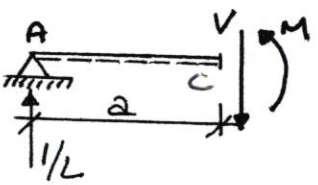


$\eta V_C = 1/L$

$\eta M_C = -b/L$

$a < x < L$



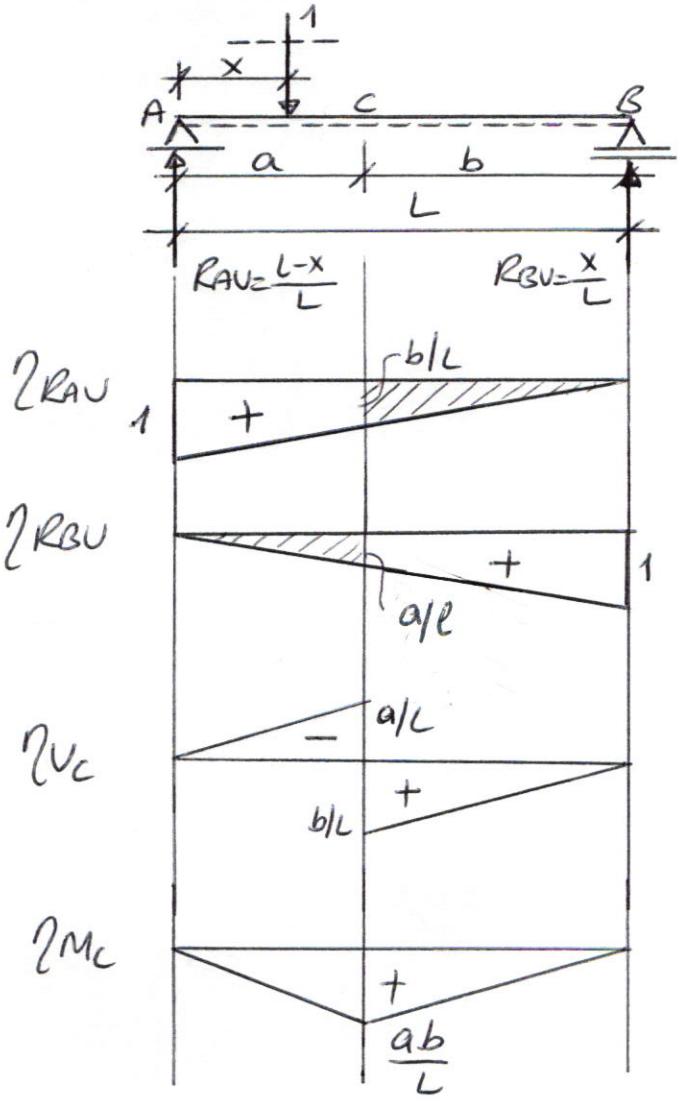


$\eta V_C = 1/L$

$\eta M_C = a/L$

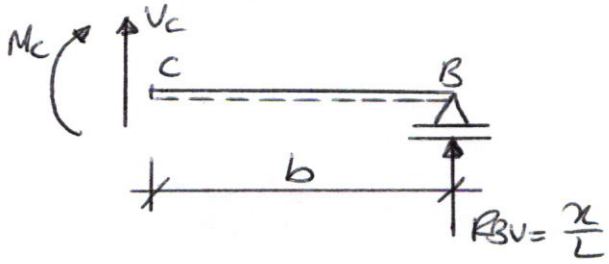
c) Tesir Çizgilerinin Mesnet Tepkileri Yardımıyla Hesabı

c.1) Birim Düşey Yük Etkisi



Birim yük A-C arasında

$0 < x < a$

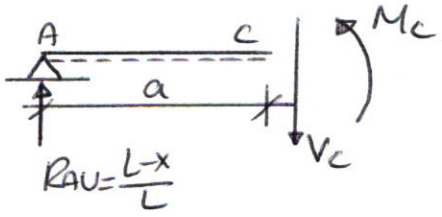


$\eta V_C = -R_{BV}$

$\eta M_C = R_{BV} \cdot b$

Birim yük C-B arasında

$a < x < L$

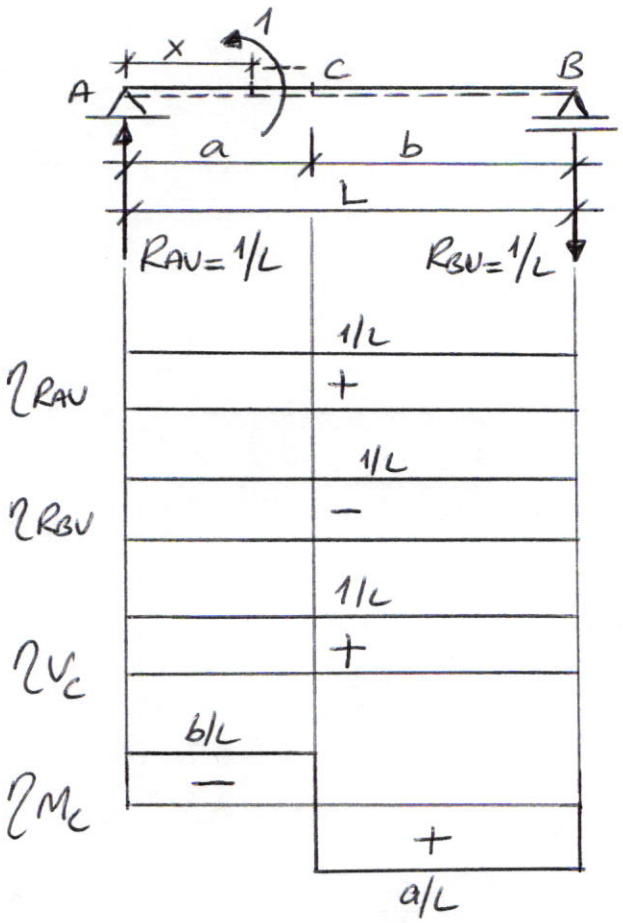


$\eta V_C = R_{AV}$

$\eta M_C = R_{AV} \cdot a$

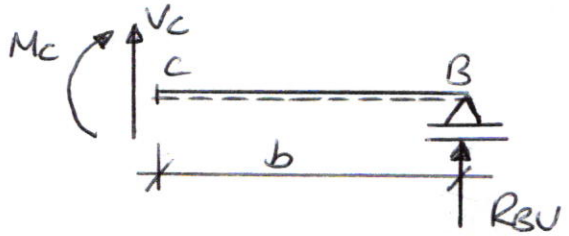
Birim yük 0-a arasında iken $\eta V_C = -R_{BV}$ olduğundan ηV_C'nin 0-a arasındaki değeri 0 ve $-\frac{a}{\ell}$ olacaktır. Benzer şekilde birim yük a-ℓ arasında iken $\eta V_C = R_{AV}$'ye eşit olduğundan ηV_C'nin a-ℓ arasındaki değeri x=a'da ya da C noktasında b/ℓ, x=ℓ'de ya da B noktasında sıfır olacaktır.

C.2) Birim moment etkisi



Birim yük A-C arasında

$0 < x < a$

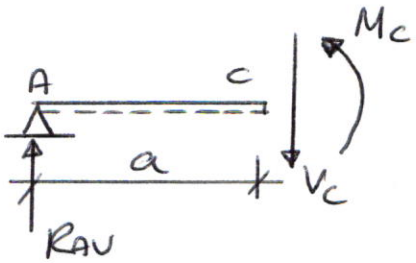


$\eta V_c = -R_{BV}$

$\eta M_c = +R_{BV} \cdot b$

Birim yük C-B arasında

$a < x < L$



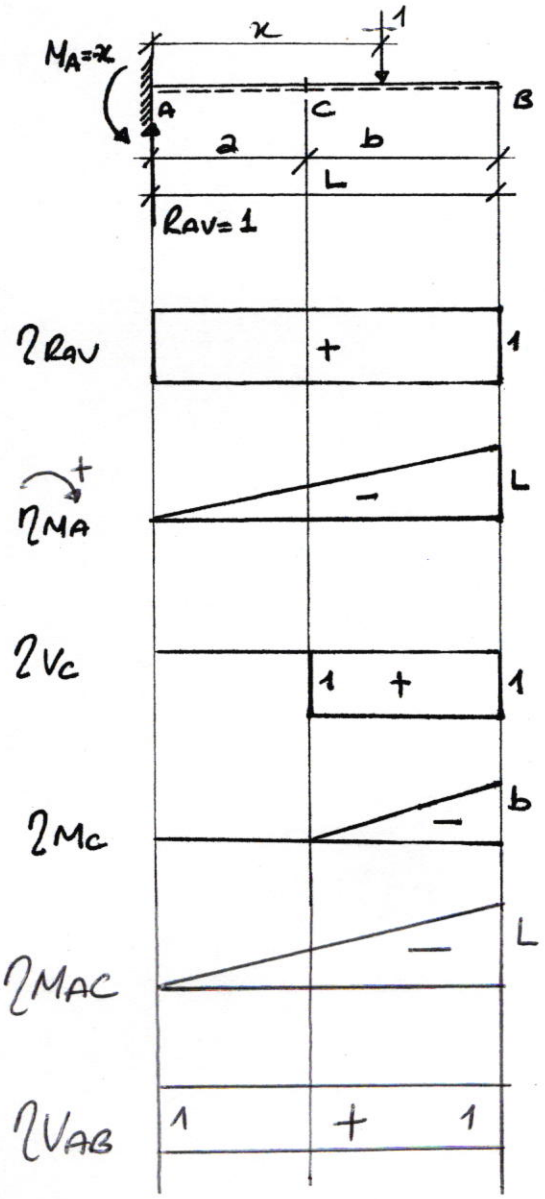
$\eta V_c = R_{AV}$

$\eta M_c = R_{AV} \cdot a$

Birim moment A-C arasında iken $\eta V_c = -R_{BV}$ olduğundan ηV_c'nin A-C arasındaki değeri $\eta V_c = -(-1/L) = 1/L$, C-B arasındaki değeri de $\eta V_c = 1/L$ olacaktır.

VII. 4) KONSOL KİRİŞ TESİR GİZGİSİ

a) Düşey Yük Etkisi



Mesnet Tepkisi Tesir Gizgisi

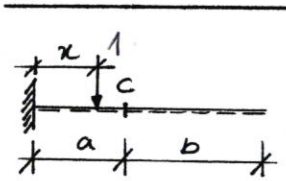
$\eta R_{AV} = 1$

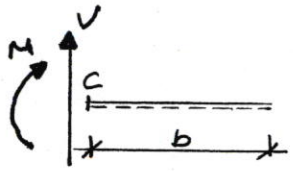
$\eta M_A = x$ (+)

$\begin{cases} x=0 \rightarrow \eta_{MA}=0 \\ x=L \rightarrow \eta_{MA}=-x \rightarrow \eta_{MA}=-L \end{cases}$

Kesit Tesiri Tesir Gizgileri

$0 < x < a$

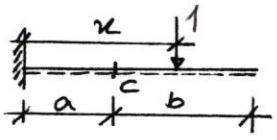


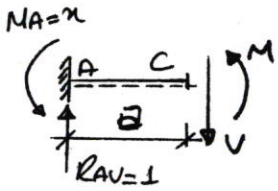


$\eta V_C = 0$

$\eta M_C = 0$

$a < x < L$





$\eta V_C = 1$

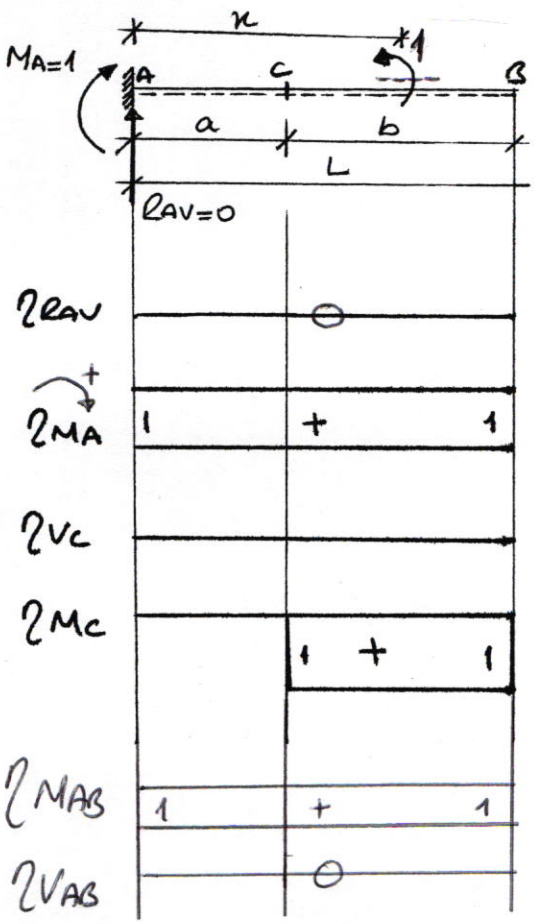
$\eta M_C = -x + a$

$\begin{cases} x=a \rightarrow \eta_{MC}=0 \\ x=L \rightarrow \eta_{MC}=-b \end{cases}$

$\eta M_{AB} = -x$

$\eta V_{AB} = 1$

b) Moment Etkisi



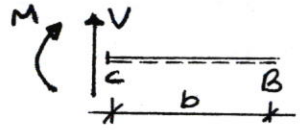
Moment Tepkisi Tesir Cizgisi

$\eta R_{AV} = 0$

$\eta M_A = 1$

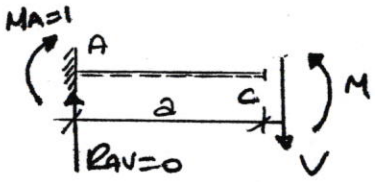
Kesit Tesiri Tesir Cizgileri

$0 < x < a$



$\eta V_C = 0$, $\eta M_C = 0$

$a < x < L$



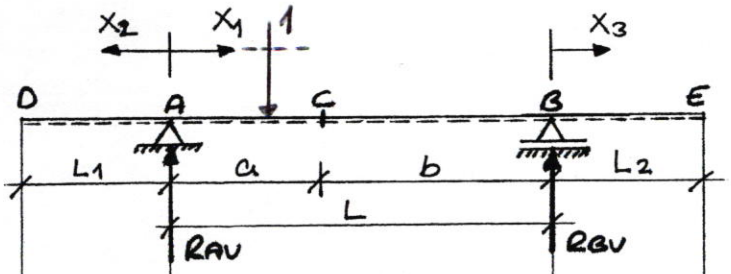
$\eta V_C = 0$, $\eta M_C = 1$

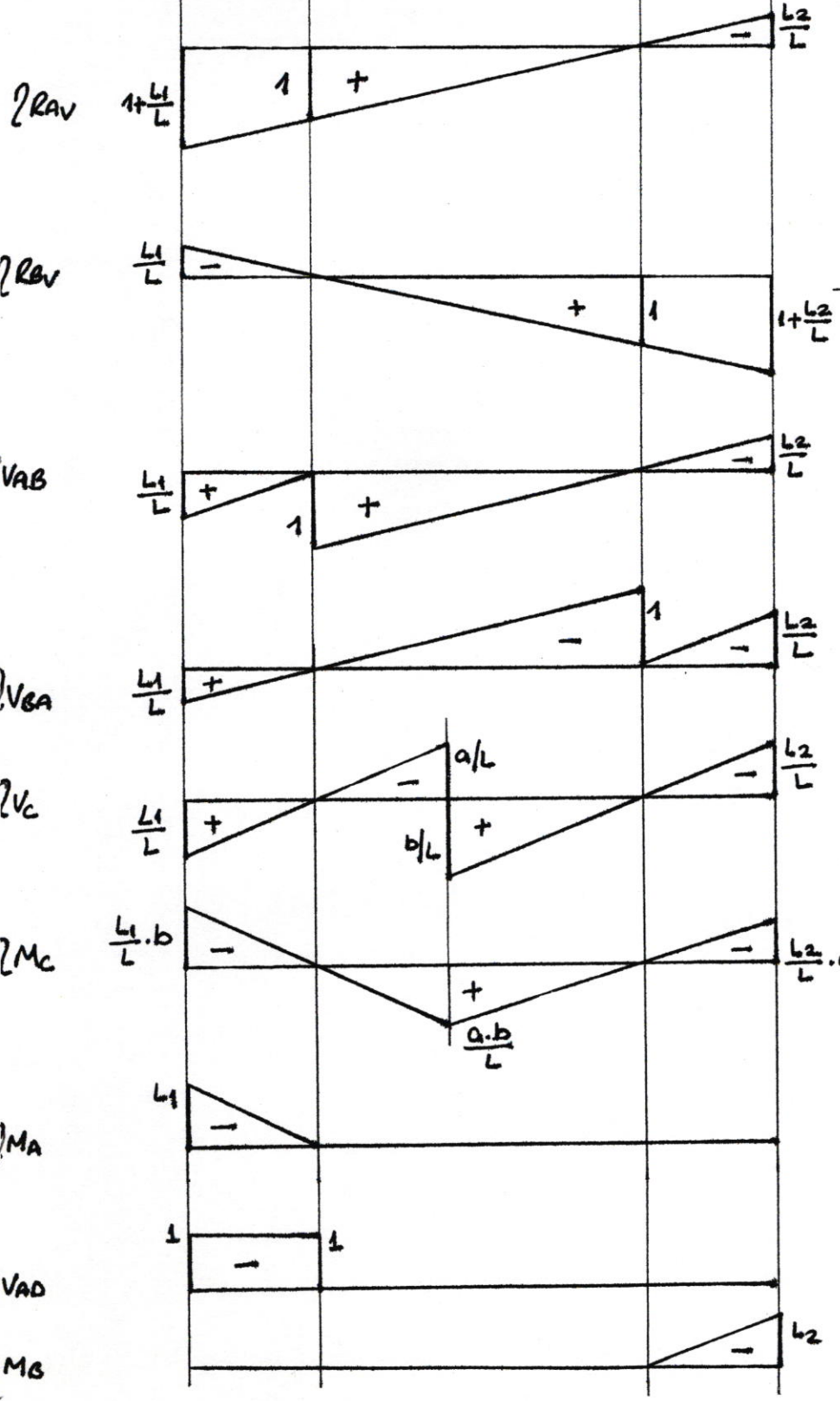
$\eta M_{AB} = 1$

$\eta V_{AB} = 0$

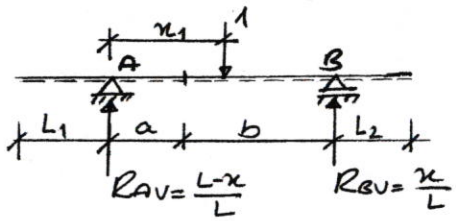
VII.5) Çıkmalı Kiriş Tesir Çizgileri

a) Düşey Yük Etkisi



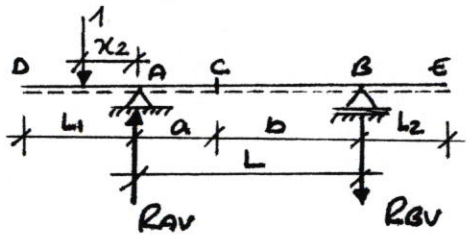


_ Birim yük A-B arasında



Not: Çıkmalı kirişin A-B arası tesir çizgisi basit kiriş tesir çizgisi gibidir. Konsol kısımlar ayrıca incelenecektir.

_ Birim yük A-D arasında



$0 < x_2 < L_1$

Mesnet Tepkisi Tesir Çizgisi

$\Sigma M_B = 0 \rightarrow R_{AV} = \frac{L + x_2}{L}$

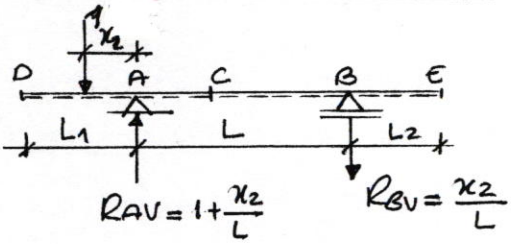
$\Sigma M_A = 0 \rightarrow R_{BV} = -\frac{x_2}{L}$

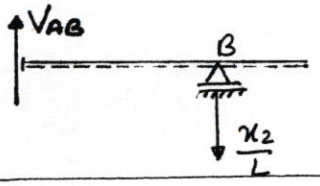
$x_2 = 0 \begin{cases} \eta_{RAV} = 1 \\ \eta_{RBV} = 0 \end{cases}$

$x_2 = L_1 \begin{cases} \eta_{RAV} = 1 + \frac{L_1}{L} \\ \eta_{RBV} = -\frac{L_1}{L} \end{cases}$

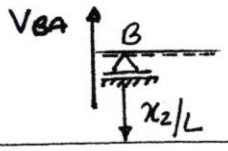
$0 < x_2 < L_1$

Kesit Tesiri Tesir Çizgileri

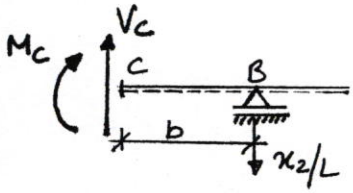




$$\eta V_{AB} = \frac{x_2}{L} \Rightarrow \begin{cases} x_2 = 0 \rightarrow \eta V_{AB} = 0 \\ x_2 = L_1 \rightarrow \eta V_{AB} = L_1/L \end{cases}$$

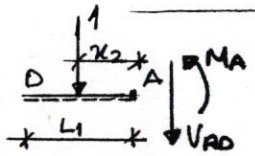


$$\eta V_{BA} = \frac{x_2}{L} \Rightarrow \begin{cases} x_2 = 0 \rightarrow \eta V_{BA} = 0 \\ x_2 = L_1 \rightarrow \eta V_{BA} = L_1/L \end{cases}$$



$$\eta V_C = \frac{x_2}{L} \Rightarrow \begin{cases} x_2 = 0 \rightarrow \eta V_C = 0 \\ x_2 = L_1 \rightarrow \eta V_C = L_1/L \end{cases}$$

$$\eta M_C = -\frac{x_2}{L}.b \rightarrow \begin{cases} x_2 = 0 \rightarrow \eta M_C = 0 \\ x_2 = L_1 \rightarrow \eta M_C = -\frac{L_1}{L}.b \end{cases}$$



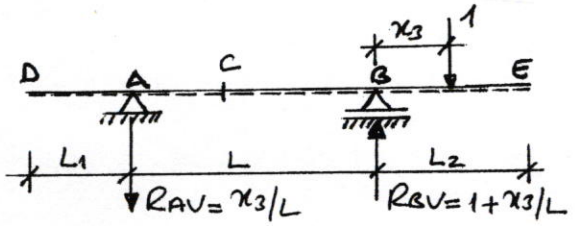
$$\eta V_{AD} = -1$$

$$\eta M_A = -x_2 \Rightarrow \begin{cases} x_2 = 0 \rightarrow \eta M_A = 0 \\ x_2 = L_1 \rightarrow \eta M_A = -L_1 \end{cases}$$

- Birim yük B-E arasında

$0 < x_3 < L_2$

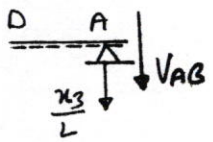
Mesnet Tepkisi Tesir Çizgisi



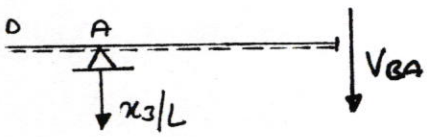
$\Sigma M_B = 0 \rightarrow R_{AV} = -x_3/L$

$\Sigma M_A = 0 \rightarrow R_{BV} = 1 + x_3/L$

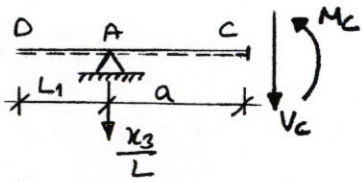
$$x_3 = 0 \begin{cases} \eta R_{AV} = 0 \\ \eta R_{BV} = 1 \end{cases} \qquad x_3 = L_2 \begin{cases} \eta R_{AV} = -L_2/L \\ \eta R_{BV} = 1 + L_2/L \end{cases}$$



$$\eta V_{AB} = -\frac{x_3}{L} \Rightarrow \begin{cases} x_3 = 0 \rightarrow \eta V_{AB} = 0 \\ x_3 = L_2 \rightarrow \eta V_{AB} = -L_2/L \end{cases}$$

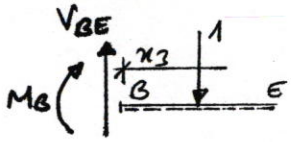


$$\eta V_{BA} = -\frac{x_3}{L} \Rightarrow \begin{cases} x_3 = 0 \rightarrow \eta V_{BA} = 0 \\ x_3 = L_2 \rightarrow \eta V_{BA} = -\frac{L_2}{L} \end{cases}$$



$$\eta V_C = -x_3/L \Rightarrow \begin{cases} x_3=0 \rightarrow \eta V_C = 0 \\ x_3 = L_2 \rightarrow \eta V_C = -\frac{L_2}{L} \end{cases}$$

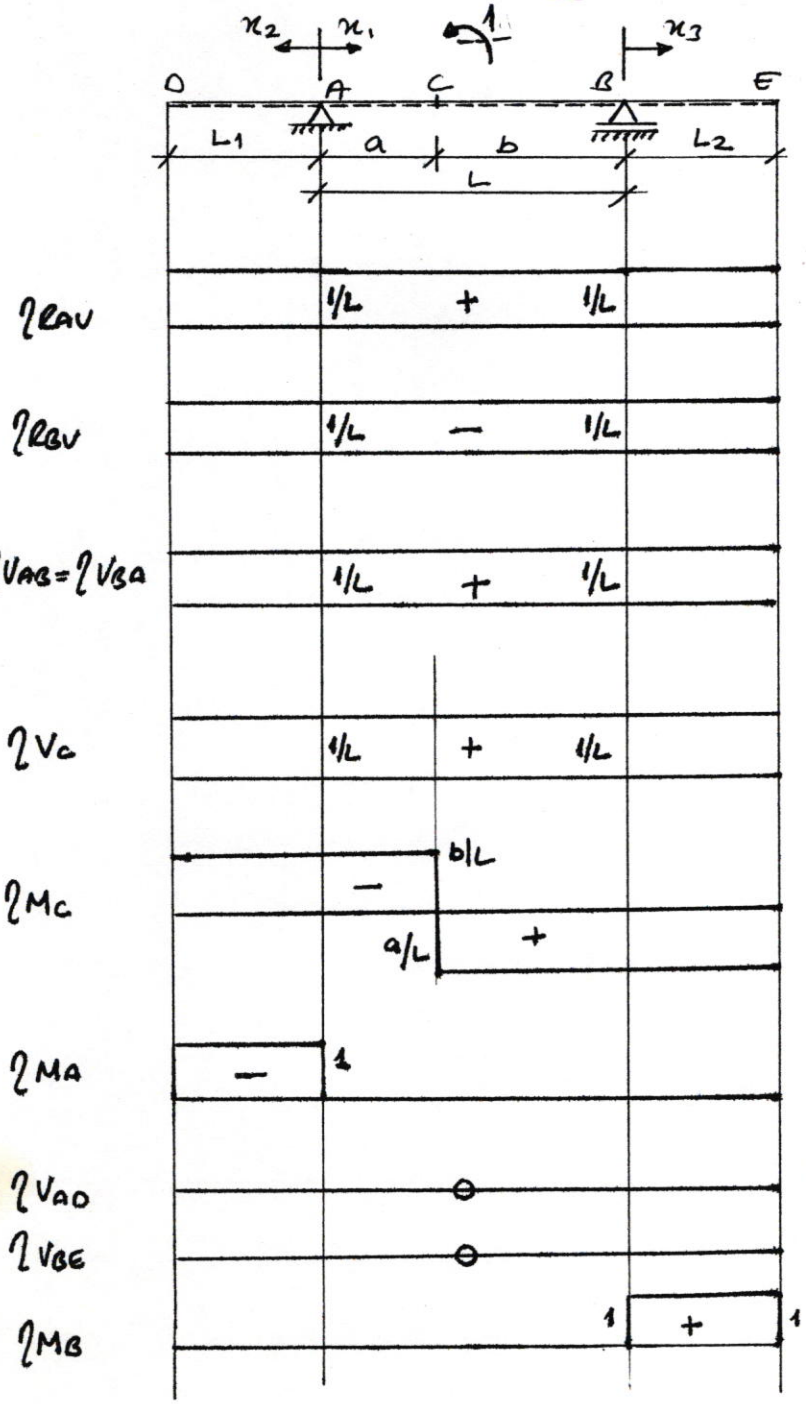
$$\eta M_C = -\frac{x_3}{L}\cdot a \Rightarrow \begin{cases} x_3=0 \rightarrow \eta M_C = 0 \\ x_3 = L_2 \rightarrow \eta M_C = -\frac{L_2}{L}\cdot a \end{cases}$$



$$\eta V_{BE} = 1$$

$$\eta M_B = -x_3 \begin{cases} x_3=0 \rightarrow \eta M_B = 0 \\ x_3 = L_2 \rightarrow \eta M_B = -L_2 \end{cases}$$

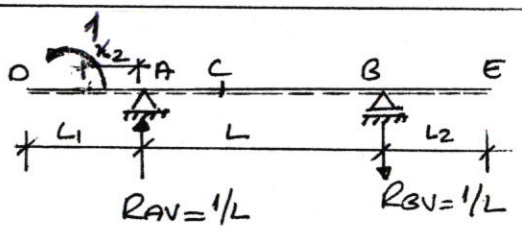
b) Birim Moment Etkisi



- Birim Moment A-B arasında

A-B arası tesir çizgisi basit kiriş tesir çizgisinin aynıdır.

- Birim moment A-D arasında

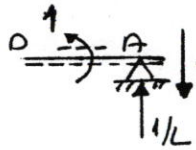


$0 < x_2 < L_1$

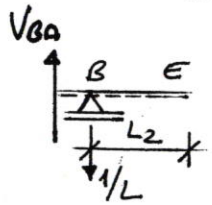
Mesnet Tepkisi Tesir Çizgisi

$\eta R_{AV} = 1/L$, $\eta R_{BV} = -1/L$

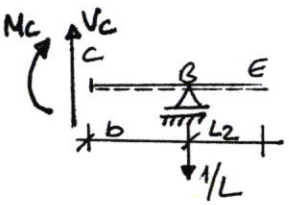
Kesit Tesiri Tesir Çizgileri



$\eta V_{AB} = 1/L$



$\eta V_{BA} = 1/L$



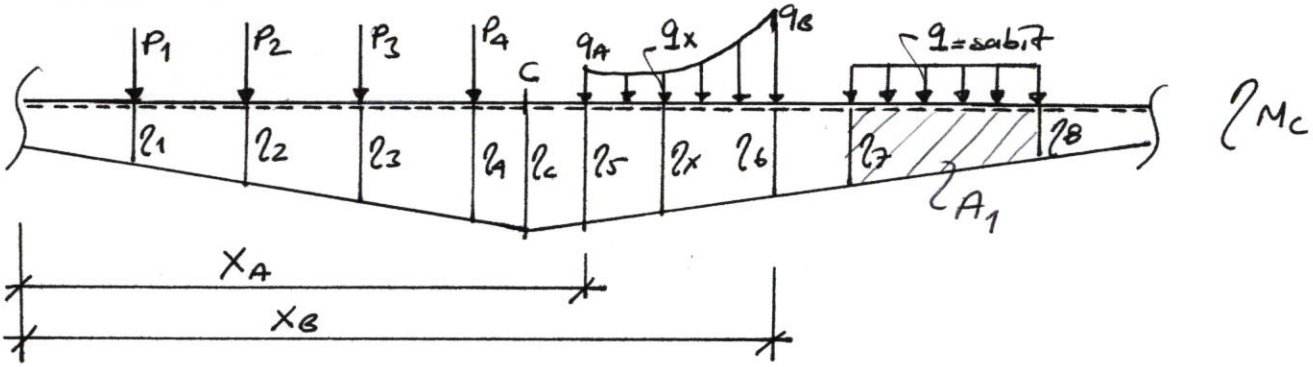
$\eta V_C = 1/L$

$\eta M_C = -b/L$

VII. 6) Tesir Gizgisi Ordinatı Boyutları

Dış Etki	TEPKİLER		Kesit Tesirleri		
	R	M	M	N	V
Düşey yük (↓)	Boyutsuz	metre	metre	Boyutsuz	Boyutsuz
Moment (↺)	1/M	Boyutsuz	Boyutsuz	1/M	1/M

VII. 7) TESİR GİZGİLERİNİN DEĞERLENDİRİLMESİ



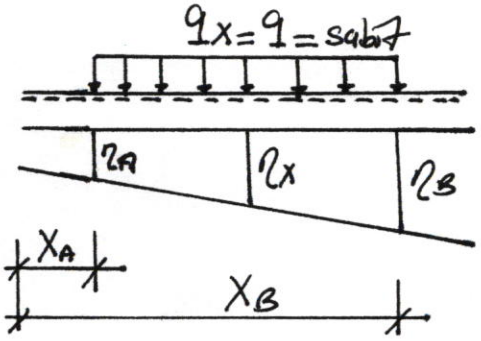
$$M_C = P_1 \cdot \eta_1 + P_2 \cdot \eta_2 + P_3 \cdot \eta_3 + P_4 \cdot \eta_4 + \int_{X_A}^{X_B} q_x \cdot \eta_x \, dx + q \cdot A_1$$

Tesir Gizgileri;

a) Verilen bir kesitte sabit yüklerden meydana gelen kesit tesirleri veya mesnet tepkilerinin hesaplanması,

b) Verilen bir kesitte kesit tesirleri veya mesnet tepkilerini minimum veya maksimum yapan katar durumunu belirleyerek statik büyüklüklerin en büyük ve en küçük değerlerinin hesaplanması için kullanılır.

Dolayısıyla sabit yüklerin etkimesi durumunda tekil yüklerde, tekil yükün etkidiği noktadaki tesir gizgisi değeri ile tekil yükün çarpımı, yayılı yüklerde yayılı yükle altındaki tesir gizgisinin integrasyonu sonucunda kesit tesiri değeri hesaplanır. Ancak düzgün yayılı yük durumunda düzgün yayılı yükle yükün altında yer alan tesir gizgisi alanı çarpılır.



$$M = \int_{X_A}^{X_B} q_x \cdot \eta_x \, dx$$

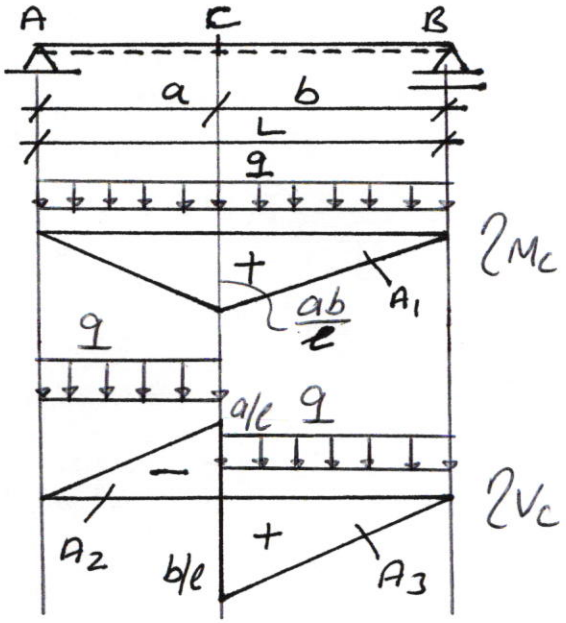
$$M = \int_{X_A}^{X_B} q \cdot \eta_x dx = q \int_{X_A}^{X_B} \eta_x \cdot dx$$

Alan (A)

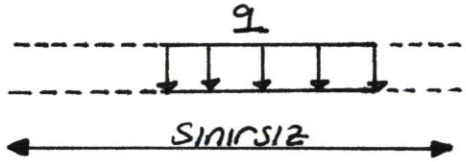
$$M = q \cdot A$$

VII_8) MAKSİMUM VE MİNİMUM DEĞERLERİ VEREN HAREKETLİ YÜK DURUMLARI

1) Düzgün Yayılı Yük Durumu



Yayılı hareketli yük



$$A_1 = \frac{ab}{\ell} \cdot \frac{\ell}{2}$$

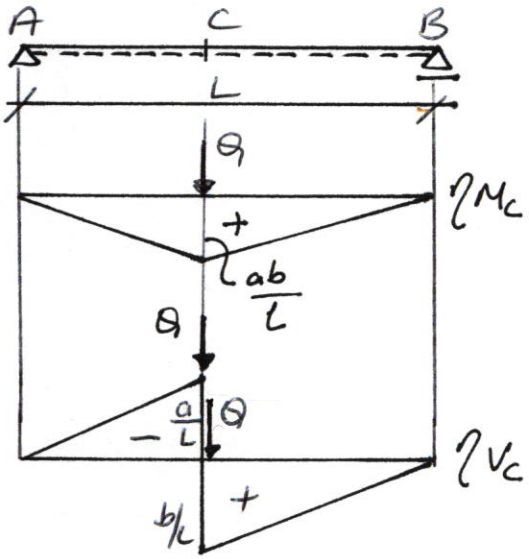
$$A_2 = -\frac{a}{\ell} \cdot \frac{a}{2}$$

$$A_3 = \frac{b}{\ell} \cdot \frac{b}{2}$$

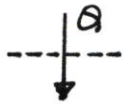
$$\max M_{c_q} = q \cdot A_1 \qquad \min M_{c_q} = 0$$

$$\max V_{c_q} = q \cdot A_3 \qquad \min V_{c_q} = -q \cdot A_2$$

2) Tekil Yük Durumu



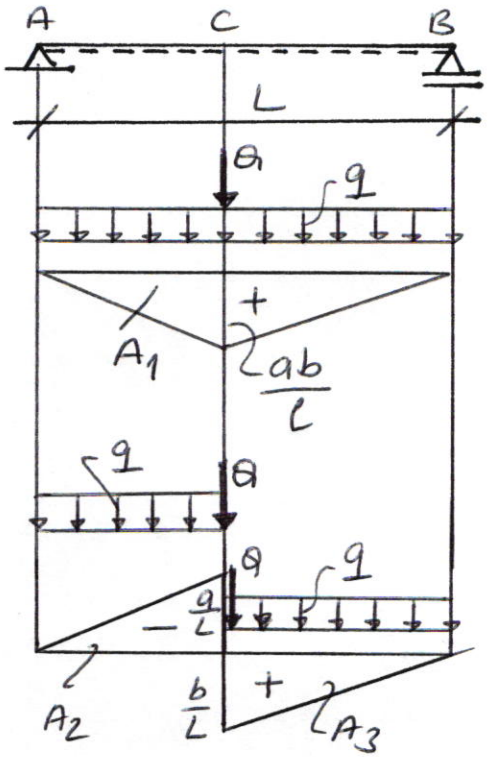
Tekil Hareketli Yük



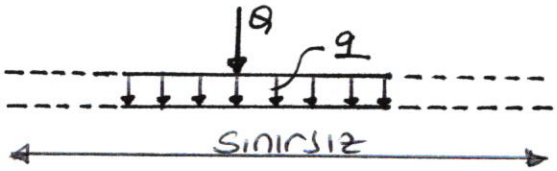
$$\min M_{cq} = 0, \quad \max M_{cq} = Q \cdot \frac{ab}{L}$$

$$\min V_{cq} = -Q \cdot \frac{a}{L} \; ; \; \max V_{cq} = +Q \cdot \frac{b}{L}$$

3) Düzgün Yayılı Yük + Tekil Yük Durumu



Yayılı hareketli yük + Tekil Yük



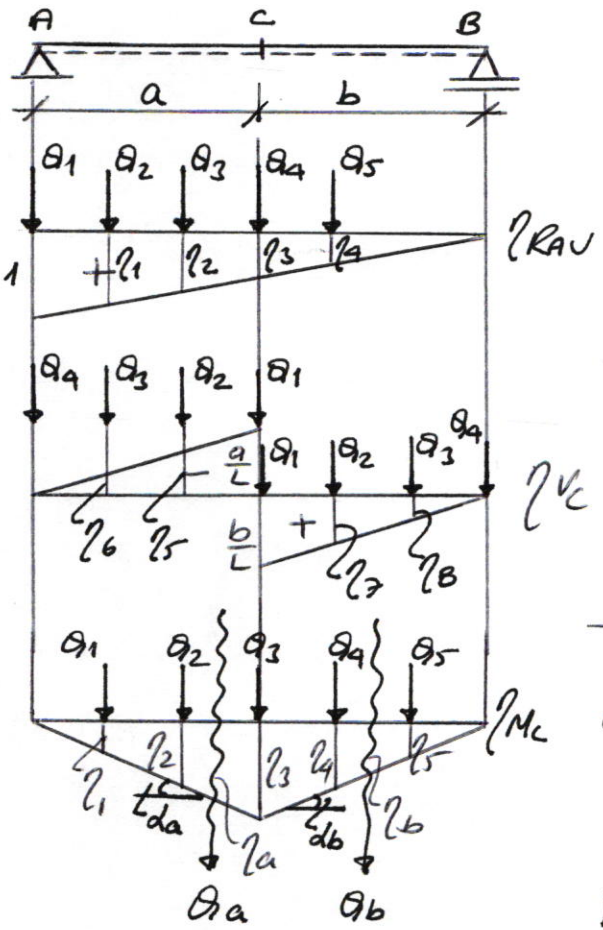
$$\min M_{cq} = 0$$

$$\max M_{cq} = q \cdot A_1 + Q \cdot \frac{ab}{L}$$

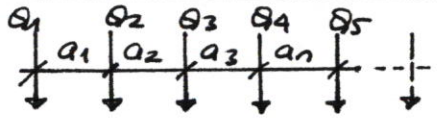
$$\min V_{cq} = -q \cdot A_2 - Q \cdot \frac{a}{L}$$

$$\max V_{cq} = +q \cdot A_3 + Q \cdot \frac{b}{L}$$

4) Aralık ve Şiddetleri Sabit (Katar Yükü) Yük Durumu



Katar Yükü



$q_1 > q_2 > q_3 > q_4 > q_5$

$$RAV_q = q_1 \cdot 1 + q_2 \cdot \eta_1 + q_3 \cdot \eta_2 + q_4 \cdot \eta_3 + q_5 \cdot \eta_4$$

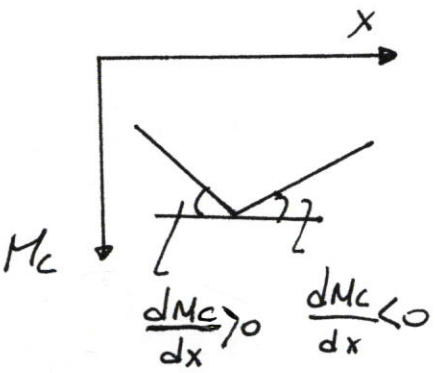
$$V_{cq_1} = -q_1 \cdot \frac{a}{L} - q_2 \cdot \eta_5 - q_3 \cdot \eta_6$$

$$V_{cq_2} = q_1 \cdot \frac{b}{L} + q_2 \cdot \eta_7 + q_3 \cdot \eta_8$$

$q_1 \cdot \eta_1 + q_2 \cdot \eta_2 + q_3 \cdot \eta_3 = q_a \cdot \eta_a \rightarrow$ a bölgesi

$q_4 \cdot \eta_4 + q_5 \cdot \eta_5 = q_b \cdot \eta_b \rightarrow$ b bölgesi

$$M_c = q_a \cdot \eta_a + q_b \cdot \eta_b$$



$$\frac{dM_c}{dx} = q_a \cdot \frac{d\eta_a}{dx} + q_b \cdot \frac{d\eta_b}{dx}$$

$$\frac{d\eta_a}{dx} = -tg\alpha_a \; ; \; \frac{d\eta_b}{dx} = tg\alpha_b$$

$$tg\alpha_a = \frac{b}{l} \quad ; \quad tg\alpha_b = \frac{a}{l}$$

$$\frac{dM_c}{dx} = -q_a \cdot tg\alpha_a + q_b \cdot tg\alpha_b$$

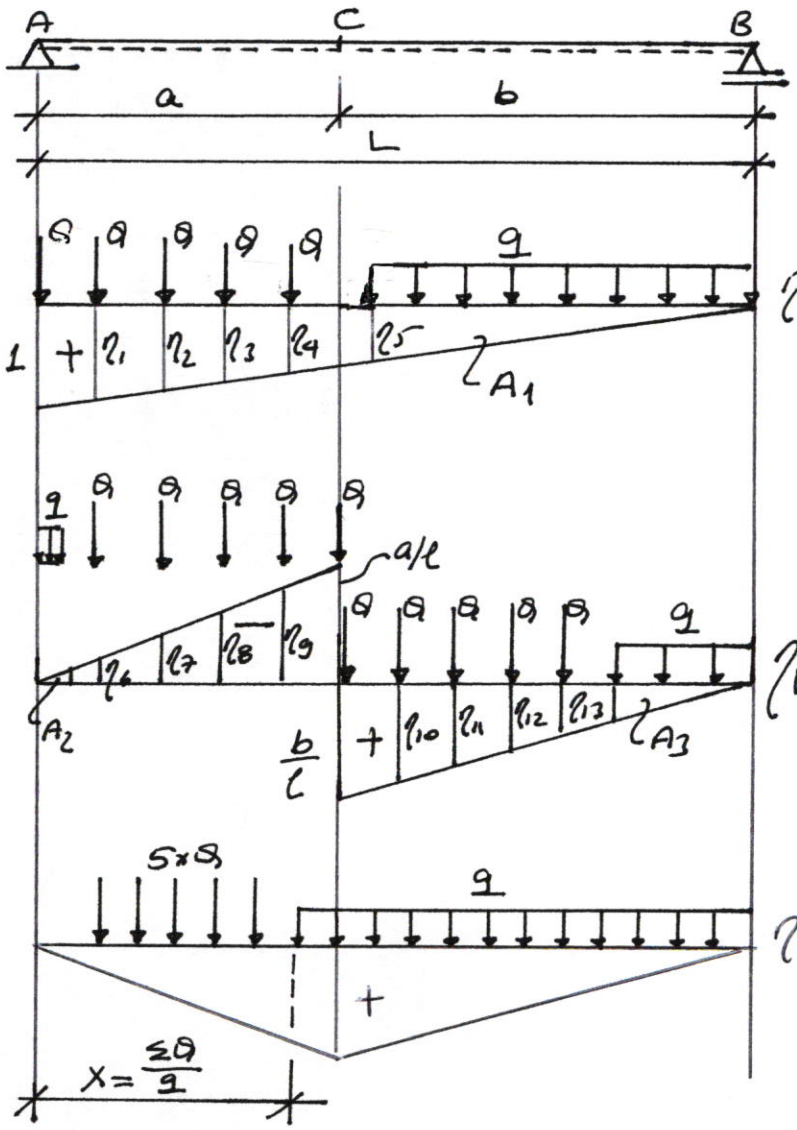
$$\frac{dM_c}{dx} > 0 \; ; \; -q_a \, tg\alpha_a + q_b \cdot tg\alpha_b > 0 \rightarrow \frac{q_a}{a} < \frac{q_b}{b}$$

$$\frac{dM_c}{dx} < 0 \; ; \; -q_a \cdot tg\alpha_a + q_b \cdot tg\alpha_b < 0 \rightarrow \frac{q_a}{a} > \frac{q_b}{b}$$

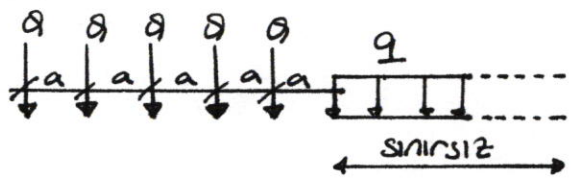
bitraz önce $\frac{q_a}{a} < \frac{q_b}{b}$

bitraz sonra $\frac{q_a}{a} > \frac{q_b}{b}$

5) Tekil ve Yayılı Yükler Oluşan Yük Katarı



Yük Katarı



$$R_{AV_q} = Q_1(1+\Sigma\eta) + q \cdot A_1$$

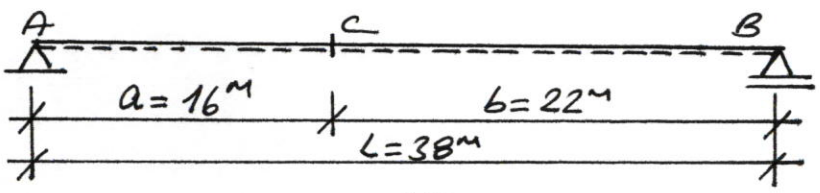
$$V_{C_{min}} = -q \cdot A_2 - Q(\Sigma\eta)$$

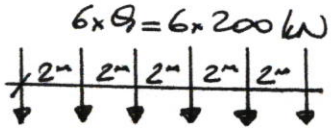
$$V_{C_{max}} = q \cdot A_3 + Q(\Sigma\eta)$$

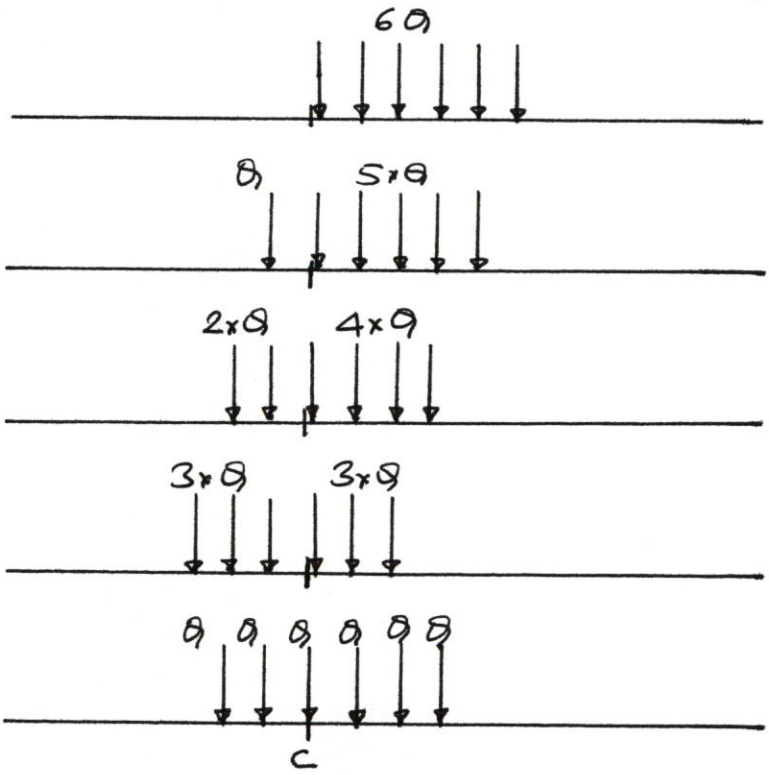
1) $\frac{\Sigma Q}{q} = X < a$ ise yükleme sisteme direk yapılır.

2) $\frac{\Sigma Q}{q} = X > a$ ise yükleme tam araştırma yapılır.

ÖRNEK:1)







$\frac{0}{16} < \frac{6\times200}{22}$

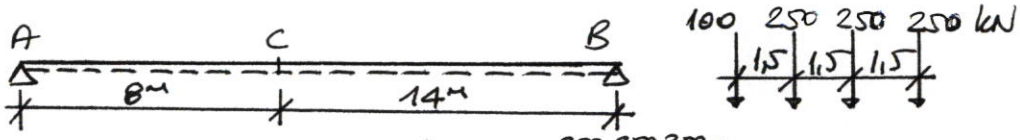
$\frac{200}{16} < \frac{5\times200}{22} \rightarrow 12,5 < 45,45$

$\frac{2\times200}{16} < \frac{4\times200}{22} \rightarrow 25 < 36,36$

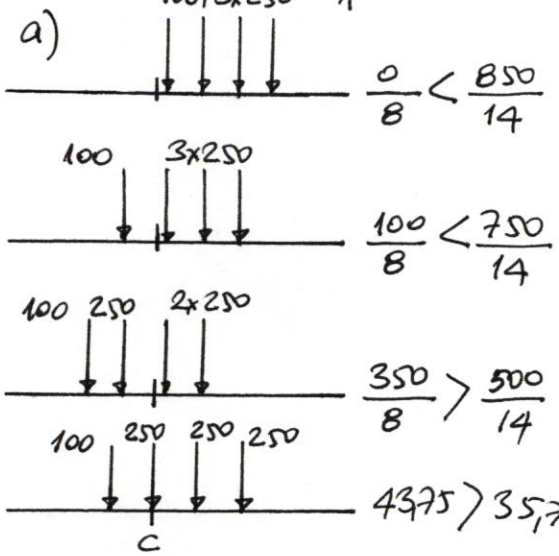
$\frac{3\times200}{16} > \frac{3\times200}{22} \rightarrow 37,5 > 27,27$

SONUÇ YÜKLEME

ÖRNEK:2)



a)



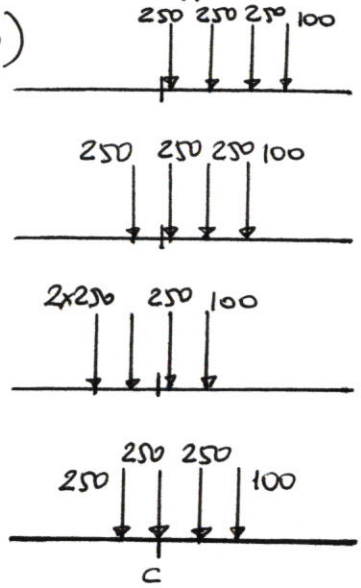
$\frac{0}{8} < \frac{850}{14}$

$\frac{100}{8} < \frac{750}{14}$

$\frac{350}{8} > \frac{500}{14}$

43,75 > 35,71

b)



$\frac{0}{8} < \frac{850}{14}$

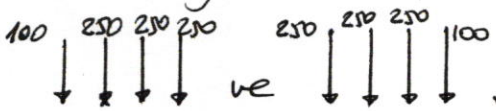
$\frac{250}{8} < \frac{600}{14}$

$\frac{500}{8} > \frac{350}{14}$

625 > 25 ✓

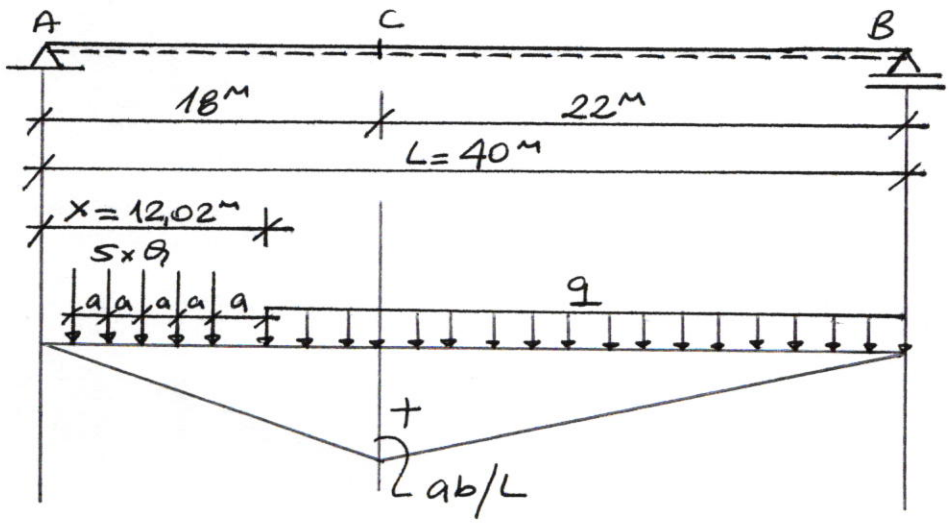
SONUÇ YÜKLEME

* Katar yükü verilen kirişi üzerinde



ve şekilde gezdirilmiştir. Birinci yüklemede sonuç 43,75 > 35,71, 2. yüklemede 625 > 25 olarak bulunmuştur.

ÖRNEK:3)



M_C

$a = 1{,}6^{M}$

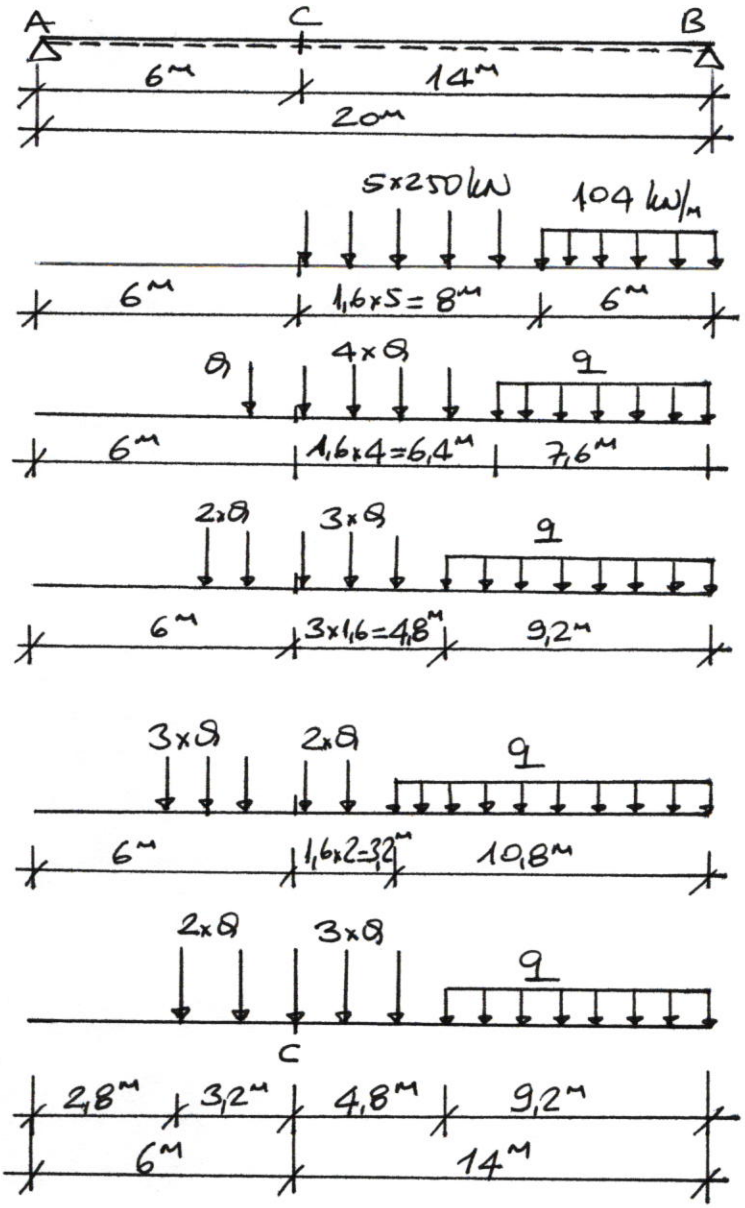
$Q = 250\ kN$

$q = 104\ kN/m$

$x = \frac{5 \times 250}{104} = 12{,}02^{M}$

$X = 12{,}02^{M} < a = 18^{M}$

ÖRNEK:4)



$\frac{\Sigma Q}{q} = \frac{5 \times 250}{104} = 12{,}02^{m} > a = 6^{m}$

$\frac{Q_a}{a} < \frac{Q_b}{b} \rightarrow \frac{Q_a}{a} > \frac{Q_b}{b}$

$\frac{0}{6} < \frac{5 \times 250 + 104 \times 6}{14} = 133{,}86$

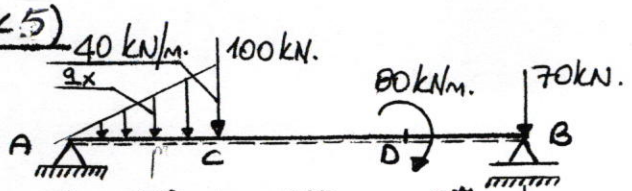
$\frac{250}{6} < \frac{4 \times 250 + 104 \times 7{,}6}{14} = 127{,}89$

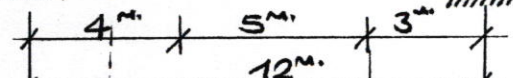
$\frac{2 \times 250}{6} = 83{,}33 < \frac{3 \times 250 + 104 \times 9{,}2}{14} = 121{,}91$

$\frac{3 \times 250}{6} = 125 > \frac{2 \times 250 + 104 \times 10{,}8}{14} = 115{,}94$

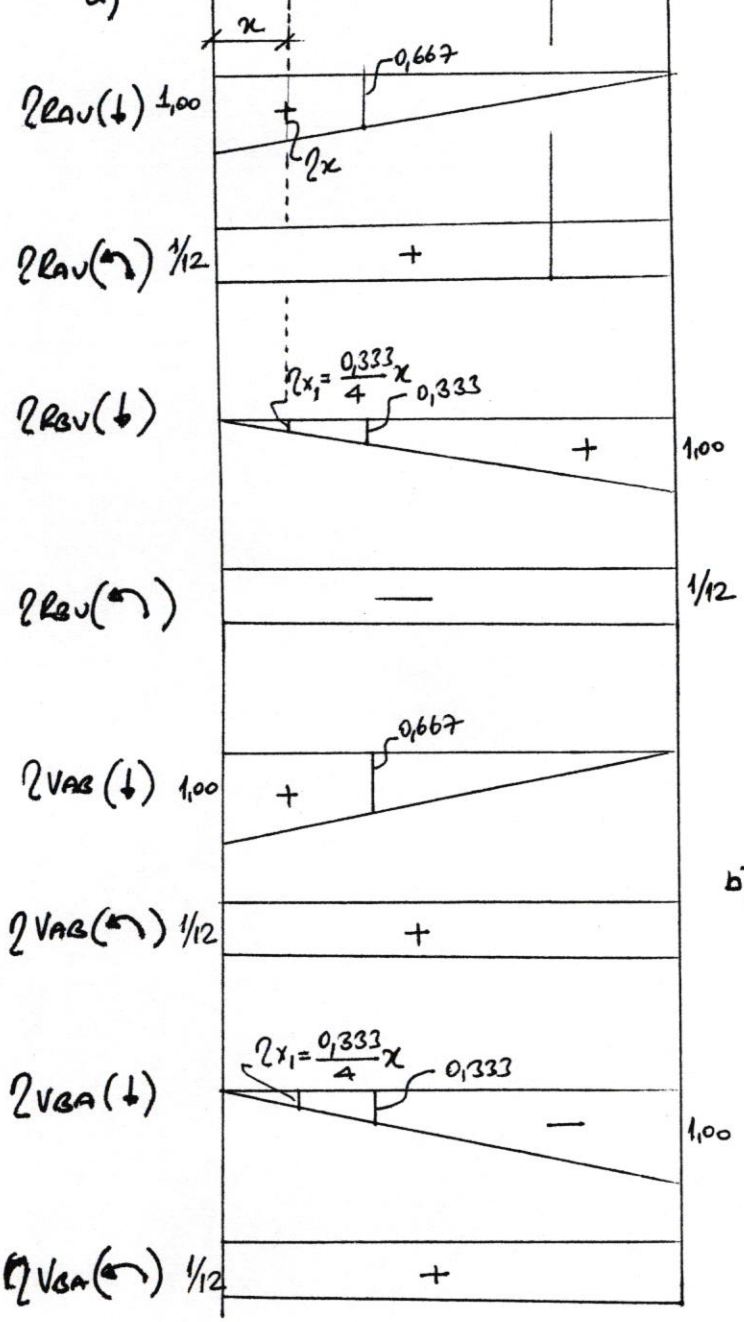
SONUÇ YÜKLEME

ÖRNEK 5)





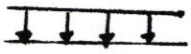
a)

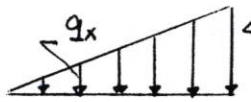


Şekilde verilen basit kirişin;

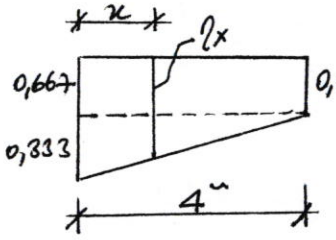
a) $\eta_{R_{AV}}$, $\eta_{R_{BV}}$, $\eta_{V_{AB}}$, $\eta_{V_{BA}}$, $\eta_{V_{CA}}$ ve η_{M_M} tesir çizgilerini çizerek.

b) R_{AV}, R_{BV}, V_{AB}, V_{BA}, V_{CA}, V_{CB} ve M_C kesit tesirlerini tesir çizgilerinden faydalanarak bulunuz.

c)  $q = 20$ kN/m hareketli yükten dolayı oluşan $\overset{max}{min}\ V_{CA}$ değerlerini bulunuz.



$9x = \frac{40}{4} x = 10x$



$\frac{0,333}{4} = \frac{\eta x'}{4-x}$

$\eta x = \left(\frac{0,333(4-x)}{4} + 0,66\right.$

b)

$$R_{AV} = \int_0^4 9(x).\ \eta(x) + 100.0,667 - 80.1/12 =$$

$$= \int_0^4 10x\left(\frac{0,333(4-x)}{4} + 0,667\right)dx + 60,033$$

$$= 122,27 \text{ kN.}$$

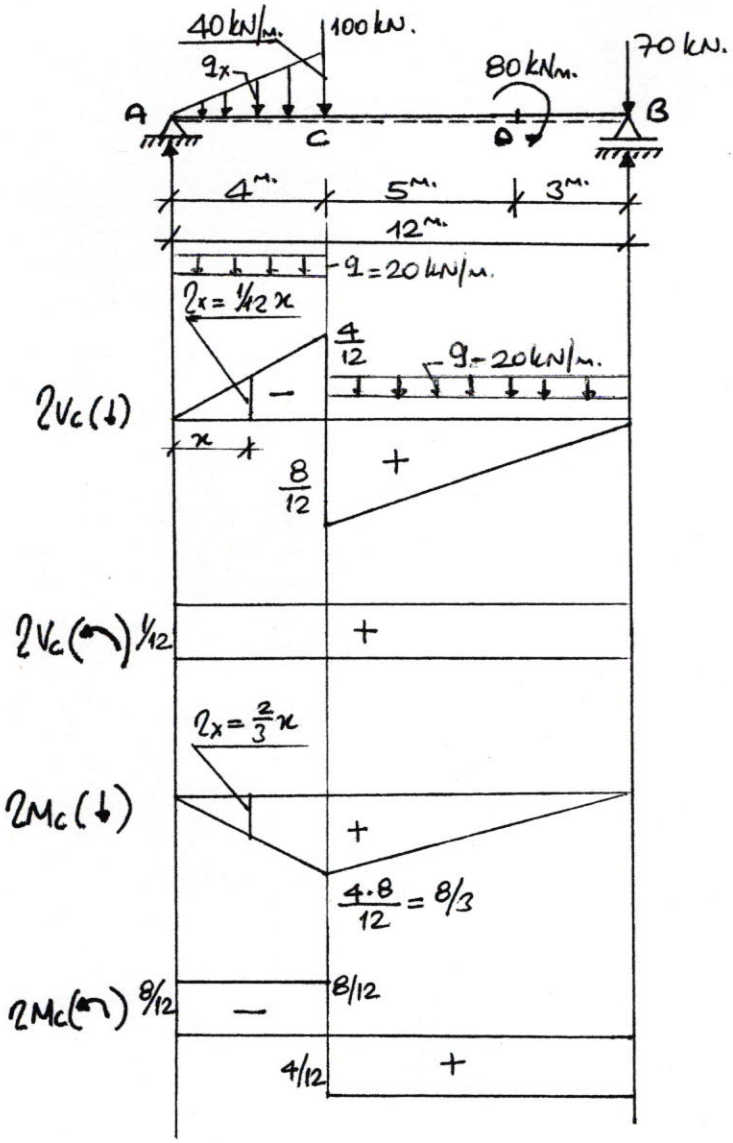
$$R_{BV} = \int_0^4 10x.\frac{0,333}{4}\,x\,dx + 100.0,333 + 80.1/12 + 70.1,00$$

$$= 127,73 \text{ kN.}$$

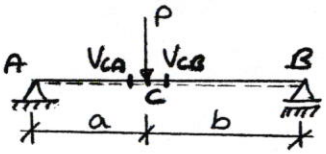
$$V_{AB} = R_{AV} = 122,27 \text{ kN.}$$

$$V_{BA} = -\int_0^4 10x.\frac{0,333}{4}\,x\,dx - 100.0,333 - 80.1/12 - 70.1,00 =$$

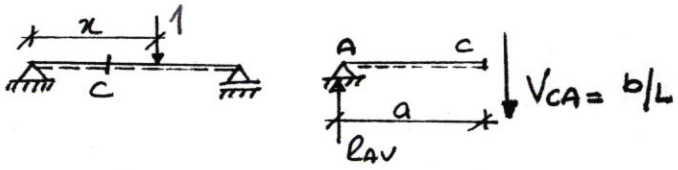
$$V_{BA} = -127,73 \text{ kN.}$$



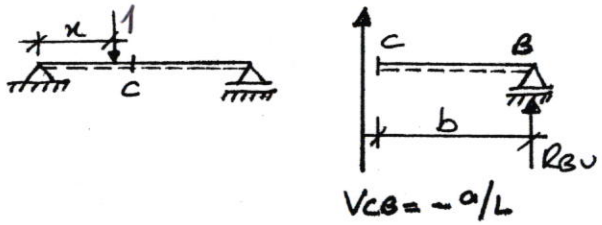
NOT:



- V_{CA} kesme kuvveti değeri sisteme soldan yaklasım durumunda yükün dahil olmadığı durumdur. Yani yükün C-B arasında ($a<x<L$) olması durumudu



- V_{CB} kesme kuvveti değeri sisteme soldan yaklasım durumunda yükün dahil olduğu durumdur. Yani yükün A-C ($0<x<a$) arasında olması durumudur



---

c) $V_{CA_g} = 42,22$ kN.

$V_{CA_{g_1}} = -20 \cdot \left(\frac{\frac{4}{12} \cdot 4}{2}\right) = -13,33$ kN.

$V_{CA_{g_2}} = 20 \cdot \left(\frac{8/12 \cdot 8}{2}\right) = 53,33$ kN.

$\min V_{CA} = 42,22 - 13,33 = 28,89$ kN.

$\max V_{CA} = 42,22 + 53,33 = 95,55$ kN.

b) $V_{CA} = -\int_0^4 10x \cdot \frac{1}{12} x\,dx + 100 \cdot \frac{8}{12} - 80 \cdot \frac{1}{12}$

$V_{CA} = 42,22$ kN.

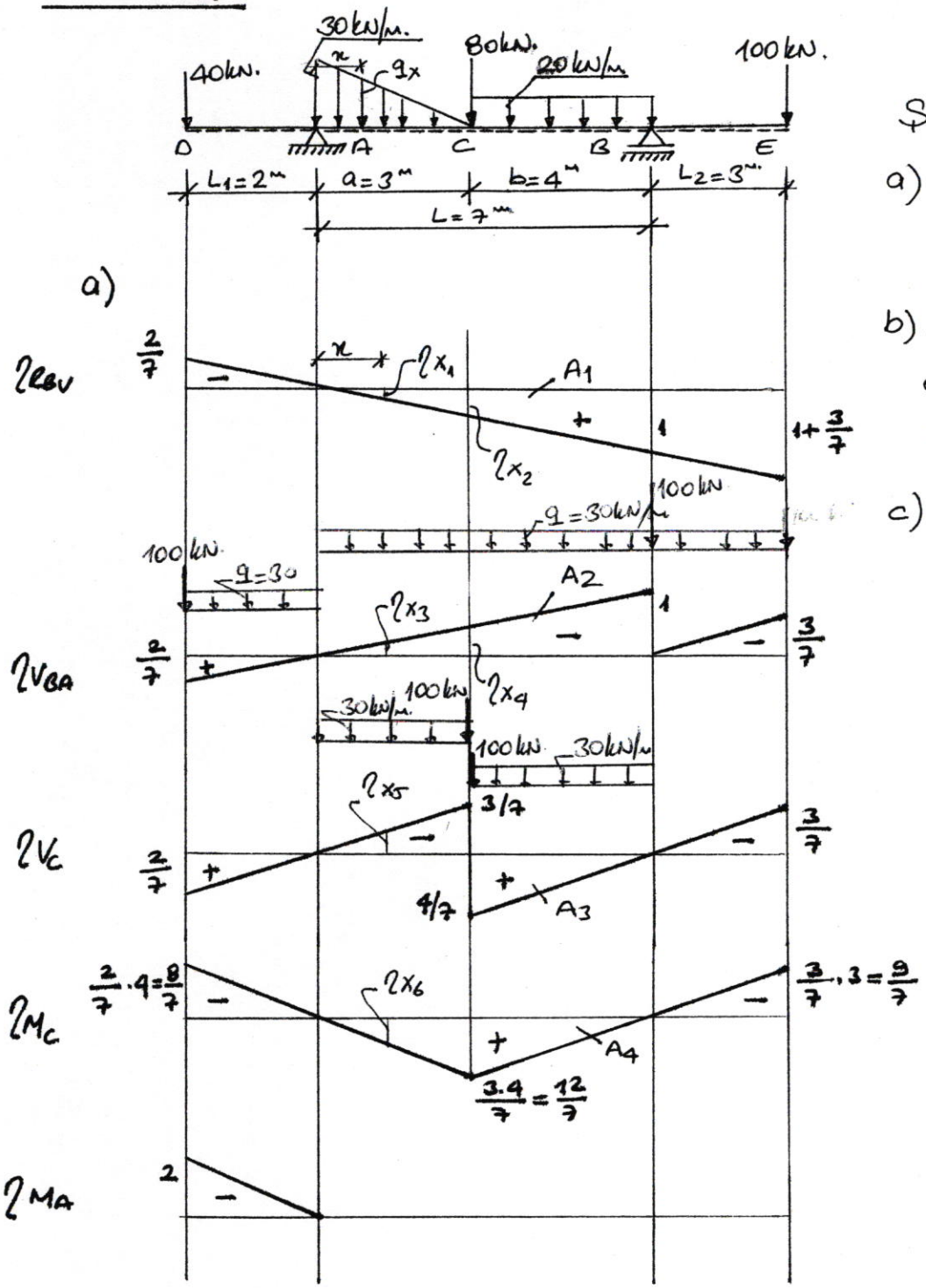
$V_{CB} = -\int_0^4 10x \cdot \frac{1}{12} x\,dx - 100 \cdot \frac{4}{12} - 80 \cdot \frac{1}{12}$

$V_{CB} = -57,78$ kN.

$M_C = \int_0^4 10x \cdot \frac{2}{3} x\,dx + 100 \cdot \frac{8}{3} - 80 \cdot \frac{4}{12}$

$M_C = 382,22$ kNm.

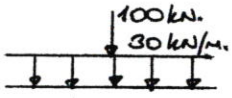
ÖRNEK: 6)



Şekilde verilen cisimli kirişin;

a) $\eta_{R_{BV}}$, $\eta_{V_{BA}}$, η_{V_C}, η_{M_C}, η_{M_A} tesir çizgilerini çiziniz.

b) Şekildeki sabit yüklerden oluşan R_{BV}, V_{BA}, V_C, M_C, M_A kesit tesirlerini bulunuz.

c) 

Hareketli yüklerden oluşan $\min_{max} V_{BA}$, $\min_{max} V_{CA}$ değerlerini bulunuz

$$\frac{30}{3} = \frac{q(x)}{3-x}$$

$$q(x) = 10(3-x)$$

$$\frac{1}{7} = \frac{\eta x_1}{x} \Rightarrow \eta x_1 = \frac{1}{7}x$$

$$\eta x_2 = \frac{3}{7}$$

$$\eta x_3 = \frac{1}{7}x$$

$$\eta x_4 = \frac{3}{7}$$

$$\frac{3/7}{3} = \frac{\eta x_5}{x} \Rightarrow \eta x_5 = \frac{1}{7}x$$

$$\frac{12/7}{3} = \frac{\eta x_6}{x} \Rightarrow \eta x_6 = \frac{4}{7}x$$

$$A_1 = \left(1+\frac{3}{7}\right)\cdot\frac{4}{2} = \frac{20}{7}$$

$$A_2 = \left(1+\frac{3}{7}\right)\cdot\frac{4}{2} = \frac{20}{7}$$

$$A_3 = \left(\frac{4}{7}\cdot 4\right)\cdot\frac{1}{2} = \frac{8}{7}$$

$$A_4 = \left(\frac{12}{7}\cdot 4\right)\cdot\frac{1}{2} = \frac{24}{7}$$

b) $$R_{BV} = \int_0^3 10(3-x)\cdot\frac{x}{7}\,dx - 40\cdot\frac{2}{7} + 80\cdot\frac{3}{7} + 20\cdot\frac{20}{7} + 100\cdot\left(1+\frac{3}{7}\right)$$

$$R_{BV} = 229{,}29 \text{ kN.}$$

$$V_{BA} = -\int_0^3 10(3-x)\cdot\frac{x}{7}\,dx + 40\cdot\frac{2}{7} - 80\cdot\frac{3}{7} - 20\cdot\frac{20}{7} - 100\cdot\frac{3}{7}$$

$$V_{BA} = -129{,}29 \text{ kN.}$$

$$V_{CA} = -\int_0^3 10(3-x).\frac{x}{7}\,dx + 40.\frac{2}{7} + 80.\frac{4}{7} + 20.\frac{8}{7} - 100.\frac{3}{7} = 30{,}71\ kN.$$

$$V_{CB} = -\int_0^3 10(3-x).\frac{x}{7}\,dx + 40.\frac{2}{7} - 80.\frac{3}{7} + 20.\frac{8}{7} - 100.\frac{3}{7} = -49{,}29\ kN.$$

$$M_C = \int_0^3 10(3-x).\frac{4}{7}x\,dx - 40.\frac{8}{7} + 80.\frac{12}{7} + 20.\frac{24}{7} - 100.\frac{9}{7} = 57{,}14\ kNm.$$

$$M_A = -40.2 = -80\ kNm.$$

c) $$V_{BAq_1} = -30.\left(\frac{1.7}{2} + \frac{3/7.3}{2}\right) - 100.1 = -224{,}29\ kN.$$

$$V_{BAq_2} = 30.\left(\frac{2/7.2}{2}\right) + 100.\frac{2}{7} = 37{,}14\ kN.$$

$$\min V_{BA} = -129{,}29 - 224{,}29 = -353{,}58\ kN.$$

$$\max V_{BA} = -129{,}29 + 37{,}14 = -92{,}15\ kN.$$

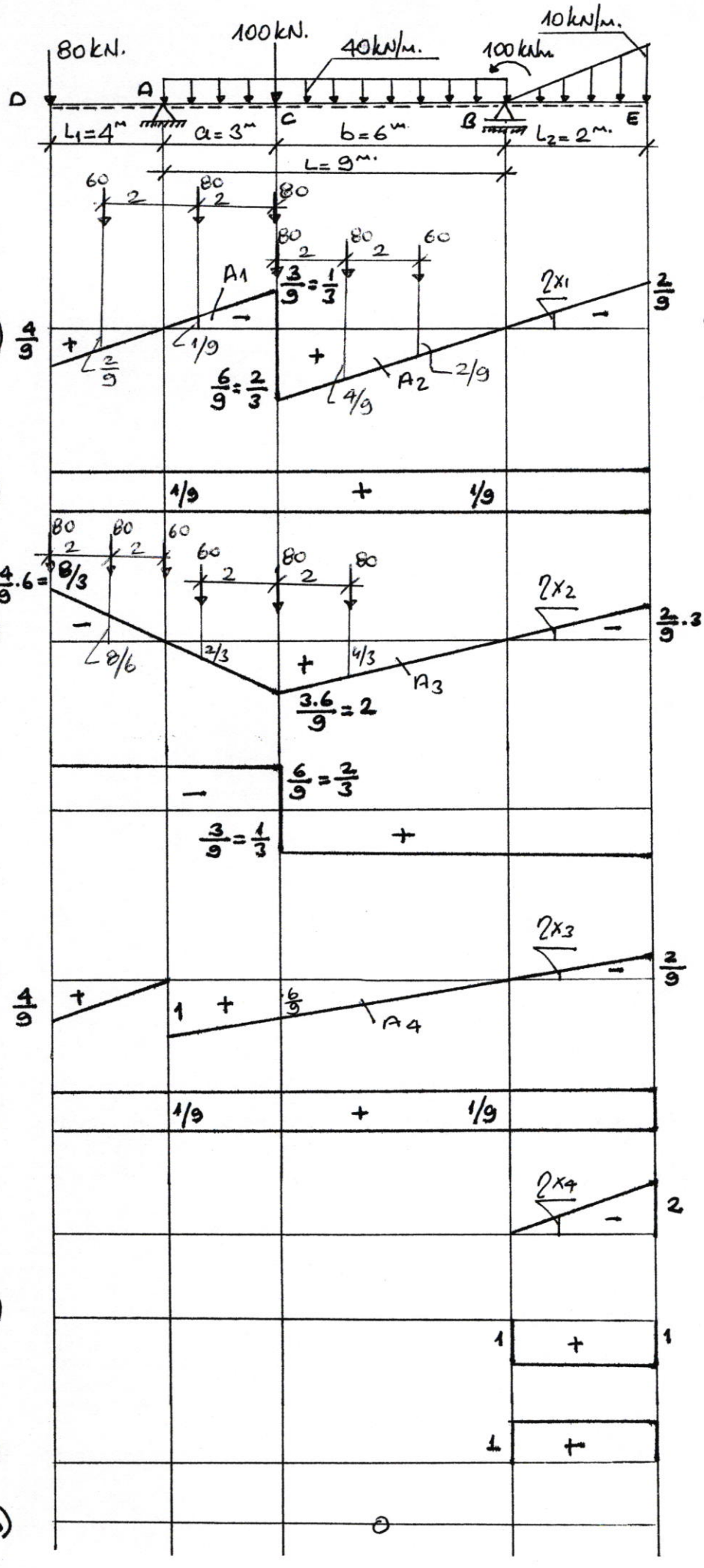
$$V_{Cq_1} = -30.\left(\frac{3/7.3}{2}\right) - 100.\frac{3}{7} = -62{,}14\ kN.$$

$$V_{Cq_2} = 30.\left(\frac{4/7.4}{2}\right) + 100.\frac{4}{7} = 91{,}43\ kN.$$

$$\min V_{CA} = 30{,}71 - 62{,}14 = -31{,}43\ kN.$$

$$\max V_{CA} = 30{,}71 + 91{,}43 = 122{,}14\ kN.$$

ÖRNEK:7)

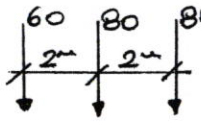


Şekilde verilen kirişin;

a) ηV_C, ηM_C, ηV_{AB}, ηM_B, ηV_{BE} tesir çizgilerini çiziniz.

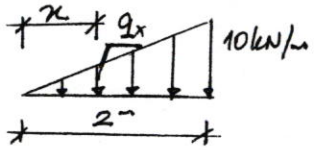
b) V_C, M_C, V_{AB}, M_B, V_{BE} kesit tesirlerini bulunuz.

c)



Hareketli yük için;

$\max_{\min} V_C$, $\max_{\min} M_C$ değerlerini hesaplayınız.



$$\frac{10}{2} = \frac{q(x)}{x} \rightarrow \underline{q(x) = 5x}$$

$$\frac{2/9}{2} = \frac{\eta x_1}{x} \rightarrow \eta x_1 = \frac{1}{9}x$$

$$\frac{2/3}{2} = \frac{\eta x_2}{x} \rightarrow \eta x_2 = \frac{1}{3}x$$

$$\eta x_3 = \frac{1}{9}x$$

$$\frac{2}{2} = \frac{\eta x_4}{x} \rightarrow \eta x_4 = 1x$$

$$A_1 = -\frac{1/3 \cdot 3}{2} = -0{,}5$$

$$A_2 = \frac{2/3 \cdot 6}{2} = 2$$

$$A_3 = \frac{2 \cdot 9}{2} = 9$$

$$A_4 = \frac{1 \cdot 9}{2} = 4{,}5$$

b)

$$V_{CA} = 80 \cdot \frac{4}{9} - 40 \cdot 0{,}5 + 100 \cdot \frac{2}{3} + 40 \cdot 2 - \int_0^2 5x \cdot \frac{1}{9} x \, dx + 100 \cdot \frac{1}{9} = 171{,}85 \text{ kN}$$

$$V_{CB} = 80 \cdot \frac{4}{9} - 40 \cdot 0{,}5 - 100 \cdot \frac{1}{3} + 40 \cdot 2 - \int_0^2 5x \cdot \frac{1}{9} x \, dx + 100 \cdot \frac{1}{9} = 71{,}85 \text{ kN}.$$

$$M_C = -80 \cdot \frac{8}{3} + 40 \cdot 9 + 100 \cdot 2 - \int_0^2 5x \cdot \frac{1}{3} x \, dx + 100 \cdot \frac{1}{3} = 375{,}55 \text{ kNm}.$$

$$V_{AB} = 80 \cdot \frac{4}{9} + 100 \cdot \frac{6}{9} + 40 \cdot 4{,}5 - \int_0^2 5x \cdot \frac{1}{9} x \, dx + 100 \cdot \frac{1}{9} = 291{,}85 \text{ kN}.$$

$$M_{BE} = -\int_0^2 5x \cdot x \, dx = -13{,}33 \text{ kNm}$$

$$M_{BC} = -13{,}33 + 100 \cdot 1 = 86{,}67 \text{ kNm}.$$

$$V_{BE} = \frac{10 \cdot 2}{2} \cdot 1 = 10 \text{ kN}.$$

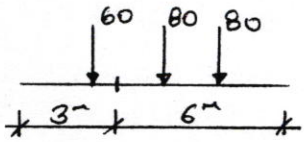
c)

$$V_{cg_1} = -80 \cdot \frac{1}{3} - 80 \cdot \frac{1}{9} + 60 \cdot \frac{2}{9} = -22{,}22 \text{ kN}.$$

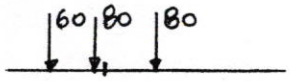
$$V_{cg_2} = 80 \cdot \frac{2}{3} + 80 \cdot \frac{4}{9} + 60 \cdot \frac{2}{9} = 102{,}22 \text{ kN}.$$

$$V_{CA_{min}} = 171{,}85 - 22{,}22 = 149{,}63 \text{ kN} \; ; \; V_{CA_{max}} = 171{,}85 + 102{,}22 = 274{,}07 \text{ kN}.$$

$$V_{CB_{min}} = 71{,}85 - 22{,}22 = 49{,}63 \text{ kN} \; ; \; V_{CB_{max}} = 71{,}85 + 102{,}22 = 174{,}07 \text{ kN}.$$



$$\frac{q_a}{a} < \frac{q_b}{b} \Rightarrow \frac{60}{3} < \frac{160}{6} \Rightarrow 20 < 26{,}67 \; \checkmark$$



$$\frac{q_a}{a} > \frac{q_b}{b} \Rightarrow \frac{140}{3} > \frac{80}{6} \Rightarrow 46{,}67 > 13{,}33 \; \checkmark$$

$$M_{cg_1} = -80 \cdot \frac{8}{3} - 80 \cdot \frac{8}{6} = -320 \text{ kNm}.$$

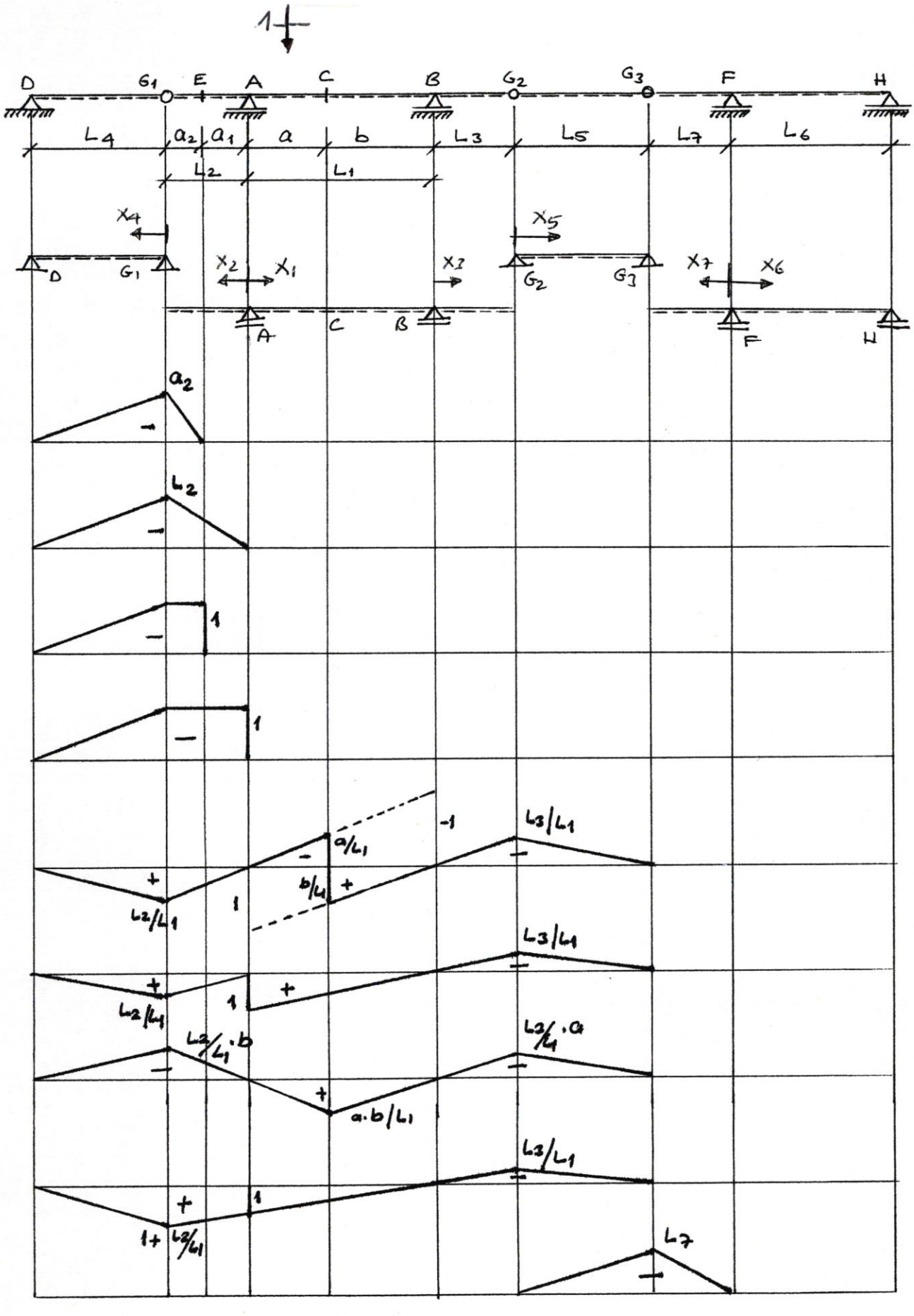
$$M_{cg_2} = 60 \cdot \frac{2}{3} + 80 \cdot 2 + 80 \cdot \frac{4}{3} = 306{,}67 \text{ kNm}.$$

$$M_{c_{min}} = 375{,}55 - 320 = 55{,}55 \text{ kNm}.$$

$$M_{c_{max}} = 375{,}55 + 306{,}67 = 682{,}22 \text{ kNm}.$$

VII.9) GERBER KİRİŞ TESİR ÇİZGİSİ

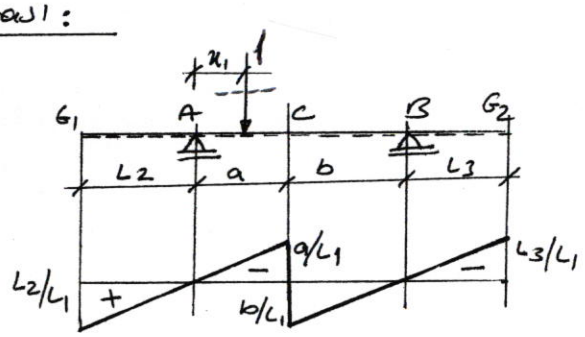
a) Düşey yük Etkisi



ηV_C kesit tesirinin bulunuşu:

Not: Gerber kirişte tesir çizgisi çizilirken önce taşıma şeması çizilir. Böylece birim yükün gezinmesinde sıfır olan bölgeler tayin edilir. Birim yük her kiriş üzerinde gezdirilerek kesit tesiri tesir çizgisi çizilmiş olur.

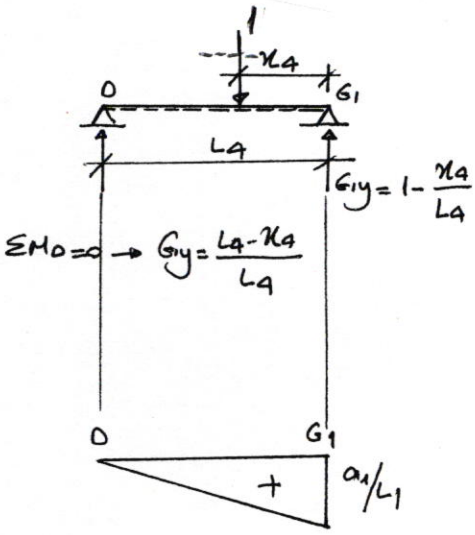
$G_1 - G_2$ arası:

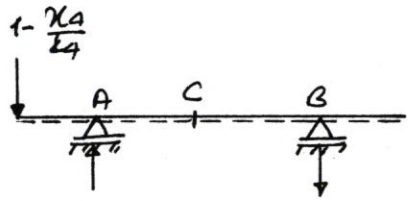


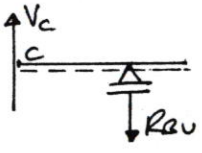
- Taşıma şemasına göre $G_1 - G_2$ kirişi taşıyıcı $D-G_1$ ve G_2-G_3 kirişi taşınan kirişlerdir. Dolayısıyla $D-G_3$ arası kirişler üzerinde birim kuvvetin gezinmesinde C noktasında kesme kuvveti oluşur.

- Birim kuvvetin $G_1 - G_3$ arasında gezinmesinde C noktasında oluşan kesme kuvveti tesir çizgisi diyagramı çıkmalı kirişlerde anlatıldığı gibi çizilir.

$D - G_1$ arası:







$$\Sigma M_B = 0 \rightarrow R_{AV} = \frac{\left(1 - \frac{x_4}{L_4}\right)(L_1 + L_2)}{L_1}$$

$$\Sigma M_A = 0 \rightarrow R_{BV} = \frac{\left(1 - \frac{x_4}{L_4}\right) \cdot a_1}{L_1}$$

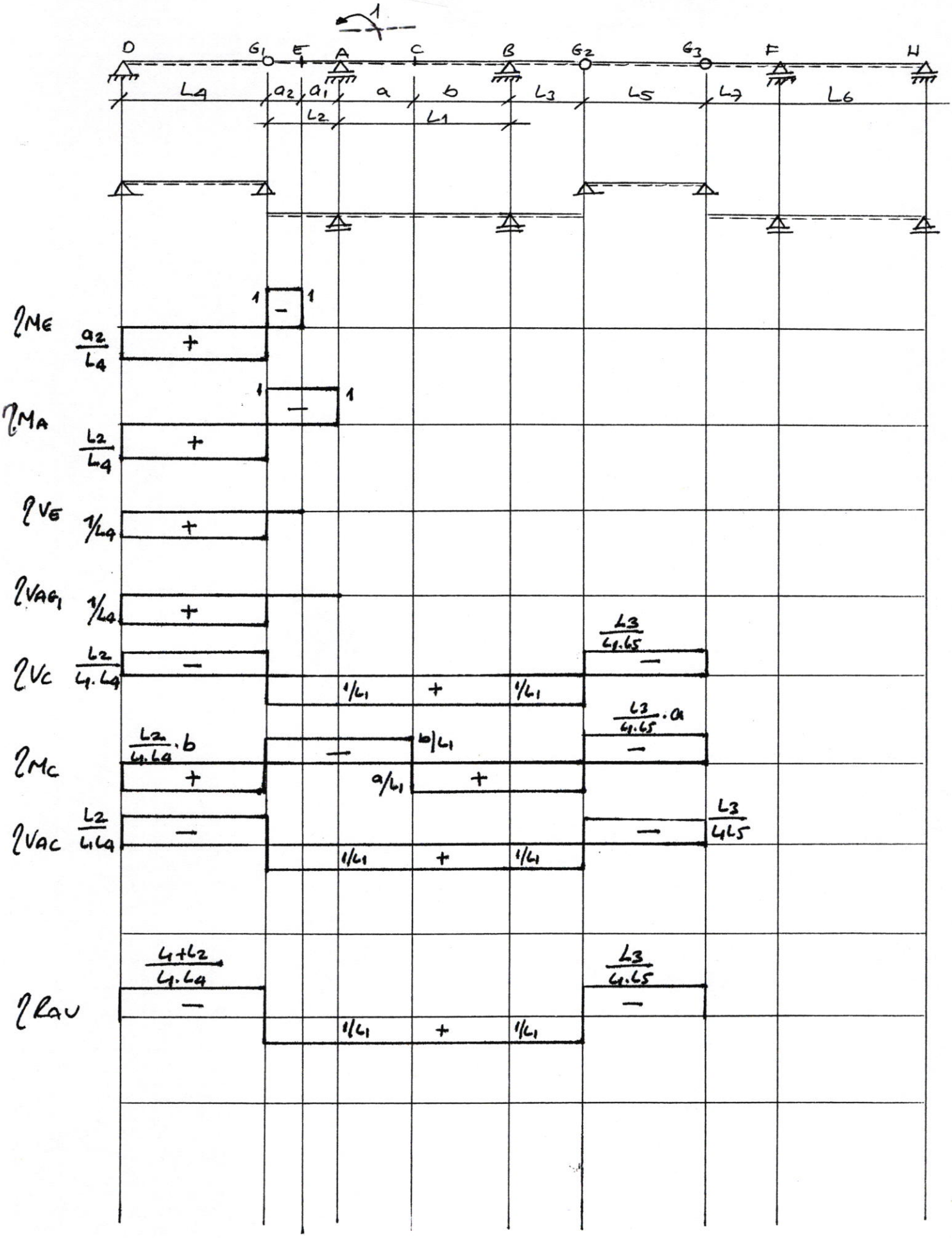
$$\eta V_C = \frac{\left(1 - \frac{x_4}{L_4}\right) \cdot a_1}{L_1}$$

$x_4 = 0 \rightarrow \eta V_C = a_1/L_1$

$x_4 = L_4 \rightarrow \eta V_C = 0$

- Benzer şekilde $G_2 - G_3$ arası tesir çizgisi değerleri elde edilir.

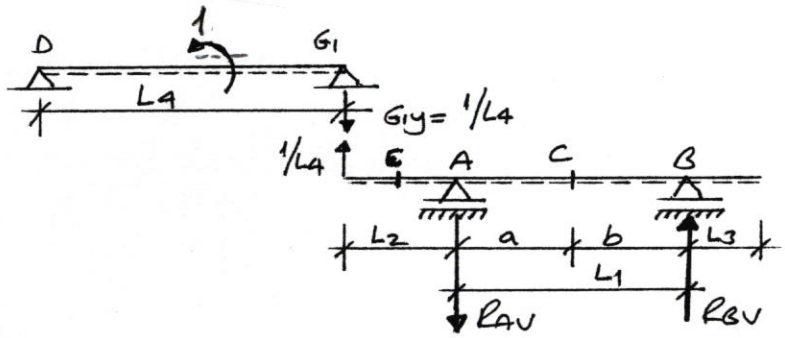
b) Moment Einfluß



Birim moment tesirinde kesit tesiri tesir çizgilerinin bulunuşu:

Not: Birim momentin $G_1 - G_2$ arasında gezinmesinden oluşan tesir çizgisi çıkmalı kirişte anlatıldığı gibidir.

D – G_1 arası:

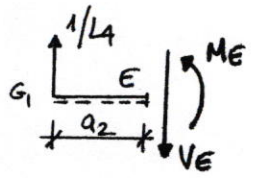


$$\Sigma M_B = 0 \rightarrow R_{AV} = \frac{1/L_4\,(L_1+L_2)}{L_1}$$

$$\zeta R_{AV} = \frac{L_1+L_2}{L_1 . L_4}$$

$$\Sigma M_A = 0 \Rightarrow R_{BV} = \frac{1/L_4 . L_2}{L_1}$$

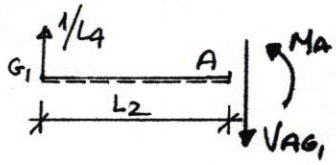
$$\zeta R_{BV} = \frac{L_2}{L_1 . L_4}$$



D-G1

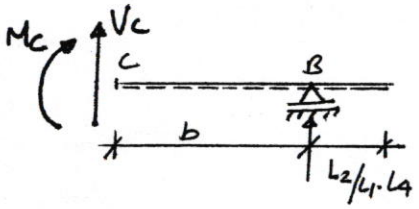
$$\zeta V_E = +1/L_4$$

$$\zeta M_E = a_2/L_4$$



$$\zeta V_{AG_1} = 1/L_4$$

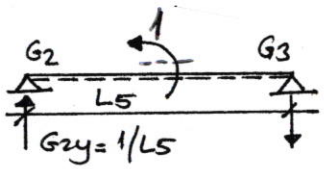
$$\zeta M_A = L_2/L_4$$



$$\zeta V_{AC} = -\frac{L_2}{L_1 . L_4}\ , \quad \zeta V_C = -\frac{L_2}{L_1 . L_4}$$

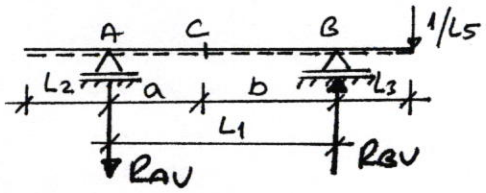
$$\zeta M_C = \frac{L_2}{L_1 . L_4} . b\ , \quad \zeta R_{AV} = \frac{L_1+L_2}{L_1 . L_4}$$

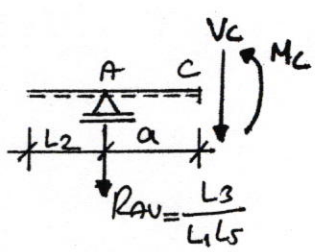
G_2 – G_3 arası:



$$\Sigma M_B = 0 \Rightarrow R_{AV} = \frac{L_3 . 1/L_5}{L_1} = \frac{L_3}{L_1 . L_5}$$

$$\Sigma M_A = 0 \Rightarrow R_{BV} = \frac{1/L_5\,(L_1+L_3)}{L_1} = \frac{L_1+L_3}{L_1 . L_5}$$





$$\zeta V_C = -\frac{L_3}{L_1 L_5} \quad ; \quad \zeta V_{AC} = -\frac{L_3}{L_1 L_5}$$

$$\zeta M_C = -\frac{L_3}{L_1 L_5} . a \quad ; \quad \zeta R_{AV} = -\frac{L_3}{L_1 L_5}$$

Gerber kirişlerde tesir çizgilerinin çizilmesinde işlem sırası;

1- Sistemin taşıma şeması tayin edilir.

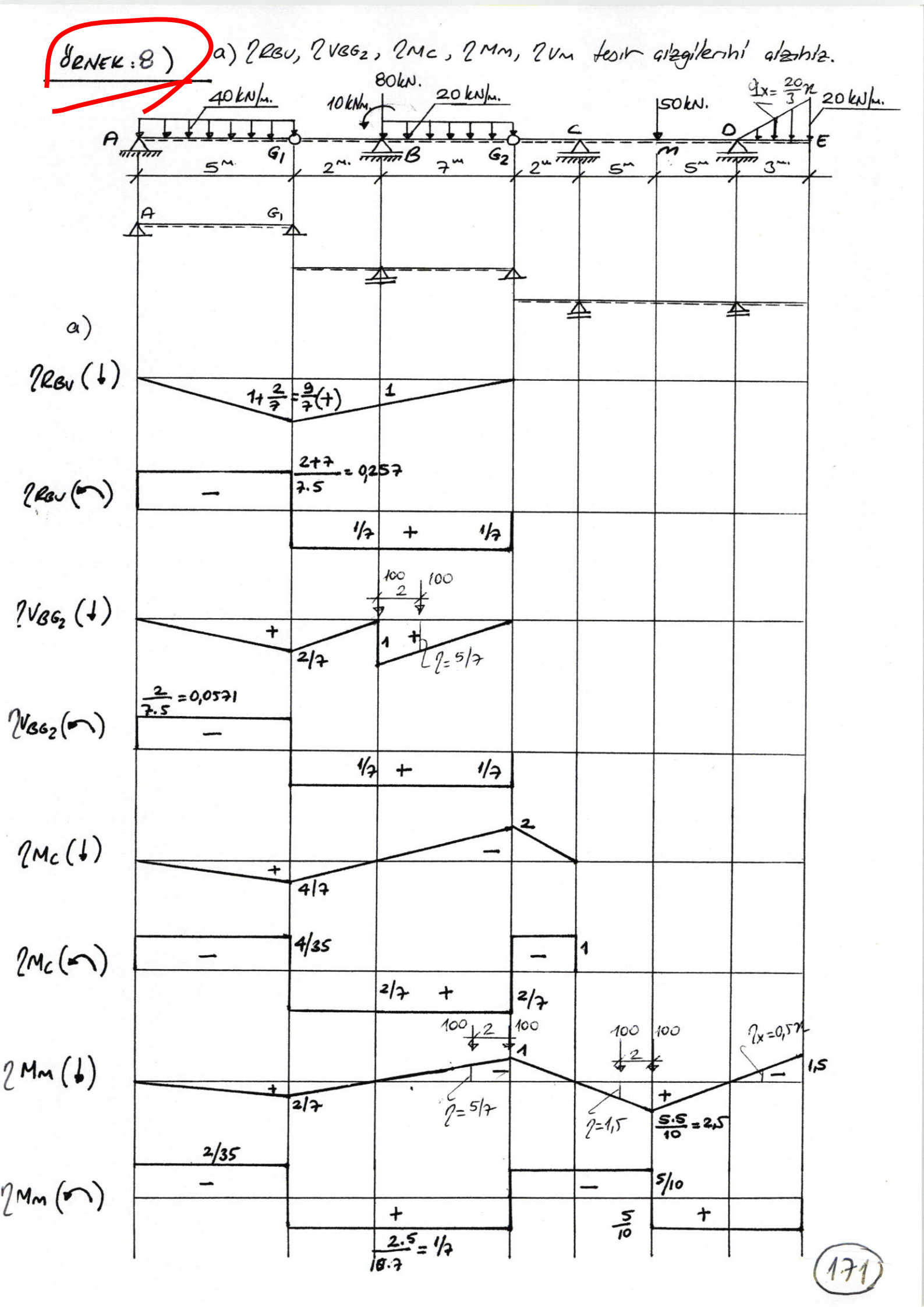
2- Bir birimlik kuvvet sistem üzerinde belirli doğru/eğride hareket ederken tesir çizgisi çizilecek olan kesitte kesit tesirinin sıfır olduğu bölgeler taşıma şeması yardımıyla tayin edilir.

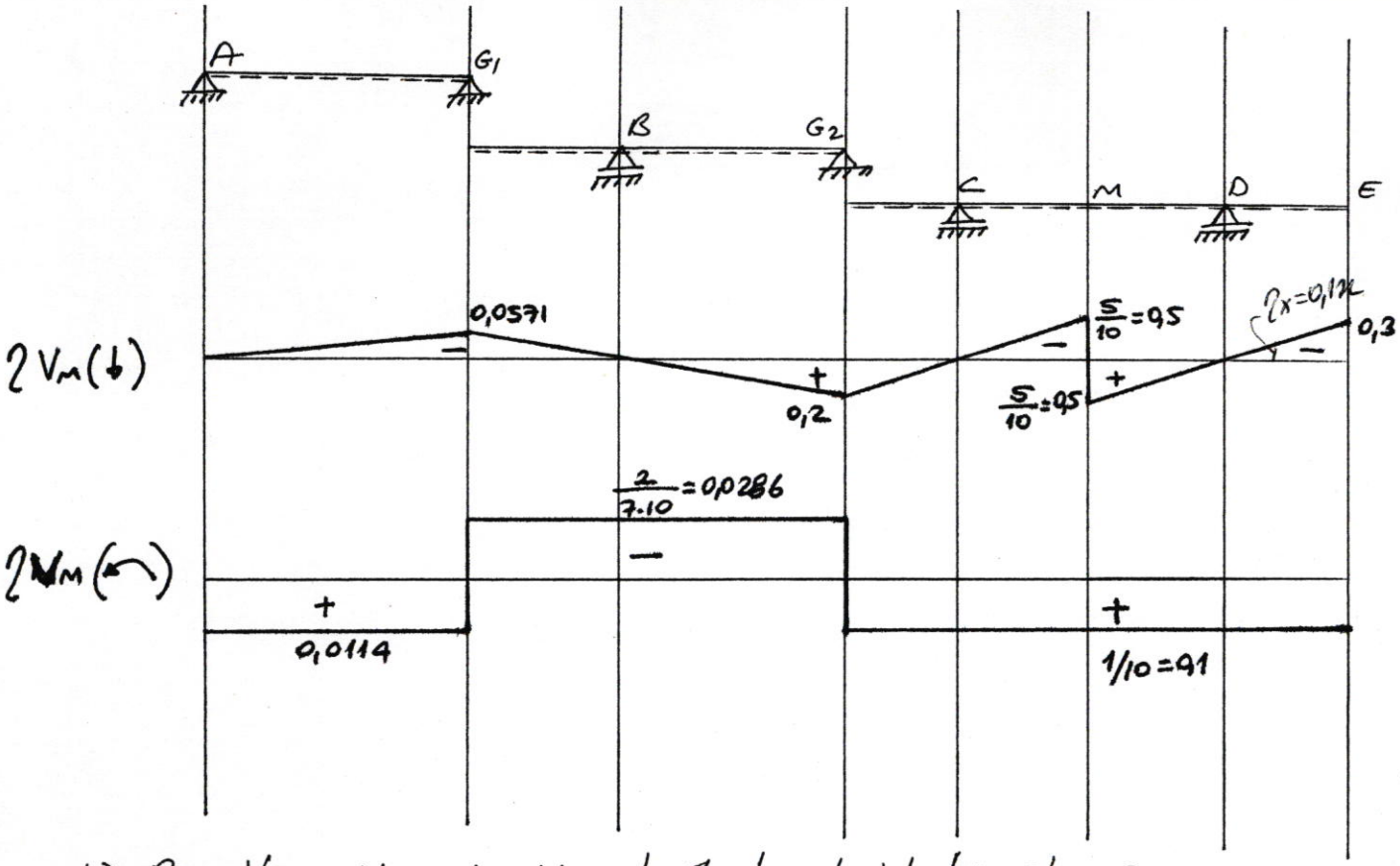
3- Tesir çizgisinin sıfır olduğu noktalar tespit edilir. Gerber kirişte bu noktalar genellikle ara mesnet hizalarıdır. Çizilen mesnet tepkisine ait tesir çizgisi diyagramı ise o mesnette tesir çizgisi sıfır olmayacaktır.

4- Tesir çizgisi tayin edilecek kesitin bulunduğu açıklıkta gerber kirişin tesir hattı ile basit kiriş veya konsol kirişin tesir çizgileri birbirinin aynıdır. Tesir çizgilerinin çıkmalar üzerindeki kısımları basit kirişin tesir çizgileri uzatılarak bulunur.

5- Tesir çizgilerinin çizilmesinde süreklilliğe dikkat edilmelidir.

ÖRNEK: 8) a) $?R_{BV}$, $?V_{BG_2}$, $?M_C$, $?M_M$, $?V_M$ tesir cizgileri cizelim.





b) R_{BV}, V_{BG_2}, M_C, M_M, V_M kesit tesirlerini hesaplayınız.

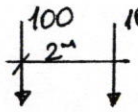
$$R_{BV} = 40.\left(\frac{9/7.5}{2}\right) + 80.1 + 20.(1.7/2) + 10.1/7 = 280\ kN.$$

$$V_{BG_2} = 40.\left(\frac{2/7.5}{2}\right) + \underbrace{80.1}_{\text{alınmaz}} + 20(1.7/2) + 10.1/7 = 100\ kN.$$

$$M_C = 40.\left(\frac{4/7.5}{2}\right) - 20.(2.7/2) + 10.2/7 = -80\ kNm.$$

$$M_M = 40.\left(\frac{2/7.5}{2}\right) - 20.(1.7/2) + 50.2,5 - \int_0^3 \frac{20}{3}x.0,5x\,dx + 10.1/7 = 55\ kNm.$$

$$V_{M_C} = -40.(0,0571.5/2) + 20(0,2.7/2) + 50.0,5 - \int_0^3 \frac{20}{3}x.0,1x\,dx - 10.0,0286 = 27\ kN.$$

c) 

yayılı yük durumunda $\max_{\min} V_{BG_2}$, $\max_{\min} M_M$ değerlerini hesaplayınız.

$$V_{BG_2 q} = 100.1 + 100.5/7 = 171,43\ kN.$$

$$\begin{cases} \min V_{BG_2} = 100\ kN. \\ \max V_{BG_2} = 100 + 171,43 = 271,43\ kN. \end{cases}$$

$$M_{Mq_1} = 100.1,5 + 100.2,5 = 400\ kN.$$

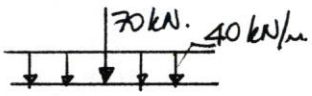
$$M_{Mq_2} = -100.1 - 100.5/7 = -171,43\ kN.$$

$$\begin{cases} \min M_M = 55 - 171,43 = -116,43\ kN. \\ \max M_M = 55 + 400 = 455\ kNm \end{cases}$$

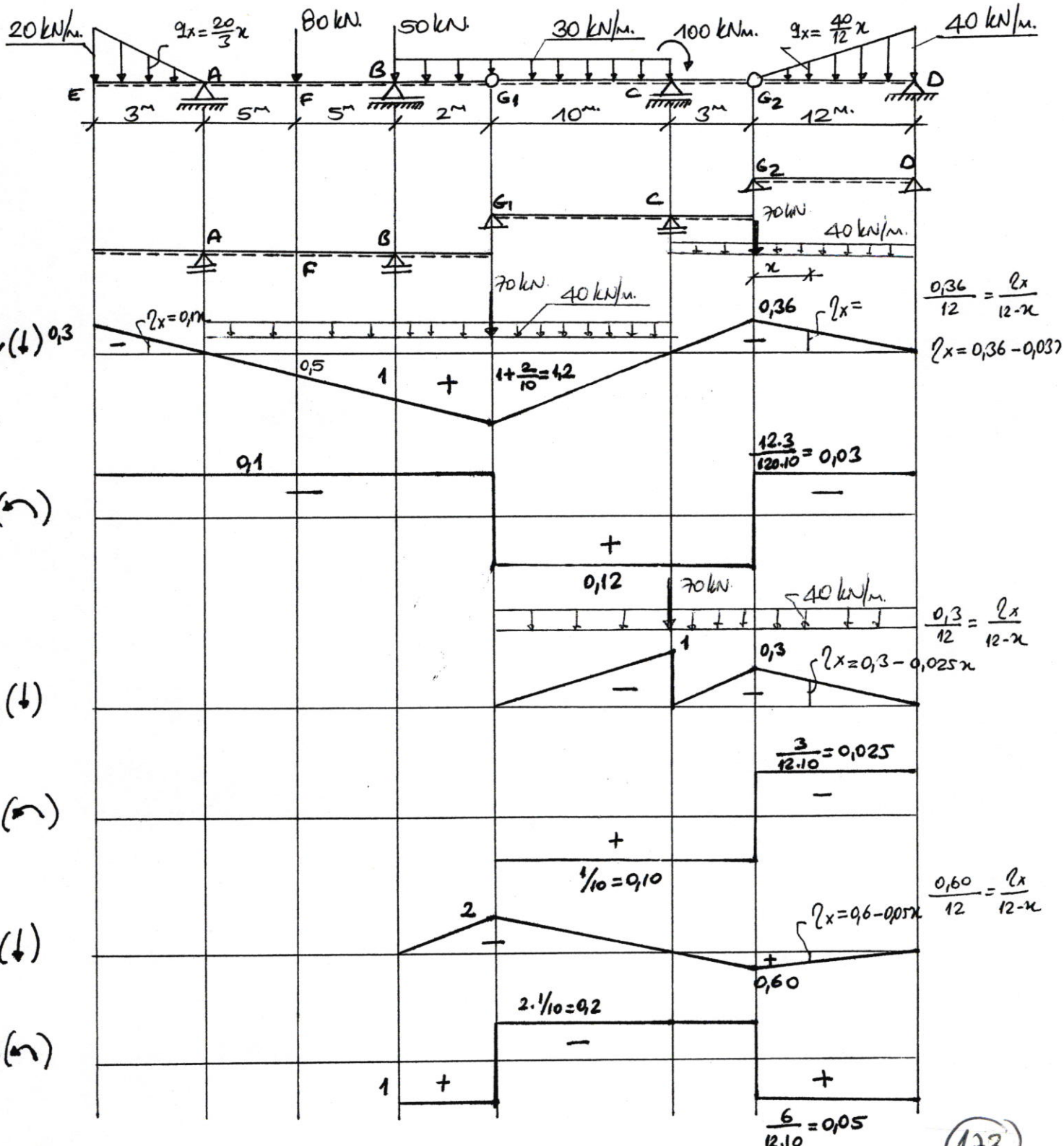
ÖRNEK : 9)

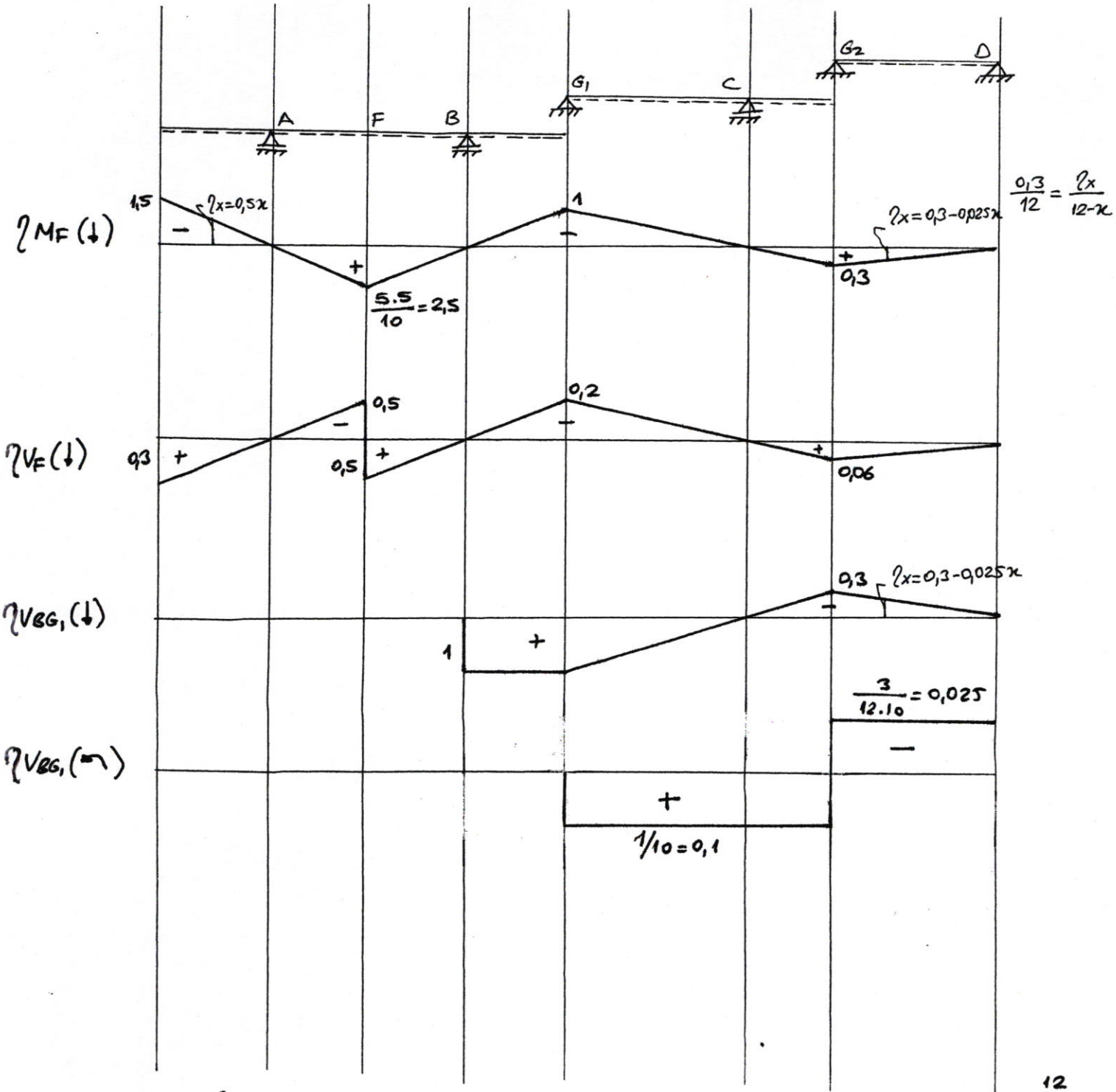
a) Şekildeki mütemadi kirişe ait η_{R_B}, $\eta_{V_{CG_1}}$, η_{M_B}, η_{M_F}, η_{V_F}, $\eta_{V_{BG_1}}$ tesir çizgilerini çiziniz.

b) Tesir çizgilerinden faydalanarak R_B, V_{CG_1}, M_B, V_{BG_1} kesit tesirlerini hesaplayınız.

c) 

Hareketli yükünden faydalanarak $\max_{\min} R_B$, $\max_{\min} V_{CG_1}$ değerlerini hesaplayınız.





b) $R_{BU} = -\int_0^3 \frac{20}{3}x \cdot 0{,}1x\,dx + 80 \cdot 0{,}5 + 50 \cdot 1 + 30 \cdot \left(\frac{1+1{,}2}{2}\right) \cdot 2 + 30 \cdot \frac{1{,}2 \cdot 10}{2} + 100(-0{,}12) - \int_0^{12} \frac{40}{12}x(0{,}36 - 0{,}03x)d$

$R_{BU} = 289{,}20$ kN.

$V_{CG_1} = -30 \cdot \frac{1 \cdot 10}{2} - 100 \cdot 0{,}10 - \int_0^{12} \frac{40}{12}x(0{,}3 - 0{,}025x)\,dx = -184{,}0$ kN.

$M_B = -30 \cdot \frac{2 \cdot 2}{2} - 30 \cdot \frac{2 \cdot 10}{2} + 100 \cdot 0{,}2 + \int_0^{12} \frac{40}{12}x(0{,}6 - 0{,}05x)\,dx = -292$ kNm.

$V_{BG_1} = \underbrace{50 \cdot 1}_{\text{katılmıyor}} + 30 \cdot \frac{2 \cdot 1}{2} + 30 \cdot \frac{1 \cdot 10}{2} - 100 \cdot 91 - \int_0^{12} \frac{40}{12}x(0{,}3 - 0{,}025x)dx = 176$ kN.

c) $R_{BVg} = 289{,}20\ kN$

$$R_{BVg_1} = 70 \cdot 1{,}2 + 40 \cdot \left(\frac{1{,}2 \cdot 22}{2}\right) = 612{,}0\ kN.$$

$$R_{BVg_2} = -70 \cdot 0{,}36 - 40 \cdot \left(\frac{0{,}36 \cdot 15}{2}\right) = -130{,}2\ kN.$$

$$\min R_{BV} = 289{,}20 - 130{,}20 = 159{,}0\ kN.$$

$$\max R_{BV} = 289{,}20 + 612{,}0 = 901{,}2\ kN.$$

$$V_{CG_1 g} = -184{,}0\ kN.$$

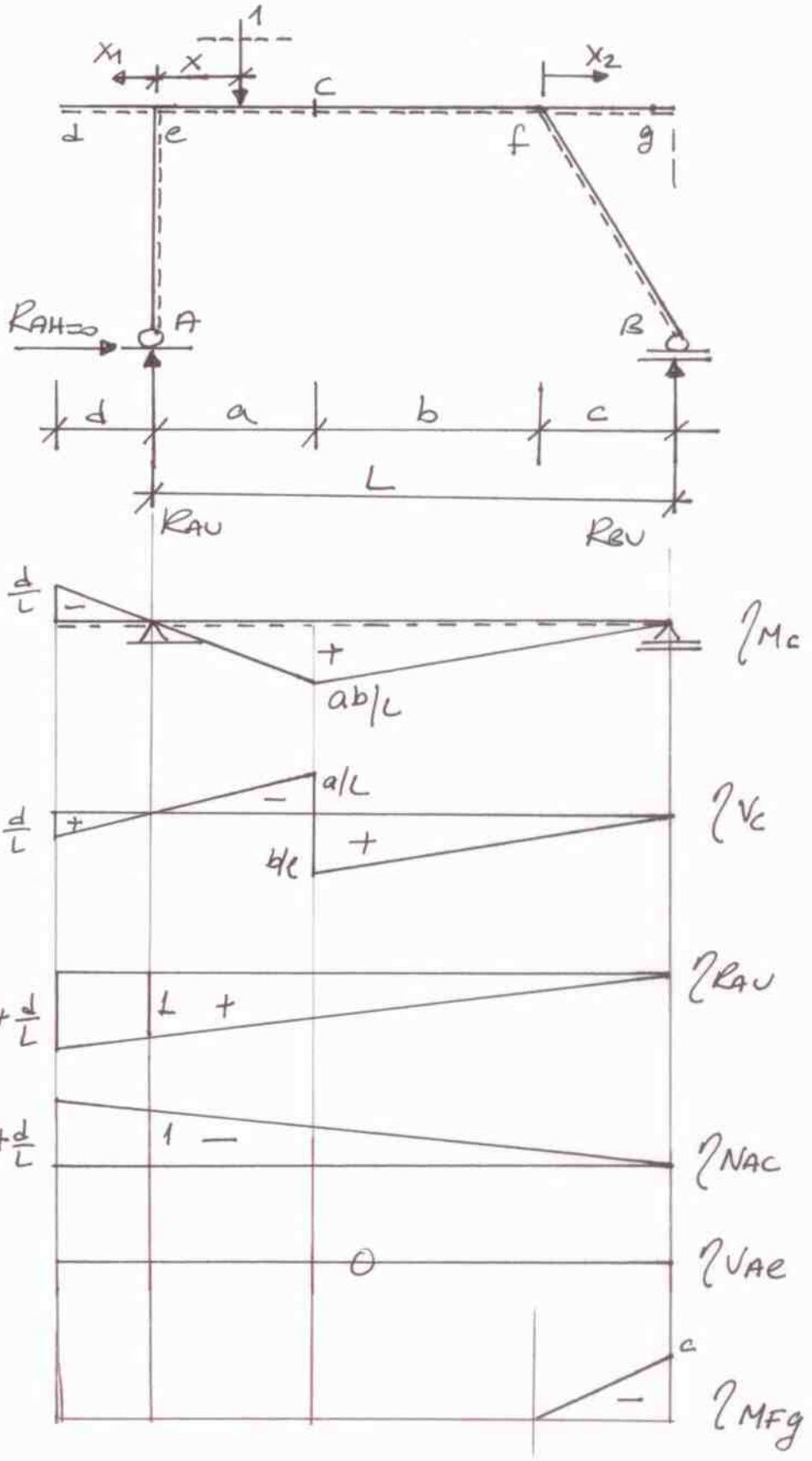
$$V_{CG_1 g_1} = -70 \cdot 1 - 40\left(\frac{1 \cdot 10}{2} + \frac{0{,}3 \cdot 15}{2}\right) = -360\ kN.$$

$$V_{CG_1 g_2} = 0$$

$$\min V_{CG_1} = -184{,}0 - 360 = -544{,}0\ kN.$$

$$\max V_{CG_1} = -184{,}0\ kN.$$

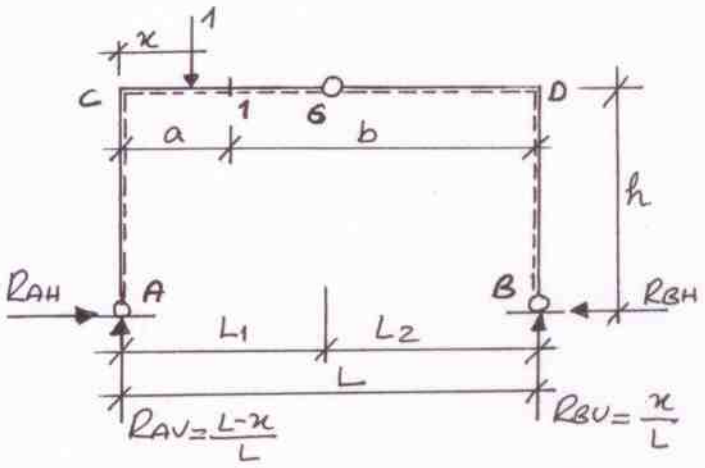
VIII-10) BASIT GERGEVEDE TESIR GIZGILERI



VIII. 12) Üç MAFSALLI GERÇEVE SİSTEMLERDE TESİR ÇİZGİLERİ

a) Düşey Yük Etkisi

Çerçeve sistemlerin birim düşey kuvvet etkisi altında tesir çizgileri diyagramları, kesit tesiri diyagramlarının çizimi bakımından faydalanılarak çizilir. Basit kiriş tesir çizgisinde olduğu gibi birim kuvvetin x'e bağlı değişimi incelenir. Burada çerçeveleri basit kiriş analojisinden faydalanarak çözeceğiz.

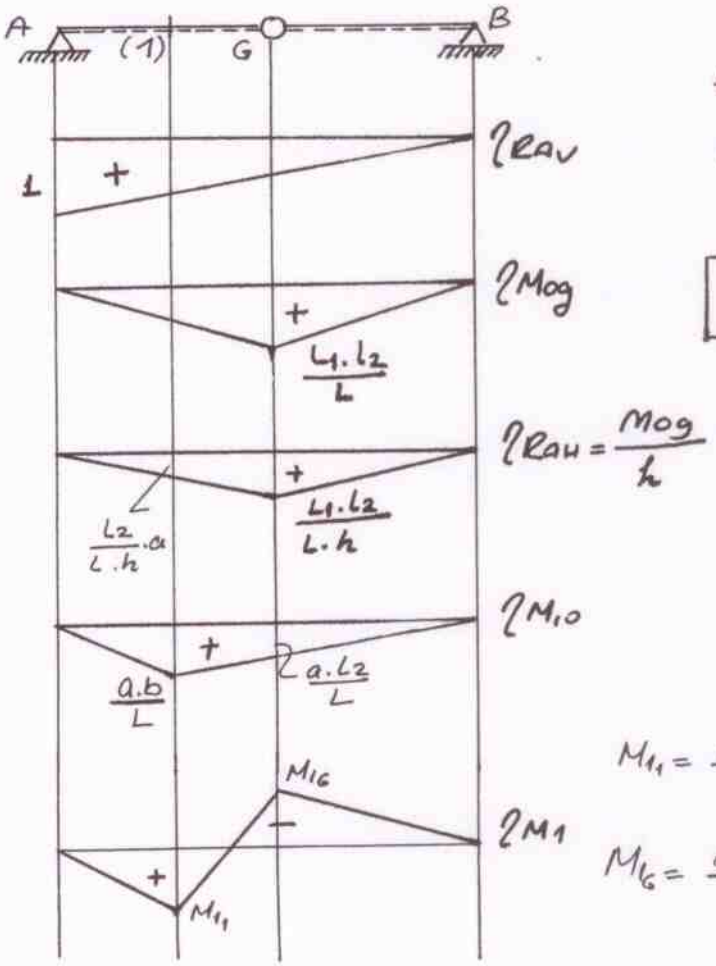


$\Sigma M_B=0 \rightarrow R_{AV}=\frac{L-x}{L}$

$\Sigma M_A=0 \rightarrow R_{BV}=x/L$

$\Sigma M_{G\,sol}=0 \rightarrow R_{AH}=\frac{\overbrace{R_{AV}.L_1-1(L_1-x)}^{M_{OG}}}{h}$

$$\boxed{R_{AH}=\frac{M_{OG}}{h}}$$



1 noktasına göre moment:

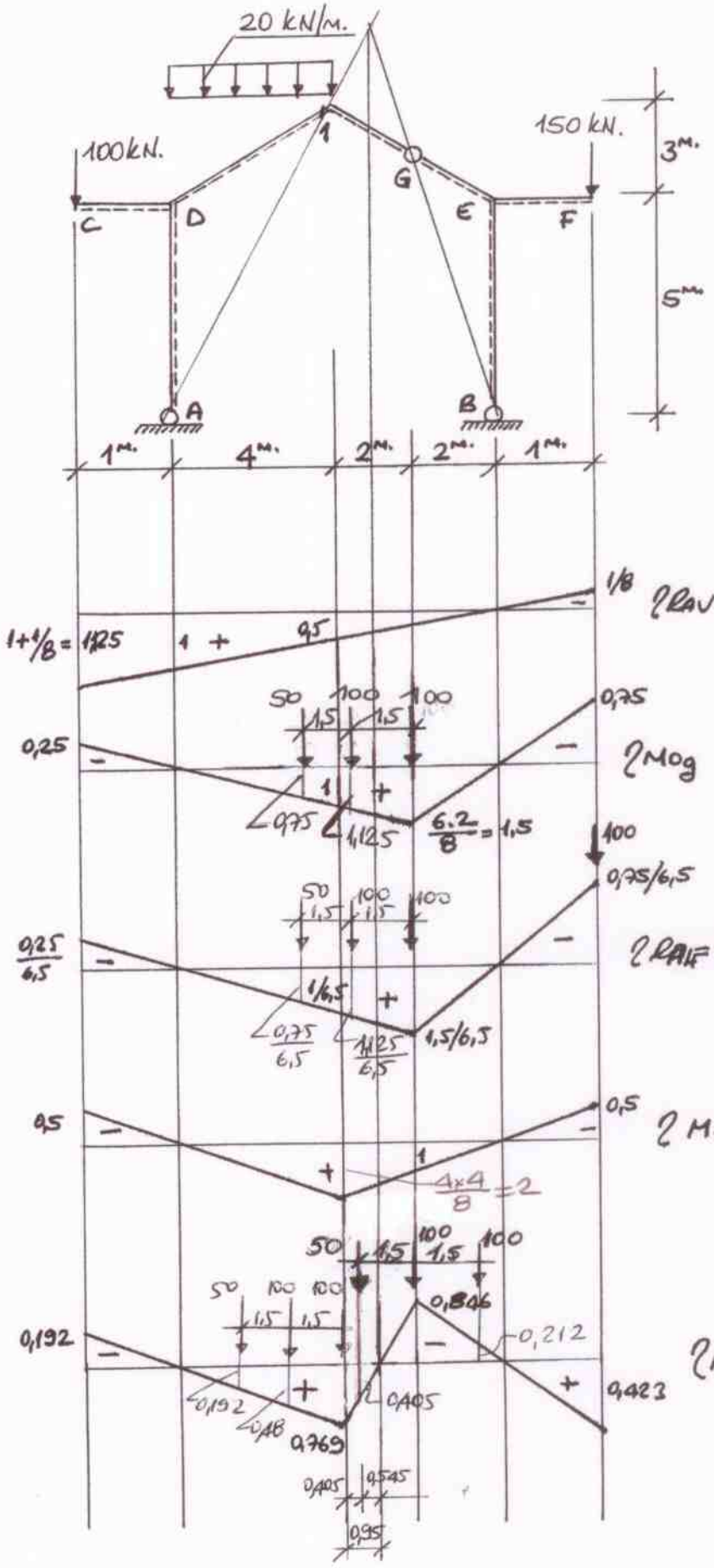
$M_1=\underbrace{R_{AV}.a-1(a-x)}_{M_{10}}-R_{AH}.h$

$$\boxed{M_1=M_{10}-R_{AH}.h}$$

$$M_{11}=\frac{a.b}{L}-\frac{L_2}{L.h}.a.h=\frac{a.b}{L}-\frac{L_2.a}{L}=\frac{a(b-L_2)}{L}$$

$$M_{16}=\frac{a.L_2}{L}-\frac{L_1.L_2}{L.h}.h=\frac{aL_2}{L}-\frac{L_1L_2}{L}=\frac{L_2(a-L_1)}{L}$$

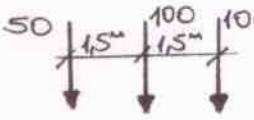
ÖRNEK:10)



a) ?RAV, ?RAH, ?MOC ve ?M1 tesir çizgilerini çiziniz.

b) RAV, RAH ve M1 değerlerini tesir çizgilerinden faydalanarak bulunuz.

c) Verilen hareketli yükten oluşacak M_{AH}^{max} ve M_{1}^{max} değerlerini bulunuz.



$?M_1 = ?M_{10} - ?R_{AH} \cdot y$

$= -0,5 + \frac{0,25}{6,5} \cdot 8 = -0,192$

$= 2 - \frac{1}{6,5} \cdot 8 = 0,769$

$= 1 - \frac{1,5}{6,5} \cdot 8 = -0,846$

$= -0,5 + \frac{0,75}{6,5} \cdot 8 = 0,423$

b) $R_{AV} = 100.1{,}125 + 20.\left(\frac{1+0{,}5}{2}\right).4 - 150.\frac{1}{8} = 153{,}75\ kN.$

$R_{AH} = -100.\frac{0{,}25}{6{,}5} + 20.\left(\frac{1/6{,}5.4}{2}\right) - 150.\frac{0{,}75}{6{,}5} = -15\ kN.$

$M_1 = -100.0{,}192 + 20.\frac{0{,}769.4}{2} + 150.0{,}423 = 75{,}00\ kNm.$

c) $R_{AHg_1} = 50.\frac{0{,}75}{6{,}5} + 100.\frac{1{,}125}{6{,}5} + 100.\frac{1{,}5}{6{,}5} = 46{,}15\ kN.$

$R_{AHg_2} = -100.\frac{0{,}75}{6{,}5} = -11{,}54\ kN.$

$min\ R_{AH} = -15 - 11{,}54 = -26{,}54\ kN$

$max\ R_{AH} = -15 + 46{,}15 = 31{,}15\ kN.$

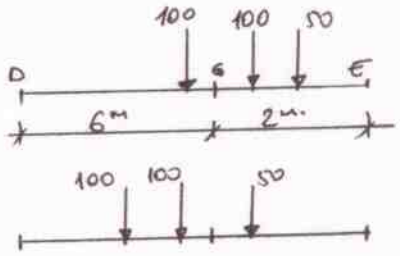
$M_{1g_1} = 50.0{,}192 + 100.0{,}48 + 100.0{,}769 = 134{,}50\ kNm.$

$M_{1g_2} = 50.0{,}405 - 100.0{,}846 - 100.0{,}212 = -85{,}55\ kNm.$

$min\ M_1 = 75{,}00 - 85{,}55 = -10{,}55\ kNm$

$max\ M_1 = 75 + 134{,}50 = 209{,}50\ kNm.$

R_{AH} tesir çizgisine hareketli yükün etkimesi:

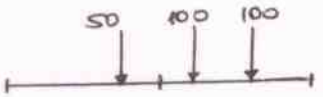


$\frac{100}{6} < \frac{150}{2}$

$16{,}67 < 75$ ✓

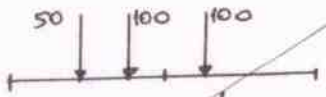
$\frac{200}{6} > \frac{50}{2}$ ✓

$33{,}31 > 25$



$\frac{50}{6} < \frac{200}{2}$

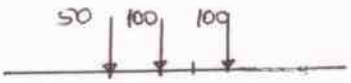
$8{,}33 < 100$ ✓



$\frac{150}{6} > \frac{100}{2}$

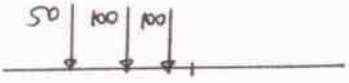
$25 > 50$

uygun değil.



$\frac{150}{6} < \frac{100}{2}$

$25 < 50$ ✓

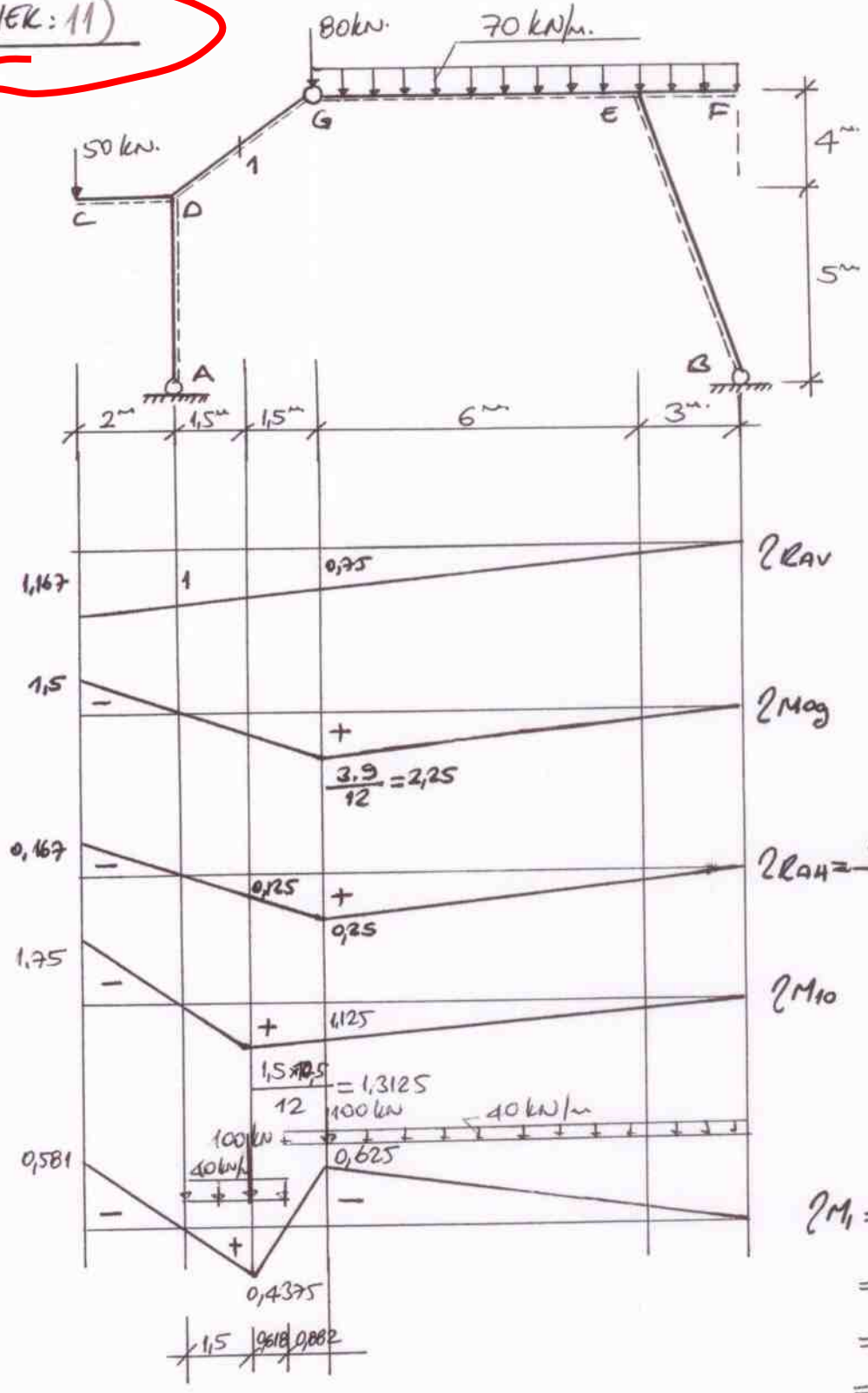


$\frac{250}{6} > 0$

$41{,}67 > 0$ ✓

Fark büyük olduğundan bu yükleme kabul edilir.

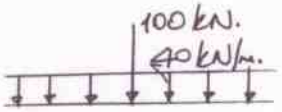
ÖRNEK: 11)



a) Şekildeki sistemin $?R_{AV}$, $?R_{AH}$ ve $?M_1$ tesir çizgilerini çizelim.

b) Tesir çizgilerinden faydalanarak R_{AV}, R_{AH} ve M_1 değerlerini bulunuz.

c) Hareketli yükten dolayı $\max_{\min} M_1$ değerini bulunuz.



$$?M_1 = ?M_{10} - ?R_{AH}\cdot y$$

$= -1,75 + 0,167 \times 7 = -0,581$

$= 1,3125 - 0,125 \times 7 = 0,4375$

$= 1,125 - 0,25 \times 7 = -0,625$

b) $R_{AV} = 50 \cdot 1,167 + 80 \cdot 0,75 + 70 \cdot \frac{0,75 \cdot 9}{2} = 354,60$ kN.

$R_{AH} = -50 \cdot 0,167 + 80 \cdot 0,25 + 70 \cdot \frac{0,25 \cdot 9}{2} = 90,4$ kN.

$M_1 = -50 \cdot 0,581 - 80 \cdot 0,625 - 70 \cdot \frac{0,625 \cdot 9}{2} = -275,93$ kNm.

c) $M_{1g_1} = 40 \cdot \left(\frac{0,4375 \cdot 2,118}{2}\right) + 100 \cdot 0,4375 = 62,28$ kNm

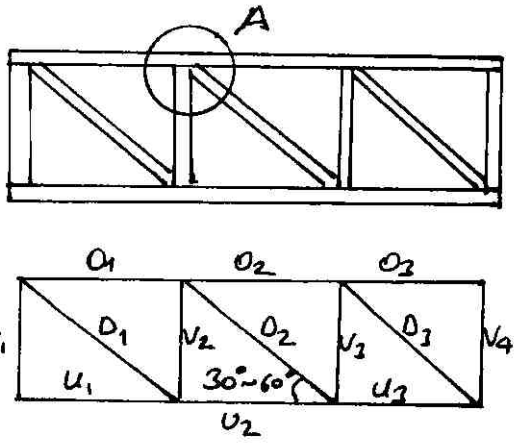
$M_{1g_2} = -40 \cdot \left(\frac{0,625 \times 9,882}{2}\right) - 100 \cdot 0,625 = -186,03$ kNm

$M_{1g} = -275,93$ kNm.

$\min M_1 = -275,93 - 186,03 = -461,96$ kNm.

$\max M_1 = -275,93 + 62,28 = -213,65$ kNm.

IX- KAFES SİSTEMLER



O: Üst bazlık

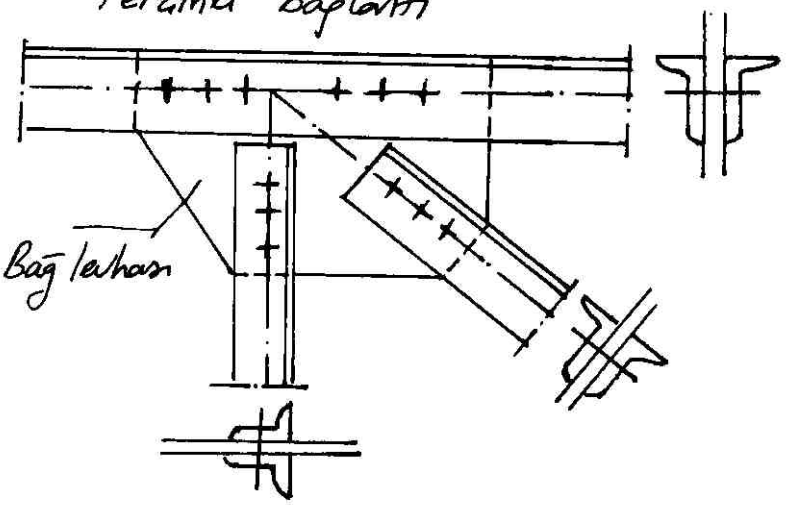
U: Alt "

V: Dikme

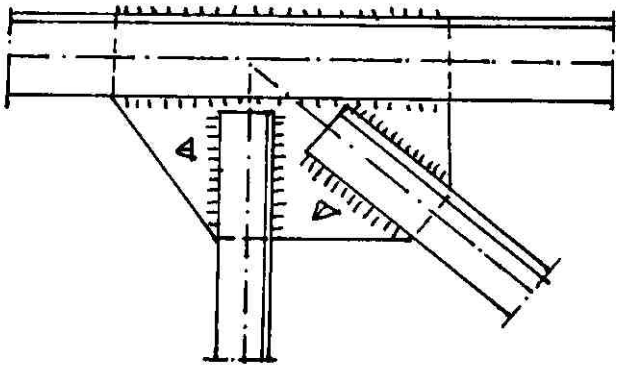
D: Diyagonal

A detayı

Perçinli bağlantı



Kaynaklı Bağlantı



Kafes kirişlerde gerilmelerin meydana gelmemesi için;

1- Çubuk eksenleri ve diş kuvvetler aynı düzlem içinde bulunmalıdır.

2- Çubuk eksenleri doğru olmalıdır. Ve bir düğümdeki çubukların eksenleri bir noktada birleşmelidir.

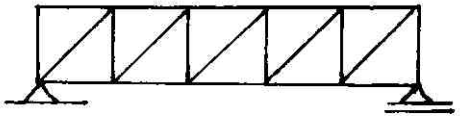
3- Diş kuvvetler düğüm noktalarında etkilenmelidir.

4- Çubukların aralarındaki açılar çok küçük olmamalıdır. (30° ~ 60°)

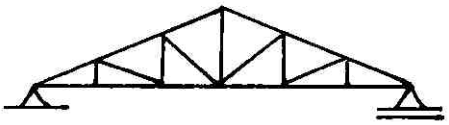
IX-1) Kafes Sistemlerin Sınıflandırılması

a) Başlık Şekillerine Göre:

a.1) Paralel Başlıklı Kafes Kiriş

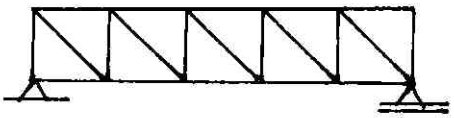


a.2) Paralel Olmayan Başlıklı Kafes Kiriş

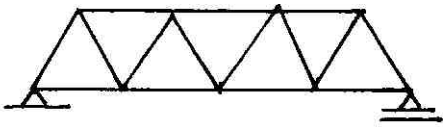


b) Diğme Çubukların Şekline Göre:

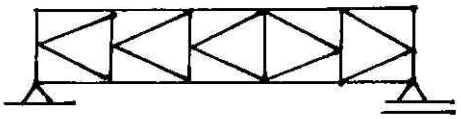
b.1) N tipi kafes kiriş



b.2) V tipi kafes kiriş

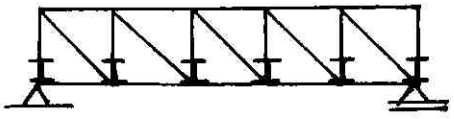


b.3) K tipi kafes kiriş

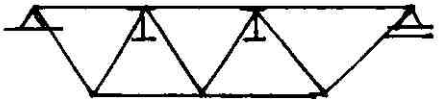


c) Yol Durumuna Göre

c.1) Yolu (tabliyesi) alt başlıkta

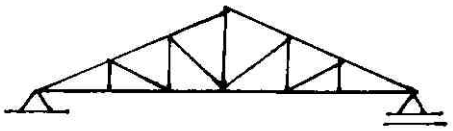


c.2) Yolu (tabliyesi) üst başlıkta

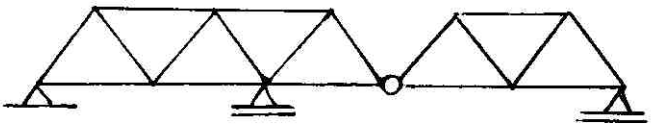


d) Statik durumlarına göre

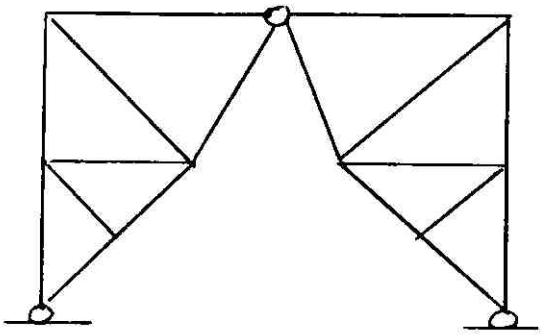
d.1) Basit Kiriş



d.2) Gerber Kiriş



d.3) Üç Mafsallı Kafes Kiriş



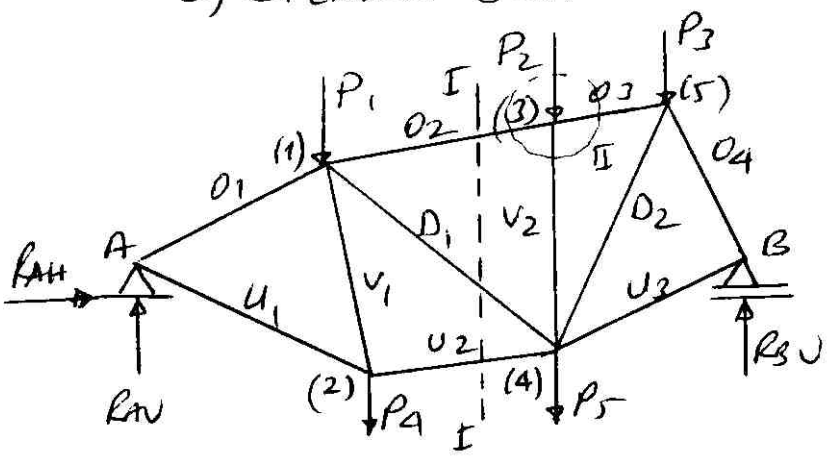
e) Kullanım amacına göre

e.1) Çatı kafes kiriş

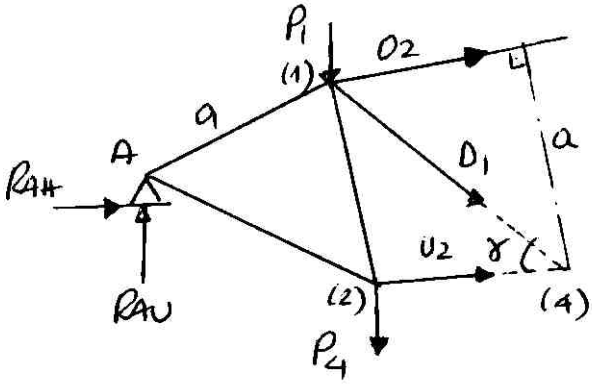
e.2) Köprü kafes kiriş

IX-2) Kafes Kirişlerde Hesap Yöntemleri

a) Düğüm Noktası Yöntemi
b) Ritter Kesim Yöntemi
c) Cremona Yöntemi



I-I Kesimi

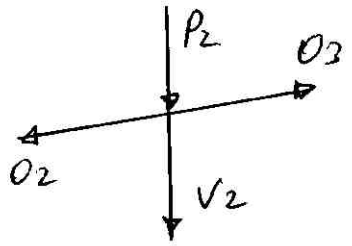


$$\Sigma M_4 = 0 \rightarrow O_2 = \frac{P_1 \cdot a_1 + P_4 \cdot a_2 - R_{AH} \cdot y - R_{AV} \cdot x_1}{a}$$

$$\Sigma M_1 = 0 \rightarrow U_2 = \frac{P_2 \cdot a_3 - R_{AH} \cdot y_1 + R_{AV} \cdot x_2}{a}$$

ΣX veya $\Sigma Y = 0 \rightarrow D_1 = !$

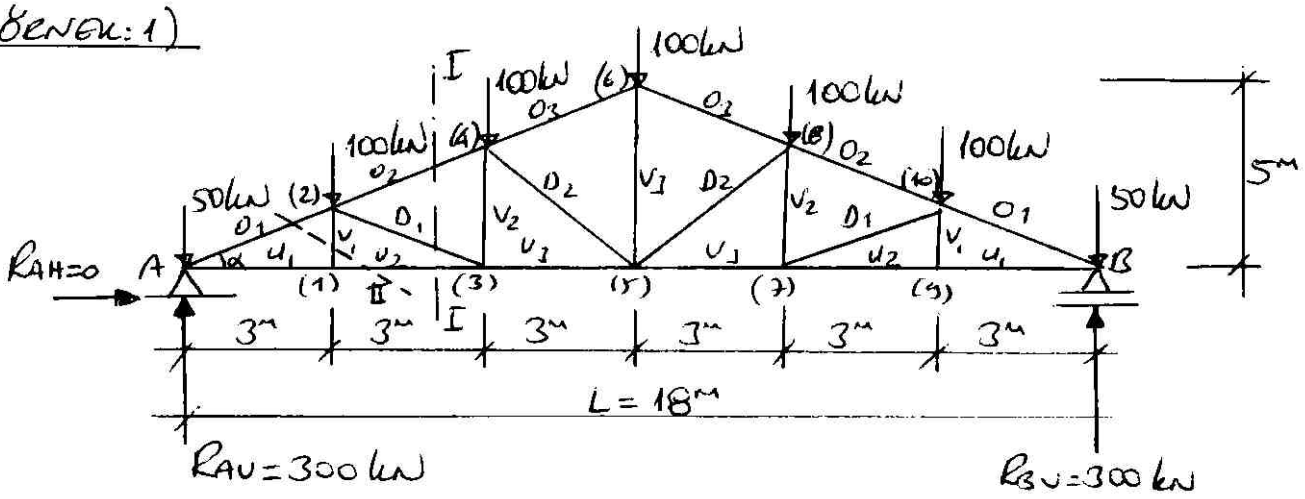
3 nolu düğüm noktası



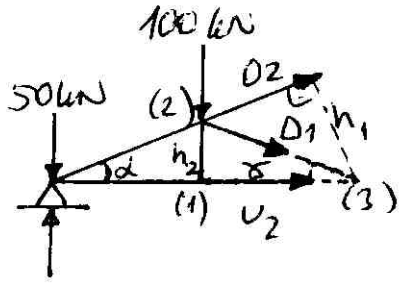
$\Sigma X = 0 \rightarrow O_3 = !$

$\Sigma Y = 0 \rightarrow V_2 = !$

ÖRNEK: 1)



I-I Kesimi



$\sin\alpha = 0{,}486$, $\cos\alpha = 0{,}874$, $tg\alpha = 0{,}556$

$h_1 = 2{,}916$ m , $h_2 = 1{,}668$ m

$\Sigma M_3 = 0 \rightarrow O_2 \cdot h_1 = 50 \cdot 6 + 100 \cdot 3 - 300 \cdot 6$

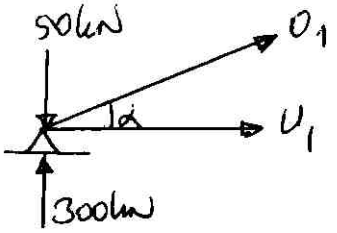
$O_2 = -411{,}52$ kN

$\Sigma M_2 = 0 \rightarrow U_2 \cdot h_2 = 300 \cdot 3 - 50 \cdot 3 = 449{,}64$ kN

$\Sigma Y = 0 \rightarrow D_1 \cdot \sin\gamma = 300 - 50 - 100 + O_2 \cdot \sin\alpha$

$D_1 = -102{,}88$ kN

A düğüm noktası



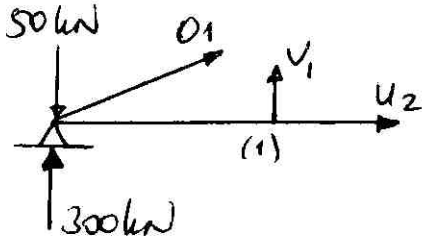
$\Sigma Y = 0 \rightarrow O_1 \cdot \sin\alpha = 50 - 300$

$O_1 = -514{,}40$ kN

$\Sigma X = 0 \rightarrow O_1 \cdot \cos\alpha = U_1$

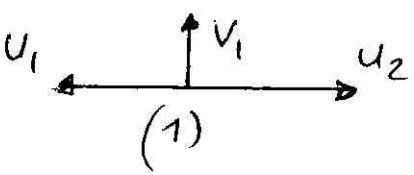
$U_1 = 449{,}64$ kN

II-II Kesimi



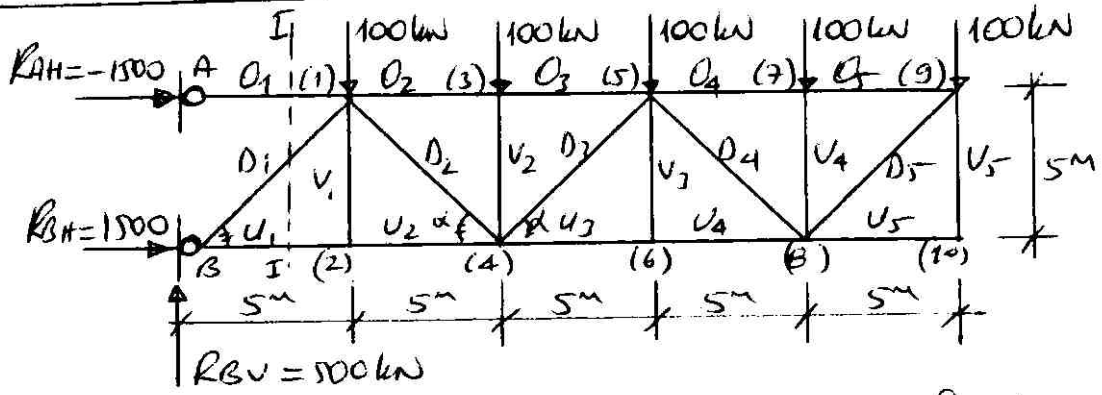
$\Sigma Y = 0 \rightarrow V_1 = -300 + 50 - O_1 \cdot \sin\alpha = 0$

veya



$\Sigma Y = 0 \rightarrow V_1 = 0$

ÖRNEK:2)



$\sin\alpha = 0{,}707$

$\cos\alpha = 0{,}707$

Mesnet Tepkilerinin hesabı

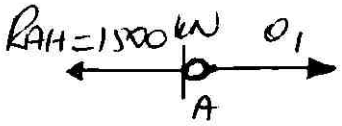
$\Sigma Y=0 \rightarrow R_{BV} = 500\ kN$

$\Sigma M_B=0 \rightarrow R_{AH} = -\dfrac{(100\cdot5+100\cdot10+100\cdot15+100\cdot20+100\cdot25)}{5} = -1500\ kN$

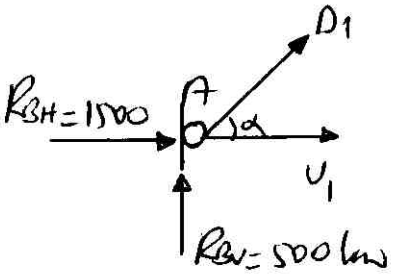
$\Sigma M_A=0 \rightarrow R_{BH} = \dfrac{100(5+10+15+20+25)}{5} = 1500\ kN$

$\Sigma X=0 \rightarrow 1500 = 1500$

Çubuk Kuvvetlerinin Hesabı



$\Sigma X=0 \rightarrow O_1 = 1500\ kN$



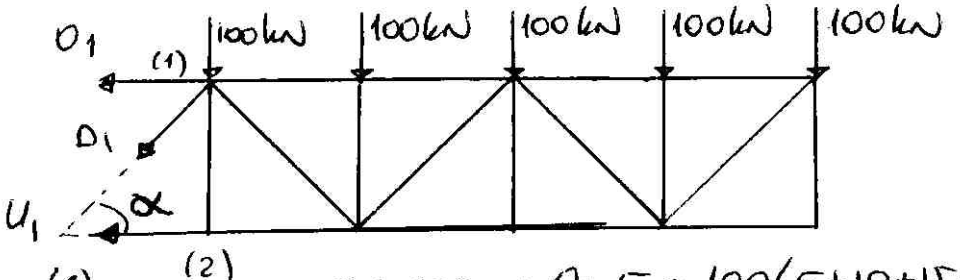
$\Sigma Y=0 \rightarrow D_1 \cdot \sin\alpha = -500$

$D_1 = -707{,}21\ kN$

$\Sigma X=0 \rightarrow U_1 = -D_1 \cdot \cos\alpha - 1500$

$U_1 = -1000\ kN$

veya I-I kesimi

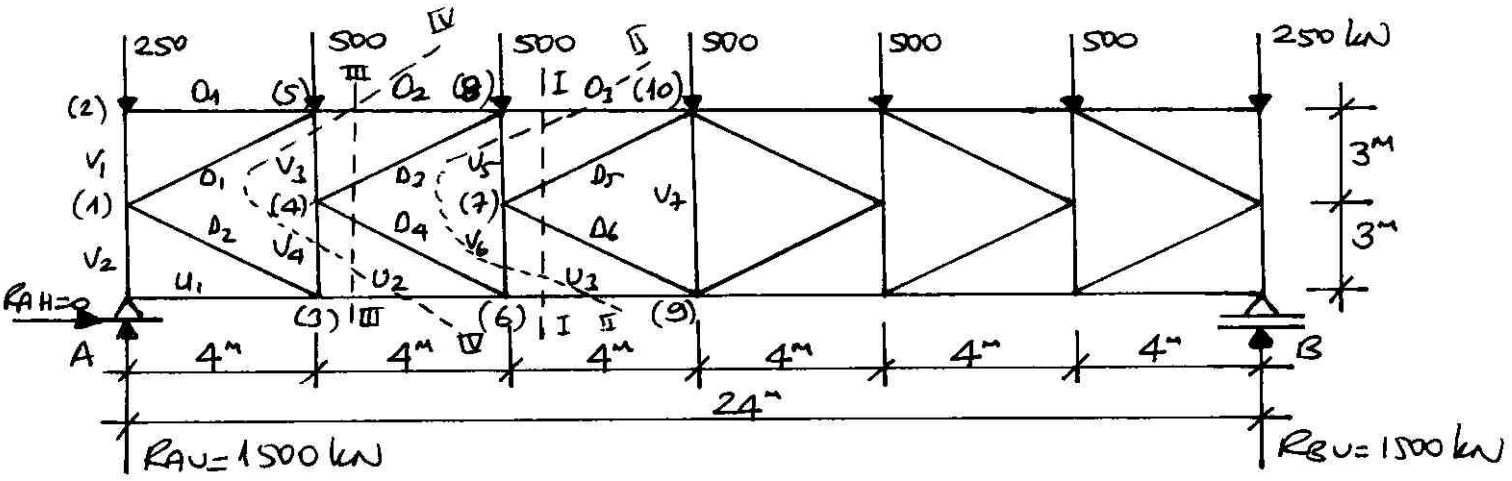


$\Sigma M_B=0 \rightarrow O_1 \cdot 5 = 100(5+10+15+20+25) \rightarrow O_1 = 1500\ kN$

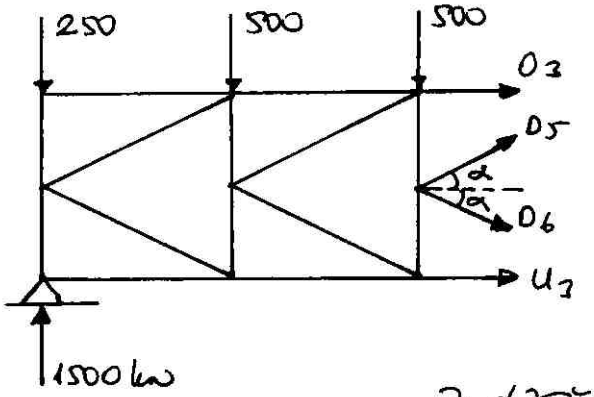
$\Sigma M_1=0 \rightarrow U_1 \cdot 5 = -100(5+10+15+20) \rightarrow U_1 = -1000\ kN$

$\Sigma Y=0 \rightarrow D_1 \cdot \sin\alpha = -500 \rightarrow D_1 = -707{,}21\ kN$

ÖRNEK: 3



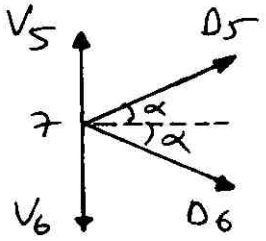
I-I KESİMİ



$\Sigma Y=0 \rightarrow D_5 \cdot \sin\alpha - D_6 \cdot \sin\alpha = -1500+1250$

$D_5 - D_6 = -\frac{250}{\sin\alpha} = -\frac{1250}{3}$ (1)

7 düğüm noktası



$\sin\alpha = 3/5$

$\cos\alpha = 4/5$

$\Sigma X=0 \rightarrow D_5 \cdot \cos\alpha + D_6 \cdot \cos\alpha = 0$

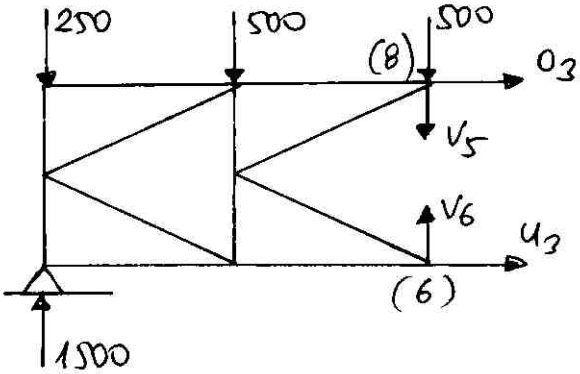
$D_5 + D_6 = 0$

$D_5 = -D_6$ (2)

(1) ve (2) denklemlerinden;

$D_6 = +208{,}33$ kN ; $D_5 = -208{,}33$ kN

II-II KESİMİ



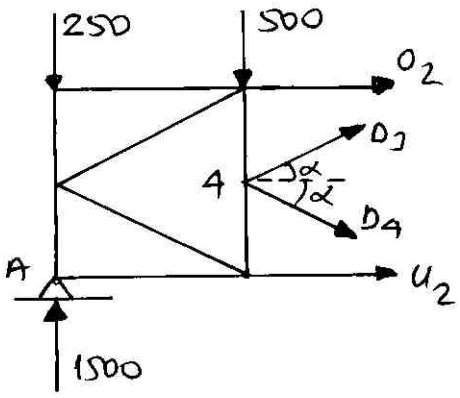
$\Sigma M_6=0 \rightarrow O_3 \cdot 6 = -1500\times 8 + 250\times 8 + 500\cdot 4$

$O_3 = -1333{,}33$ kN

$\Sigma M_8=0 \rightarrow U_3 \cdot 6 = 1500\times 8 - 250\times 8 - 500\times 4$

$U_3 = 1333{,}33$ kN

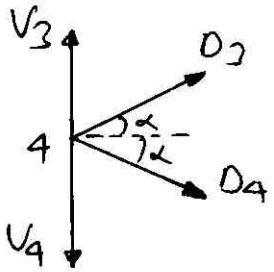
III - III KESİMİ



$\Sigma Y=0 \rightarrow O_3.\sin\alpha - O_4.\sin\alpha = -1500+750$

$O_3 - O_4 = -1250$ (3)

4 düğüm noktası



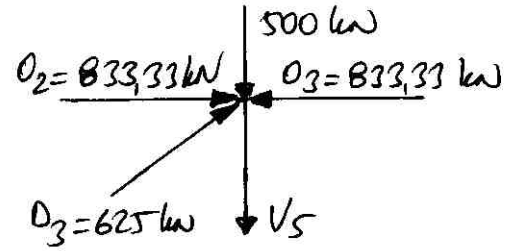
$\Sigma X=0 \rightarrow O_3.\sin\alpha = -O_4.\sin\alpha$

$O_3 = -O_4$ (4)

(3) ve (4) denklemlerinden;

$O_3 = -625$ kN ; $O_4 = +625$ kN

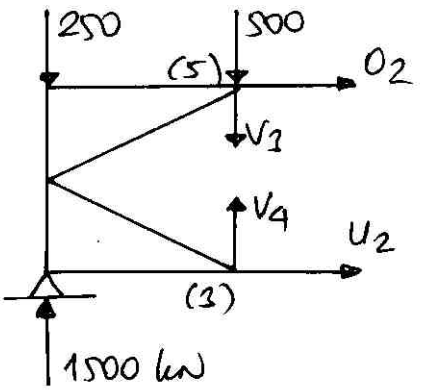
8 düğüm noktası



$\Sigma Y=0 \rightarrow V_5 = -500 + O_3.\sin\alpha$

$V_5 = -125$ kN

IV - IV KESİMİ



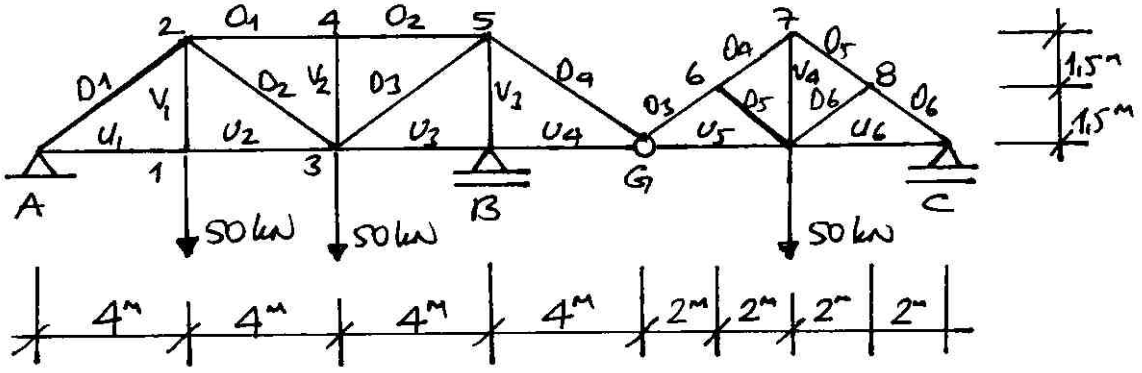
$\Sigma M_5=0 \rightarrow U_2.6 = 1500\times4 - 250\times4$

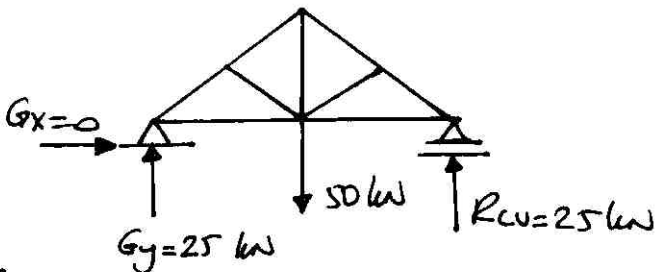
$U_2 = 833,33$ kN

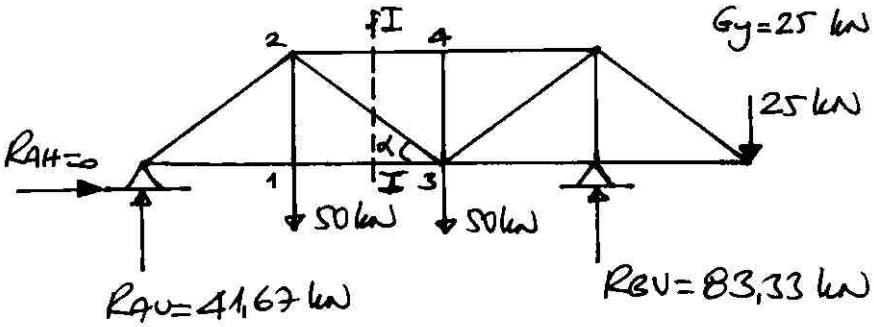
$\Sigma M_3=0 \rightarrow O_2.6 = -1500\times4 + 250\times4$

$O_2 = -833,33$ kN

ÖRNEK: 4)







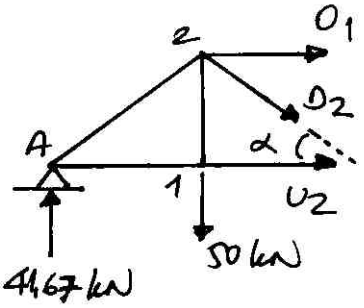
$\sin\alpha = 3/5$

$\cos\alpha = 4/5$

I-I Kesmesinden

$\Sigma M_3 = 0 \longrightarrow O_1 \cdot 3 = 50 \times 4 - 41{,}67 \times 8$

$O_1 = -44{,}45$ kN



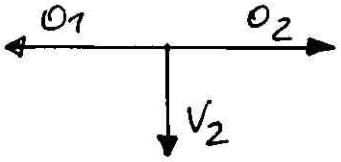
$\Sigma M_2 = 0 \longrightarrow U_2 \cdot 3 = 41{,}67 \times 4$

$U_2 = 55{,}56$ kN

$\Sigma Y = 0 \longrightarrow D_2 \cdot \sin\alpha = 41{,}67 - 50$

$D_2 = -13{,}88$ kN

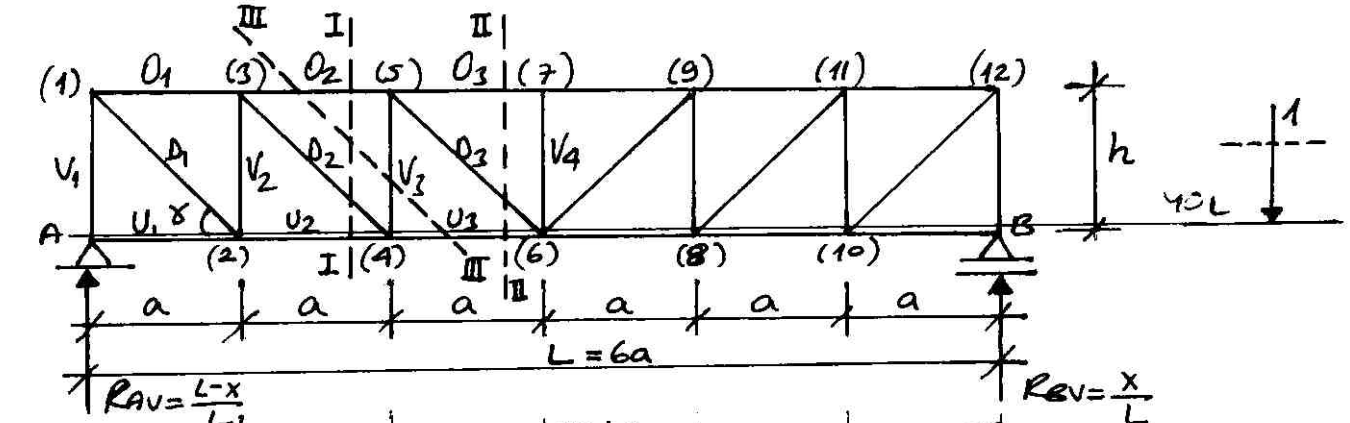
4 Düğüm Noktası

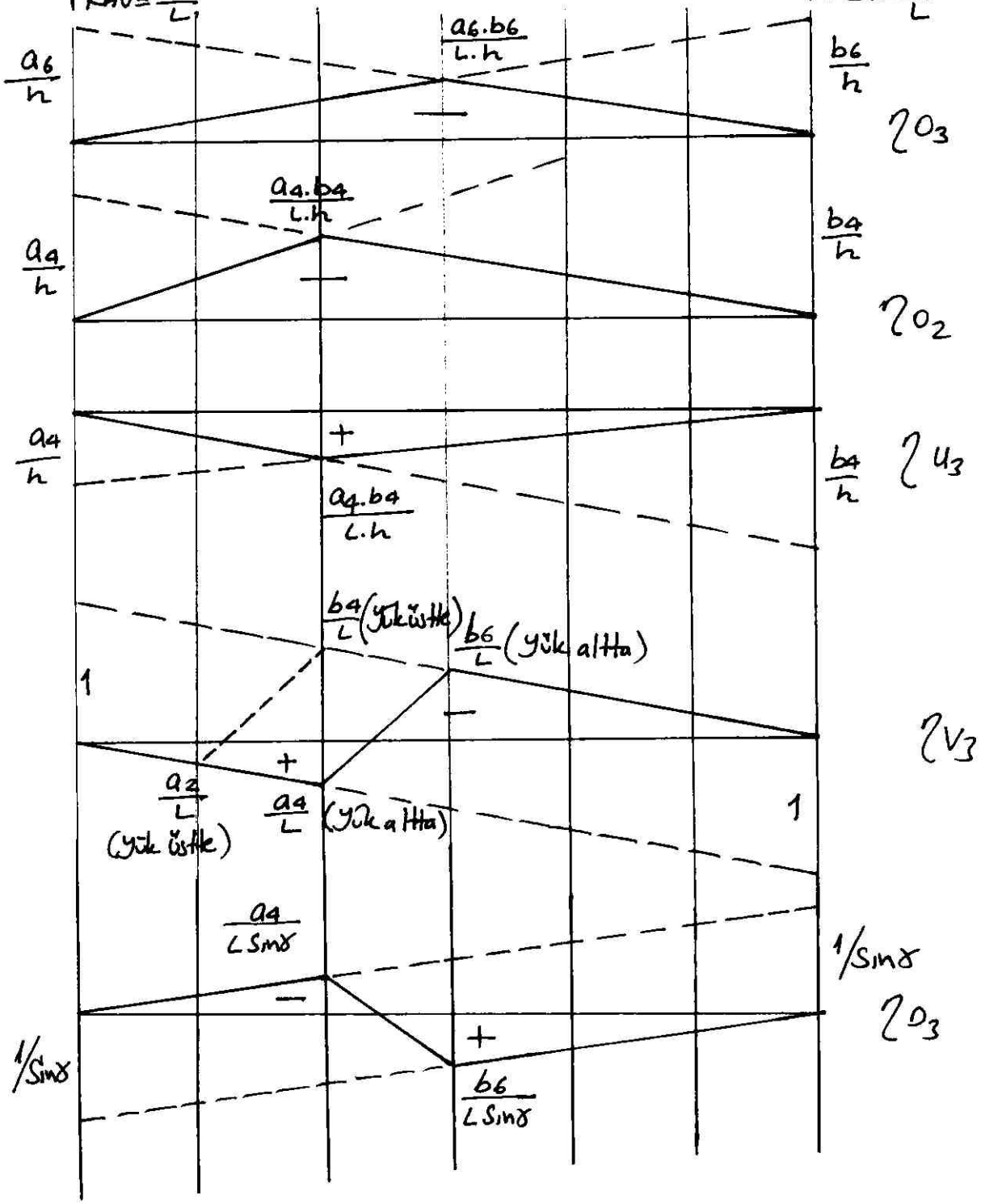


$\Sigma Y = 0 \longrightarrow V_2 = 0$

IX-3) KAFES SİSTEMLERDE TESİR ÇİZGİLERİ

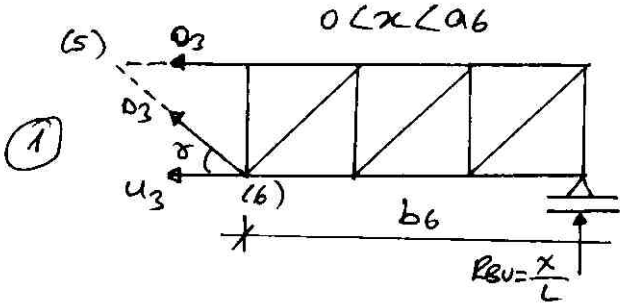
a) N TİPİ KAFES KİRİŞ





ηO_3 tesir çizgisi

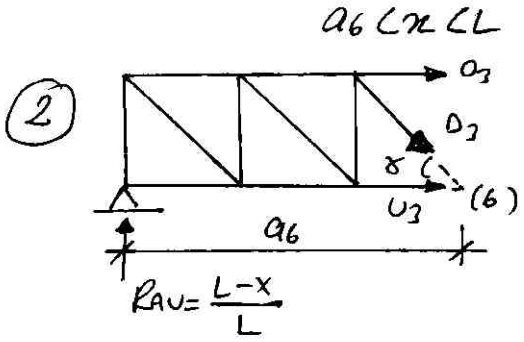
O_3 çubuk kuvvetini bulmak için (6) noktasına göre moment alınır. Çubuk kuvveti II-II kesimine göre incelenir. Süreksizlik noktaları birim yükün alt bazlıkta veya üst bazlıkta olması durumları için geçerli ve (4)-(6) veya (5)-(7) arasındadır. Bu noktalar arası çizilen tesir çizgisi hattı ile çakışmaktadır. Birim yük $0<x<a_6$ arasında ise (6) noktası ile (8) noktası arasında kalan kısım dikkate alınır ve tesir çizgisi fonksiyonu yazılır.



$$\Sigma M_6=0 \rightarrow O_3 \cdot h = -\frac{x}{L}\cdot b_6$$

$$x=0 \rightarrow \eta O_3=0$$

$$x=a_6 \rightarrow \eta O_3=-\frac{a_6 \cdot b_6}{L\cdot h}$$



$$\Sigma M_6=0 \rightarrow O_3 \cdot h = -\frac{L-x}{L}\cdot a_6$$

$$x=a_6 \rightarrow \eta O_3=-\frac{a_6 \cdot b_6}{L\cdot h}$$

$$x=L \rightarrow \eta O_3=0$$

Görüldüğü gibi birim yükün $a_6<x<L$ arasında olması durumunda sistemin A noktası ile 6 noktası arasında kalan kısmı dikkate alınır ve tesir çizgisi fonksiyonu yazılır. Aynı II-II kesimi ile U_3 çubuğu da hesaplanabilir. Ancak U_3 çubuğunun hesabı için (5) noktasına göre moment alınır. U_3 çubuğunun süreksizliği O_3 çubuğu ile aynıdır. O_3 çubuğunun hesabı için II-II kesimi kullanılır. Çubuktaki süreksizlik 4-6 noktaları arasıdır ve birim yük 4-4 arası ve yine birim yük 6-8 arasında gezinmesi durumu için fonksiyonlar hesaplanır.

Birim yük $0 < x < a_4$ arasında ise ① nolu şekil dikkate alınır ve $\Sigma Y = 0$ koşulu kullanılırsa;

$$\Sigma Y = 0 \longrightarrow D_3 \cdot \operatorname{Sin}\gamma = -\frac{x}{L} \longrightarrow D_3 = -\frac{x}{L \cdot \operatorname{Sin}\gamma}$$

$$x = 0 \rightarrow \eta D_3 = 0$$

$$x = a_4 \rightarrow \eta D_3 = -\frac{a_4}{L \operatorname{Sin}\gamma}$$

Birim yük $a_6 < x < L$ arasında ise ② nolu şekil dikkate alınır ve $\Sigma Y = 0$ koşulu kullanılırsa;

$$\Sigma Y = 0 \longrightarrow D_3 \cdot \operatorname{Sin}\gamma = \frac{L-x}{L} \longrightarrow D_3 = \frac{L-x}{L \cdot \operatorname{Sin}\gamma}$$

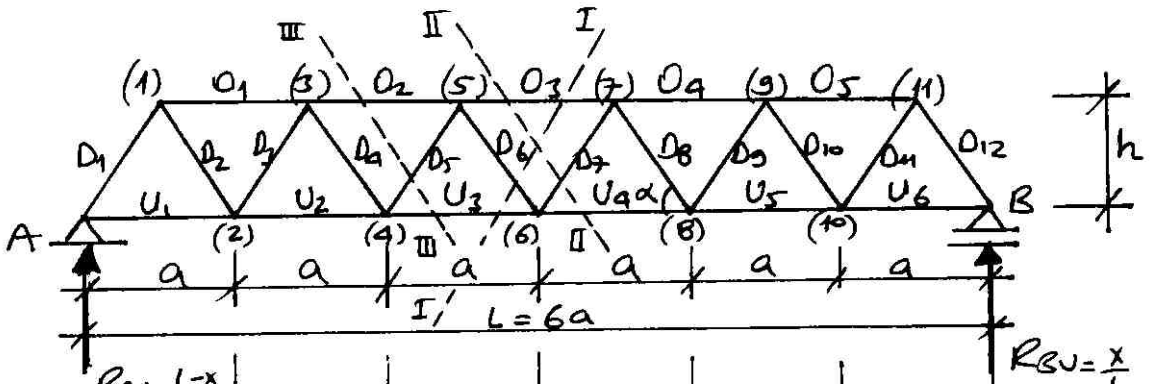
$$x = a_6 \longrightarrow \eta D_3 = \frac{b_6}{L \operatorname{Sin}\gamma}$$

$$x = L \longrightarrow \eta D_3 = 0$$

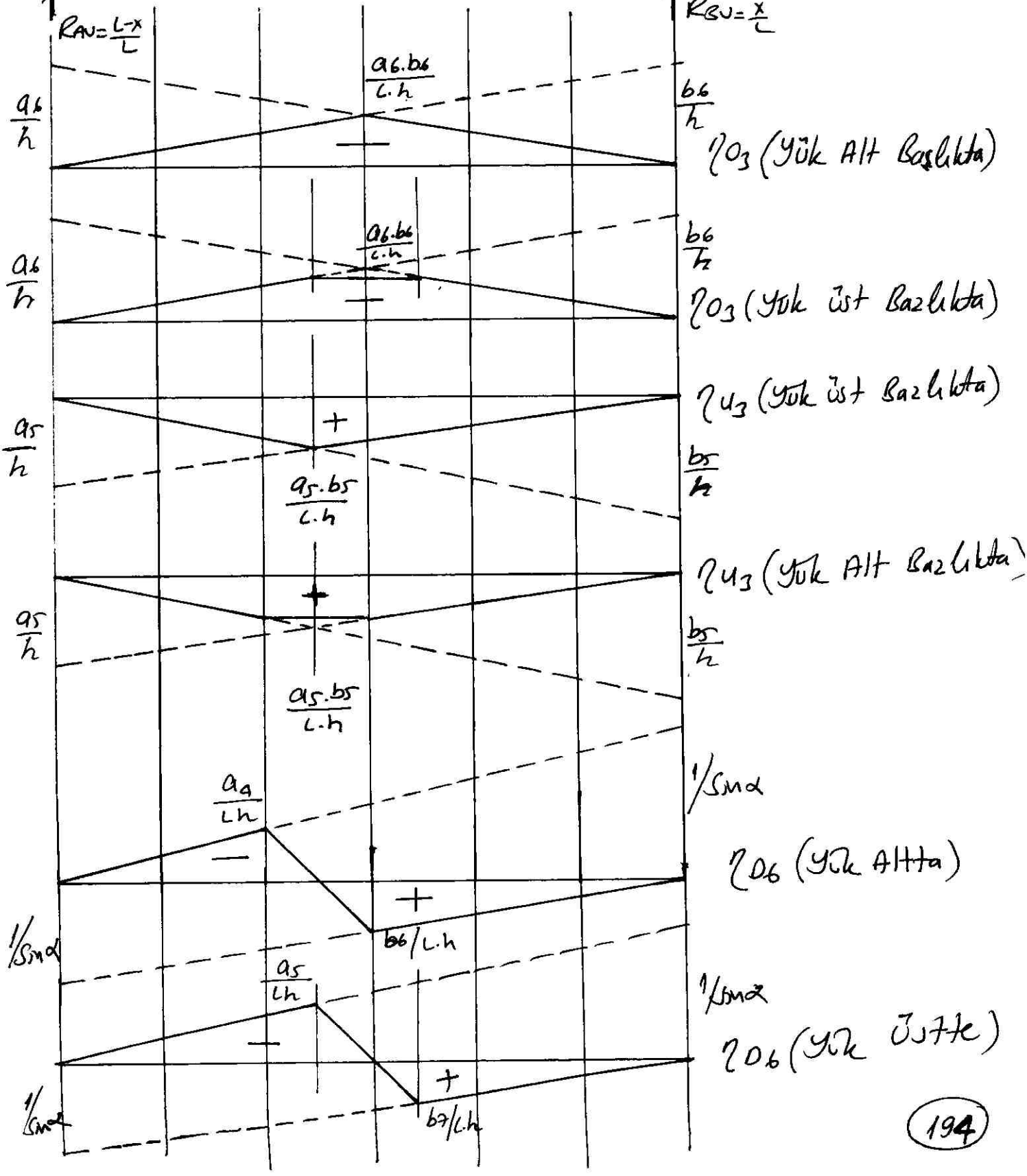
V_3 gubuğunun tesir çizgisinin çizilebilmesi için en pratik kesim III - III kesimidir. Birim yükün üst başlıkta gezinmesi durumunda süreksizlik noktaları 3-5, birim yükün alt başlıkta gezinmesi durumunda ise 4-6 arasıdır. Birim yük üst başlıkta ve $0 < x < a_3$ arasında ise sistemin 5 - 12 arası dikkate alınır ve $\Sigma Y = 0$ denklemi uygulanır. Birim yükün $a_5 < x < L$ arasında olması durumunda ise sistemin A-3 noktaları arası dikkate alınır ve $\Sigma Y = 0$ şartı uygulanır. Birim yük alt başlıkta ve A-4 arasında ise, sistemin 6-B noktaları arası, 6-B arasında ise A-4 noktaları arası dikkate alınarak $\Sigma Y = 0$ şartı uygulanır. Böylece her iki durum için de V_3 gubuğuna ait tesir fonksiyonları elde edilmiş olur.

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b) V TİPİ KAFES KİRİŞ



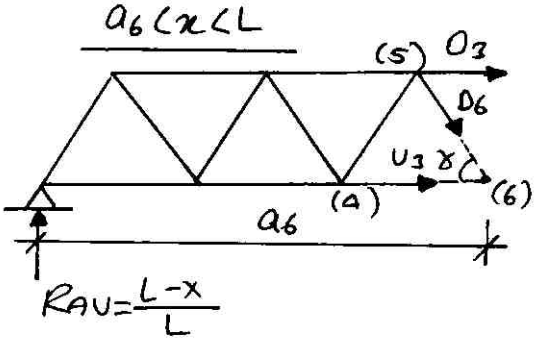
$R_{AV} = \frac{L-x}{L}$ $R_{BV} = \frac{x}{L}$



V tipi kafes kirişte tesir çizgilerinin belirlenmesi

O_3 çubuğunun tesir çizgisinin belirlenmesinde I-I veya II-II kesimi uygulanır. Sayet birim yük alt başlıkta ise tesir çizgisindeki süreksizlik 4-6 ve 6-8 noktaları arasındadır. Birim yük üst başlıkta ise tesir çizgisindeki süreksizlik 5-7 arasındadır. I-I veya II-II kesimi yapıldıktan sonra sistem 2 parça halinde ele alınır ve her bir parça için 6 noktasına göre $\Sigma M_6 = 0$ denklemi uygulanır. Birim yük A-6 arasında ise sistemin 6-B arası, birim yük 6-B arasında ise sistemin A-6 arası dikkate alınır.

$a_6 < x < L$



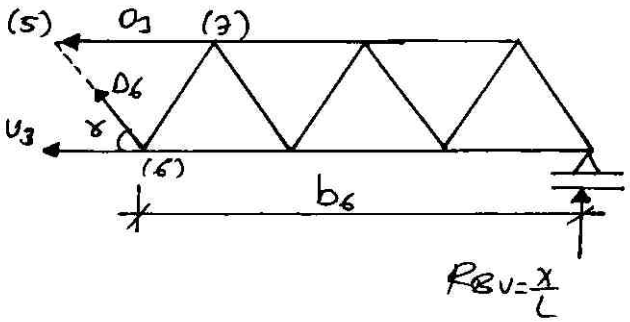
I-I KESİMİ

$$\Sigma M_6 = 0 \rightarrow O_3 \cdot h = -\frac{L-x}{L} \cdot a_6$$

$$x = a_6 \rightarrow \eta_{O_3} = -\frac{a_6 \cdot b_6}{L \cdot h}$$

$$x = L \rightarrow \eta_{O_3} = 0$$

$0 < x < a_6$



$$\Sigma M_6 = 0 \rightarrow O_3 \cdot h = -\frac{x}{L} \cdot b_6$$

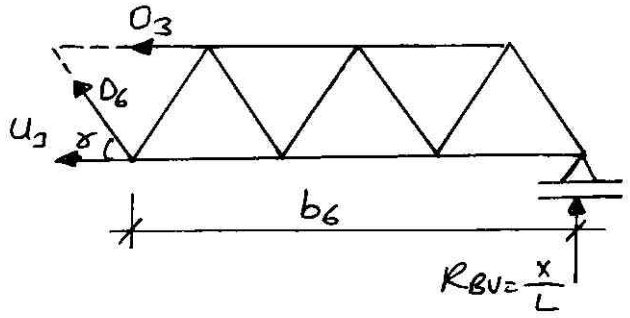
$$x = 0 \rightarrow \eta_{O_3} = 0$$

$$x = a_6 \rightarrow \eta_{O_3} = -\frac{a_6 \cdot b_6}{L \cdot h}$$

U_3 çubuğunun tesir çizgisinin belirlenmesinde I-I ve III-III kesimlerini uygulamak gerekmektedir. U_3 çubuğunun tesir çizgisindeki süreksizlikler birim yükün alt başlıkta olması halinde 4-6 arasında, birim yükün üst başlıkta olması halinde ise 3-5 ve 5-7 noktaları arasında olacaktır. Sistemin kesimi yapıldıktan sonra (5) noktasına göre moment alınarak tesir fonksiyonları elde edilir.

O_6 çubuğunun tesir alanının belirlenmesinde I-I kesiti yeterli olacaktır. I-I kesiminden sonra her iki parça üzerinde $\Sigma Y=0$ denklemi kullanarak tesir fonksiyonu elde edilecektir. Birim yükün alt bazlıkta gezinmesinden dolayı O_6 çubuğunun tesir fonksiyonunda 4-6 noktaları arasında, birim yükün üst bazlıkta gezinmesi durumunda ise 5-7 noktaları arasında süreksizlik meydana gelecektir. Bu durumda birim yük A-4 arasında iken sağdaki parça, birim yük 6-B arasında iken soldaki parça dikkate alınacaktır.

$0 < x < a_4$



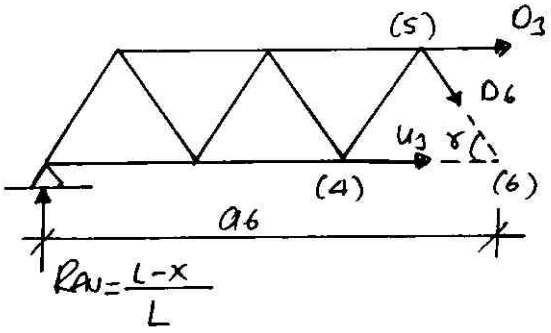
$\Sigma Y=0 \rightarrow O_6 . \sin\gamma = -\frac{x}{L}$

$x=0 \rightarrow \eta_{O_6}=0$

$x=a_4 \rightarrow \eta_{O_6}=-\frac{a_4}{L_h}$

$x=L \rightarrow \eta_{O_6}=-1/\sin\gamma$

$a_6 < x < L$



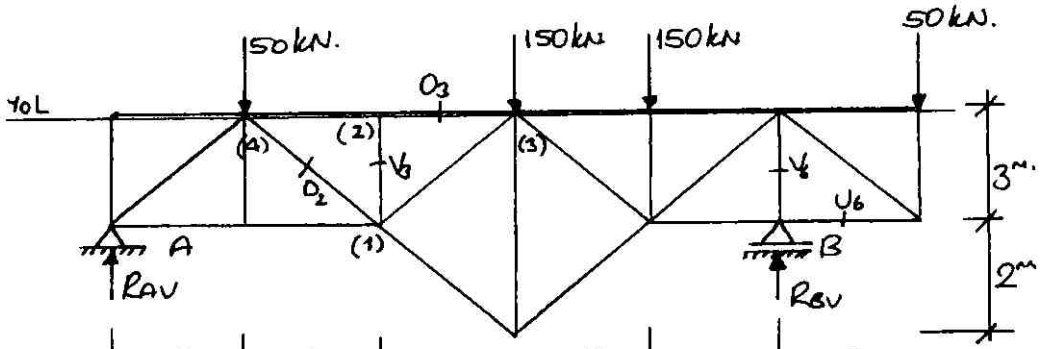
$\Sigma Y=0 \rightarrow O_6 . \sin\gamma = \frac{L-x}{L}$

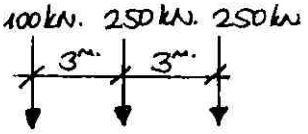
$x=a_6 \rightarrow \eta_{O_6}=\frac{b_6}{L\sin\gamma}$

$x=L \rightarrow \eta_{O_6}=0$

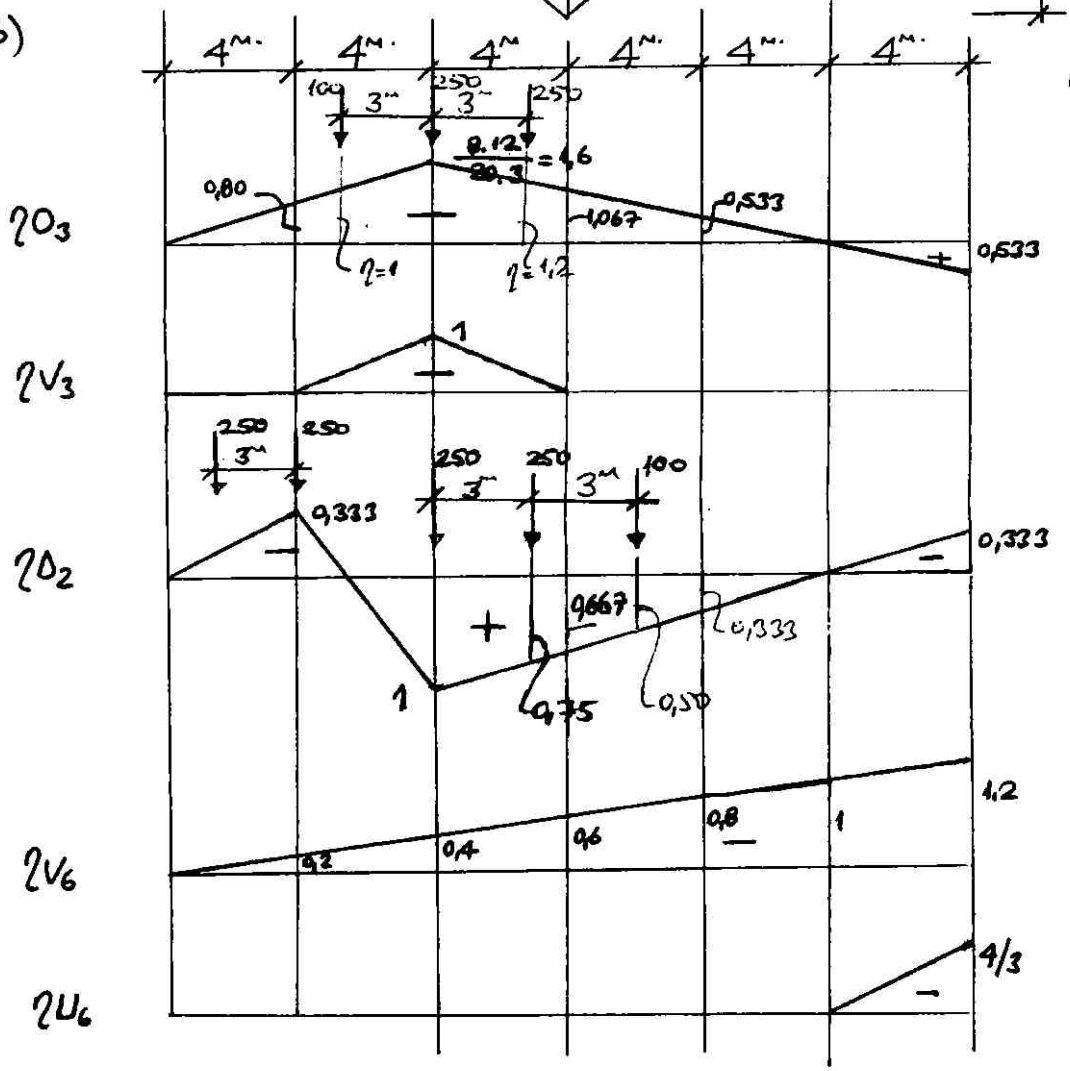
$x=0 \rightarrow \eta_{O_6}=1/\sin\gamma$

ÖRNEK : c)





b)



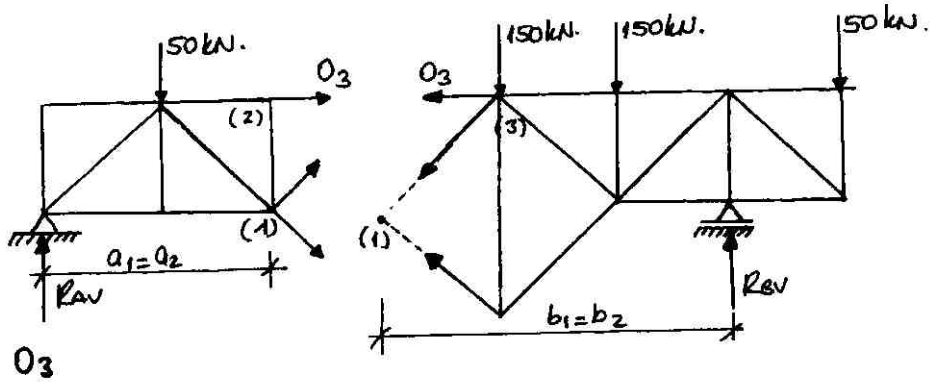
a) Şekildeki kafes sistemin O_3, D_2, V_3, V_6, U_6 çubuk kuvvetlerini hesaplayınız.

b) O_3, D_2, V_3, V_6, U_6 çubuklarının tesir çizgilerini çiziniz.

c) $\max_{\min} O_3$ ve $\max_{\min} D_2$ çubuk kuvvetlerini hesaplayınız.

a)

c)



O_3

Kesit Tesiri: $\Sigma M_B=0 \rightarrow R_{AV}=\frac{50.16+150.8+150.4-50.4}{20}=120\,kN.$

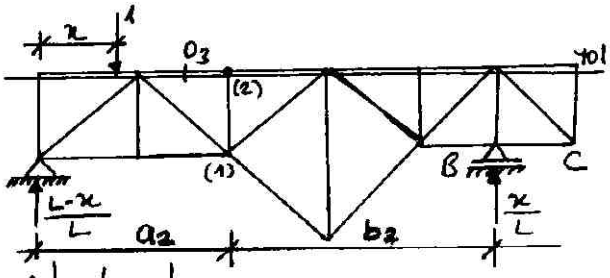
$\Sigma M_A=0 \rightarrow R_{BV}=\frac{50.4+150.12+150.16+50.24}{20}=280\,kN.$

$\Sigma Y=0 \rightarrow 50+150+150+50=120+280 \Rightarrow 400=400$ ✓

$\Sigma M_{(1)}=0 \rightarrow O_3.3=-120.8+50.4 \Rightarrow O_3=-253{,}33\,kN.$

Tesir Cizgisi: $\Sigma M_B=0 \rightarrow R_{AV}=\frac{L-x}{L}$

$\Sigma M_A=0 \rightarrow R_{BV}=\frac{x}{L}$



$0<x<a_2$

Sağ parça üzerinde (1) noktasına göre moment alırsak;

$O_3.h=-R_{BV}.b_2 \Rightarrow O_3=-\frac{x}{L}.\frac{1}{h}.b_2$

$x=0 \Rightarrow \eta_{O_3}=0$

$x=a_2 \Rightarrow \eta_{O_3}=-\frac{a_2.b_2}{L.h}=-\frac{8.12}{20.3}=1{,}6$

$x=L \Rightarrow \eta_{O_3}=-b_2/h$

$a_2<x<L$

Sol parça üzerinde (1) noktasına göre moment alırsak;

$O_3.h=-R_{AV}.a_2 \Rightarrow O_3=-\frac{L-x}{L}.\frac{a_2}{h}$

$x=a_2 \Rightarrow \eta_{O_3}=-\frac{a_2.b_2}{L.h}$

$x=L \Rightarrow \eta_{O_3}=0$

$x=0 \Rightarrow \eta_{O_3}=-a_2/h$

$B<x_1<C$

$R_{AV}=-\frac{x_1}{L}$

$R_{BV}=\frac{L+x_1}{L}$

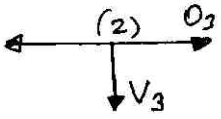
$\Sigma M_{2sol}=0 \rightarrow O_3.h=-R_{AV}.a_2 \Rightarrow O_3=\frac{x_1}{L}.\frac{a_2}{h}$

$x_1=0 \Rightarrow \eta_{O_3}=0$

$x_1=4^- \Rightarrow \eta_{O_3}=\frac{4.8}{20.3}=0{,}533$

V_3 : V_3 çubuğu çözümü için (2) nolu düğüm noktasının dengesi yazılır.

Kesit Tesiri:

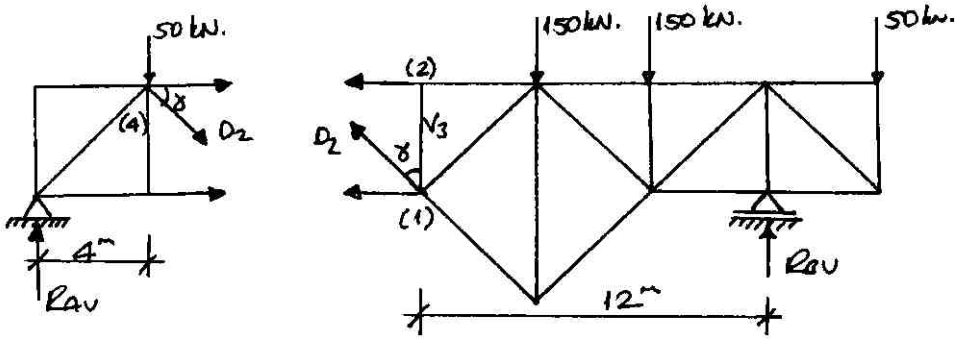


$\Sigma Y=0 \rightarrow V_3=0$

Tesir Çizgisi: : Birim yükün (4) – (3) noktaları arasında geçtiğinde V_3 çubuğu değer almaktadır. Bu noktalar dışında V_3'te kuvvet oluşmamaktadır. Birim kuvvet (2) noktası üzerinde iken $V_3=-1$; (4) ve (3) noktalarında ise $V_3=0$'dır.

- Yol altta olsaydı (Birim yük alt başlıkta geçmeseydi) $V_3=0$ olacaktı.

D_2



$\sin\gamma = 3/5$

Kesit Tesiri: $\Sigma Y_{(4)}=0 \rightarrow D_2.\sin\gamma = R_{AV} - 50$

$$D_2 = (120-50)\frac{1}{\sin\gamma} = 116{,}67 \text{ kN.}$$

Tesir Çizgisi:

$0<x<4^m$

Sağ parçanın dengesi $\Sigma Y_2=0 \Rightarrow D_2.\sin\gamma = -R_{BV} \Rightarrow D_2 = -\frac{x}{L}.\frac{1}{\sin\gamma}$

$x=0 \Rightarrow \eta D_2=0$

$x=4^m \Rightarrow \eta D_2 = -\frac{4}{20}.\frac{5}{3} = -0{,}333$

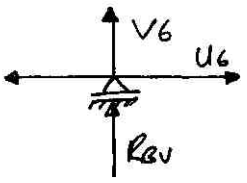
$8^m<x<20^m$

Sol parçanın dengesi $\Sigma Y_4=0 \Rightarrow D_2.\sin\gamma = \frac{L-x}{L}$

$$D_2 = \frac{L-x}{L}.\frac{1}{\sin\gamma} \begin{cases} x=8^m \Rightarrow \eta D_2 = \frac{20-8}{20}.\frac{5}{3} = 1{,}00 \\ x=20^m \Rightarrow \eta D_2 = 0 \end{cases}$$

V_6

Kesit Tesiri:



$\Sigma Y=0 \rightarrow V_6 = -R_{BV} = -280$ kN.

Tesir Cizgisi:

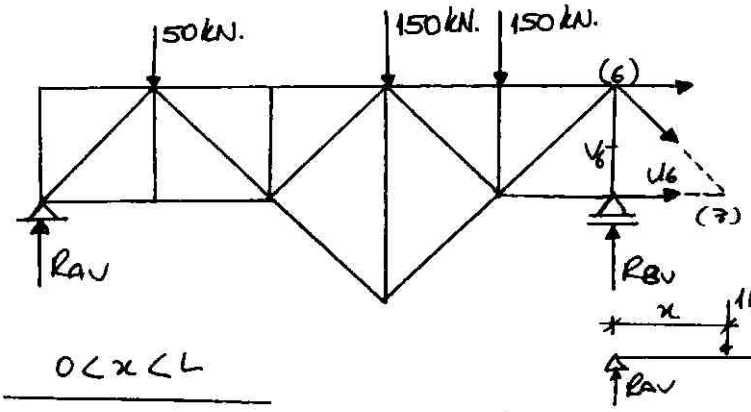
$0<x<L$

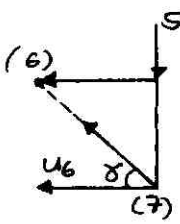
$\Sigma Y=0 \rightarrow V_6 = -R_{BV} = -\frac{x}{L} \rightarrow \begin{cases} x=0 \Rightarrow ?\ V_6=0 \\ x=L \Rightarrow ?\ V_6=-1 \end{cases}$

$L<x_1<L_1$

$R_{BV} = \frac{x_1+L}{L} \Rightarrow V_6 = -\frac{x_1+L}{L} \rightarrow \begin{cases} x_1=0 \Rightarrow ?\ V_6=-1 \\ x_1=4^m \Rightarrow ?\ V_6 = -\frac{4+20}{20} = -1{,}2 \end{cases}$

U_6





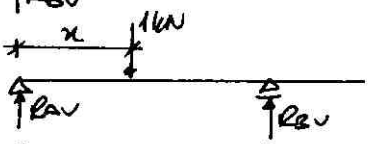
Kesit Tesiri:

$\Sigma M_6=0$

$U_6 . h = -50.4$

$U_6 = -\frac{50.4}{3} = -66{,}67$ kN.

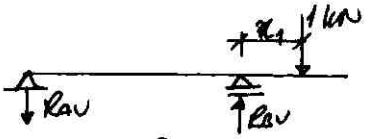
$0<x<L$



Sağ parçanın dengesi yazılır. $R_{AV} = \frac{L-x}{L}$, $R_{BV} = \frac{x}{L}$

$\Sigma M_6=0 \Rightarrow U_6 . h = 0 \Rightarrow U_6 = 0$

$L<x_1<L_1$



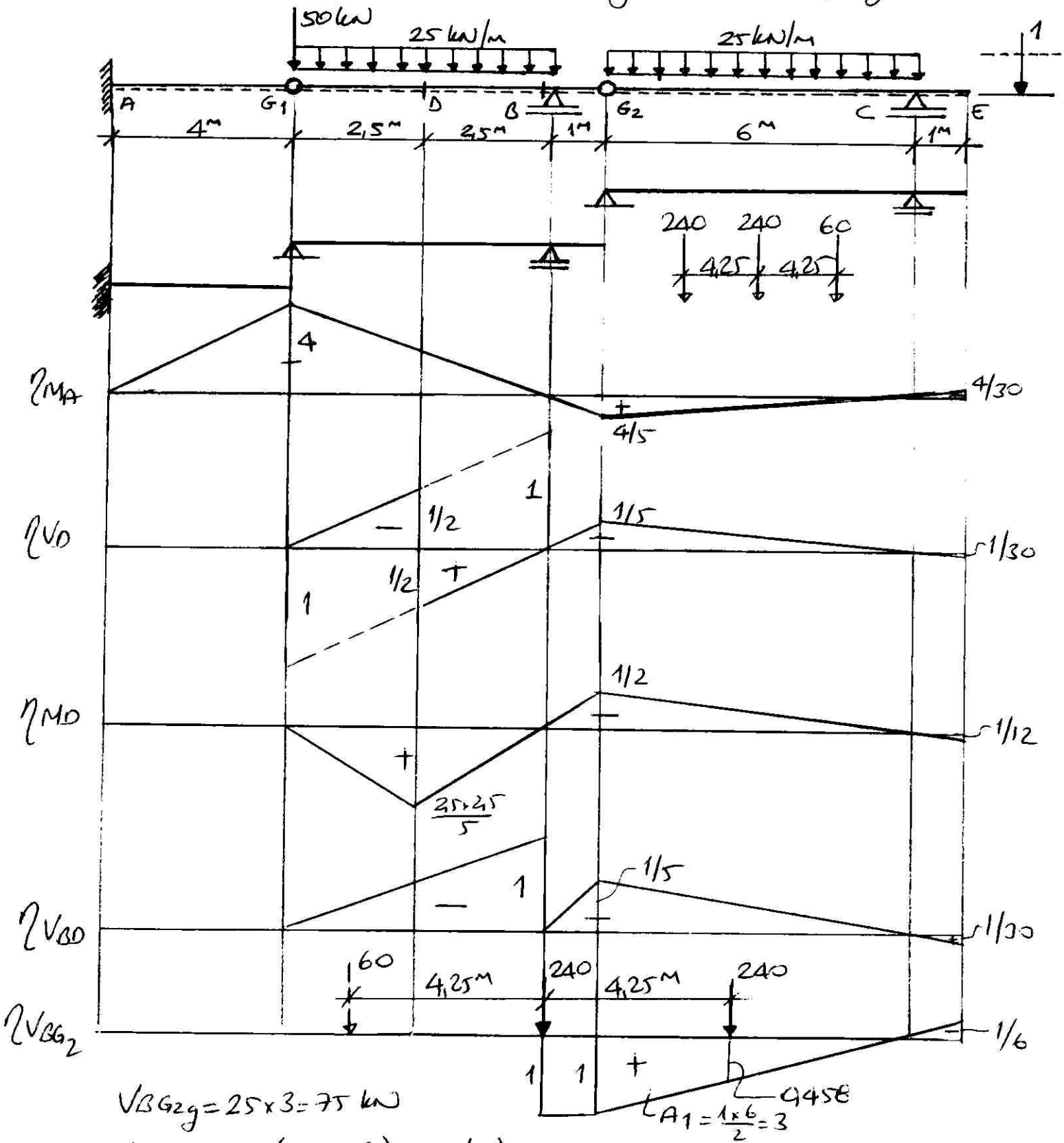
Sol parçanın dengesi yazılır: $R_{AV} = -\frac{x_1}{L}$, $R_{BV} = \frac{x_1+L}{L}$

$\Sigma M_6=0$; $R_{AV} . L + U_6 . h = 0$

$$?\ U_6 = -\frac{R_{AV} . L}{h} = -\frac{\frac{x_1}{L} . L}{h} = -\frac{x_1}{h} \quad \begin{cases} X_1=0 \Rightarrow ?\ U_6=0 \\ x_1=4^m \Rightarrow ?\ U_6=-4/3 \end{cases}$$

Örnek: 1) a) Şekilde verilen sistemin η_{MA}, η_{VD}, η_{MD} ve η_{VBD} tesir çizgilerini çizdiriz.

b) Şekilde verilen tesir çizgisi hangi büyüklüğe aittir? Sistemde verilen sabit ve hareketli yükleri dikkate alarak söz konusu kesitin maksimum değerlerini hesaplayınız.



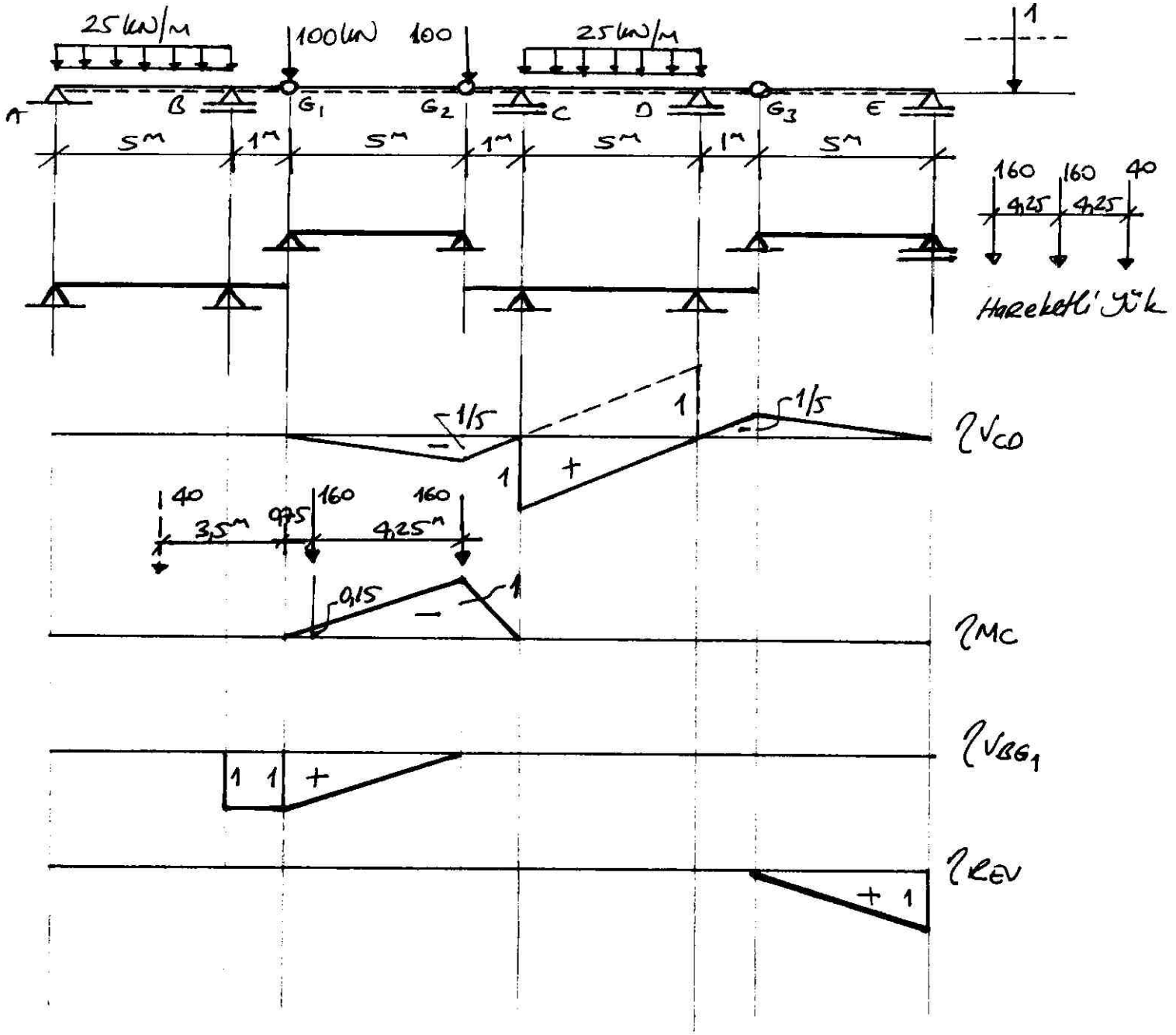
$V_{BG_2 g} = 25 \times 3 = 75$ kN

$V_{BG_2 q} = 240(1 + 0{,}458) = 350$ kN

$\max V_{BG_2} = 75 + 350 = 425$ kN

ÖRNEK:2) a) Şekilde görülen sistemin $?_{V_{CD}}$, $?_{M_C}$, $?_{V_{BG_1}}$ ve $?_{R_{EV}}$ tesir çizgilerini çiziniz.

b) Şekilde verilen sabit ve hareketli yüklerden dolayı max M_C kesit tesiri değerini hesaplayınız.

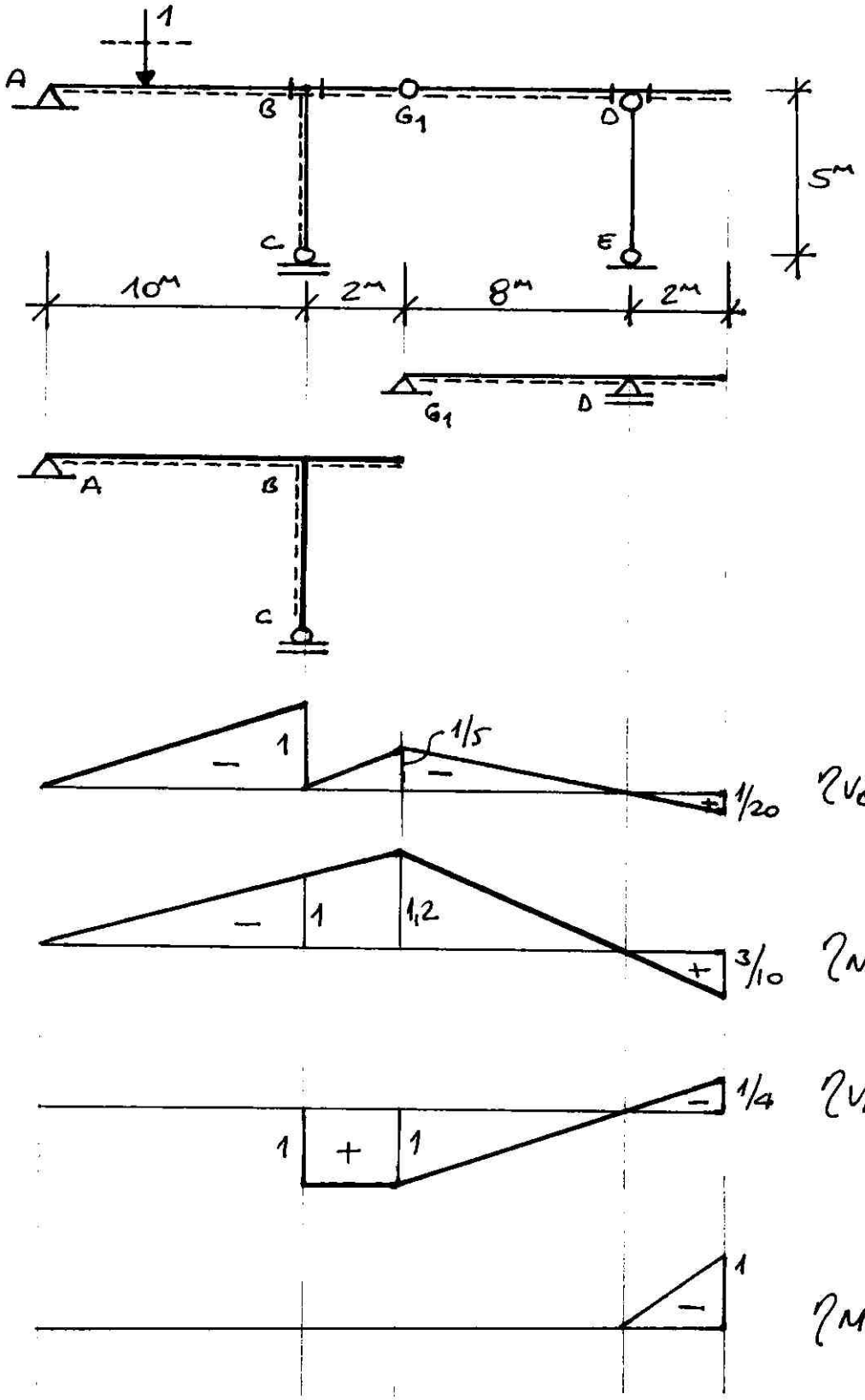


b) $M_{cg} = 100 \times (-1) = -100$ kNm

$M_{cq} = -160(0{,}15+1) = -184$ kNm

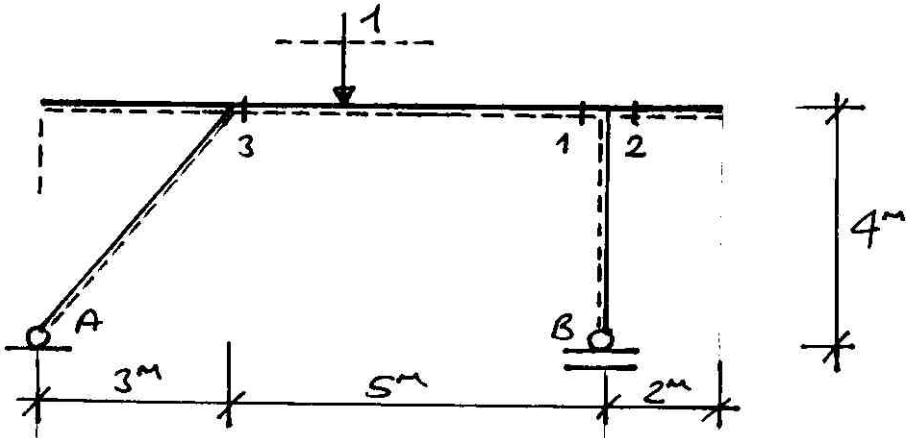
$\max M = -100 - 184 = -284$ kNm

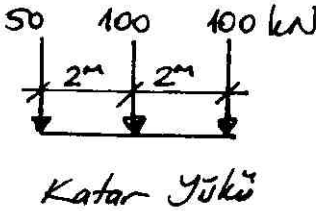
ÖRNEK: 3) Şekilde görülen sistemde $\eta_{V_{BA}}$, $\eta_{N_{CB}}$, $\eta_{V_{BG_1}}$ ve η_{M_D} tesir çizgilerini çiziniz.



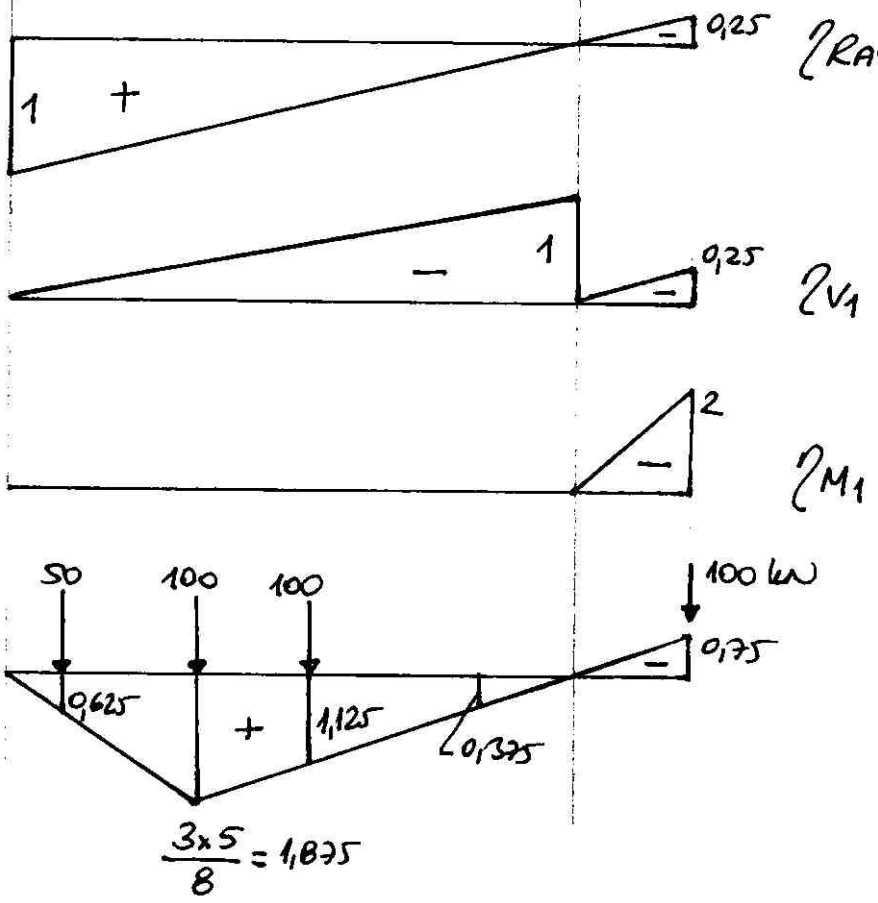
ÖRNEK: 4) a) $?_{R_{AV}}$, $?_{V_1}$, $?_{M_2}$, $?_{M_3}$ tesir çizgilerini çizdiriniz.

b) Verilen katar yükünden dolayı $\max_{\min} M_3$ değerini hesaplayınız.





Katar yükü

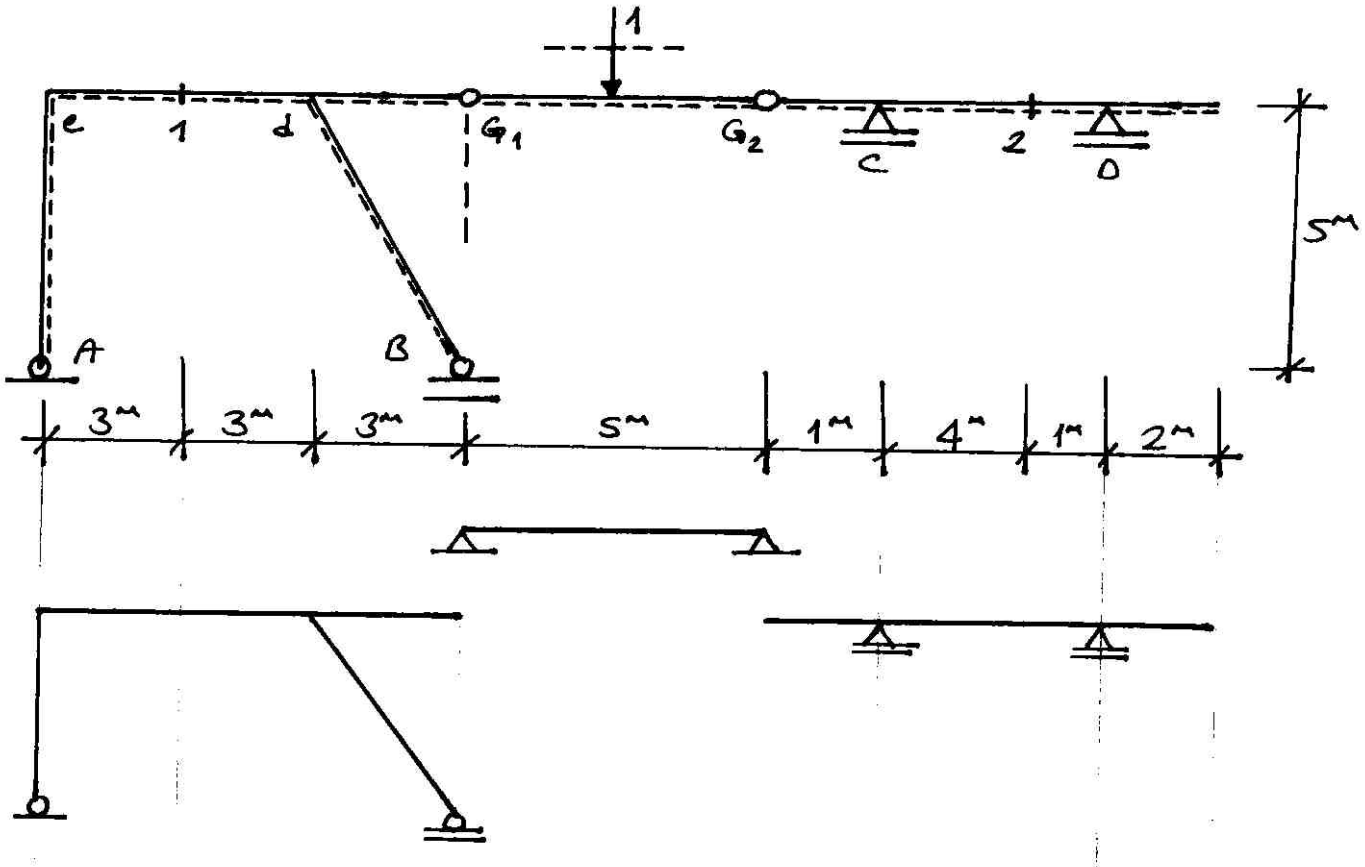


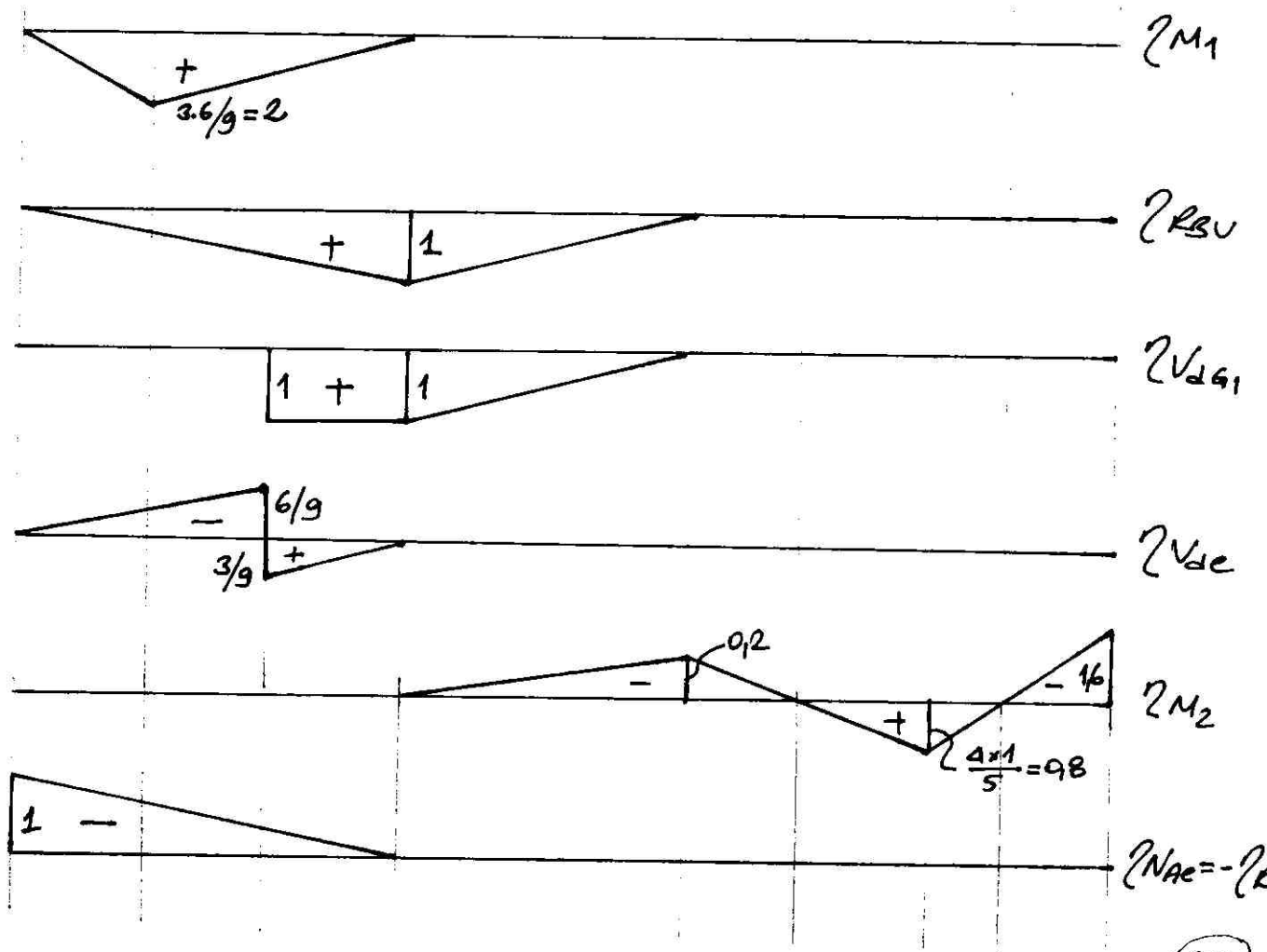
$\frac{3 \times 5}{8} = 1{,}875$

$\max M_3 = 50 \times 0{,}625 + 100(1{,}875 + 1{,}125) = 331{,}25 \text{ kNm}.$

$\min M_3 = -100 \times 0{,}75 = -75 \text{ kNm}$

ÖRNEK: 5) Şekilde görülen sistemde ηM_1, ηR_{BV}, ηV_{dG_1}, ηV_{de}, ηM_2 ve ηN_{Ae} tesir çizgilerini çiziniz.

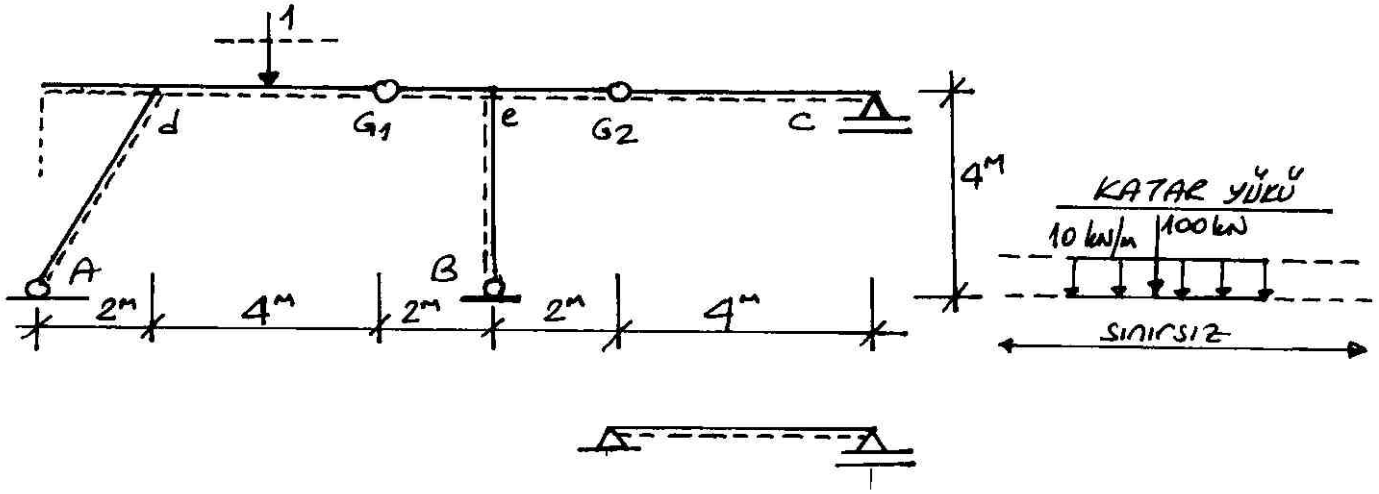


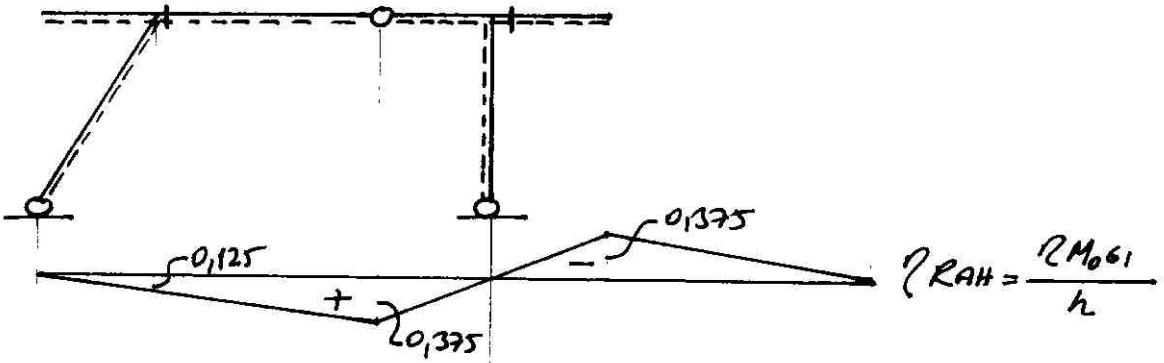


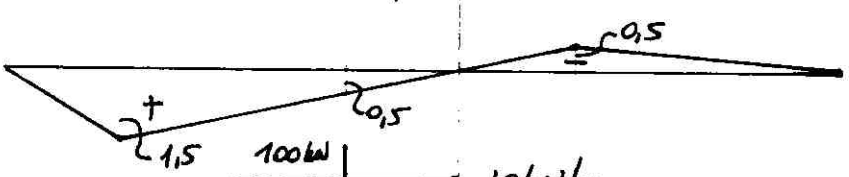
ÖRNEK:6) Şekilde boyutları ve yükleme durumu ile verilmiş olan sistemde,

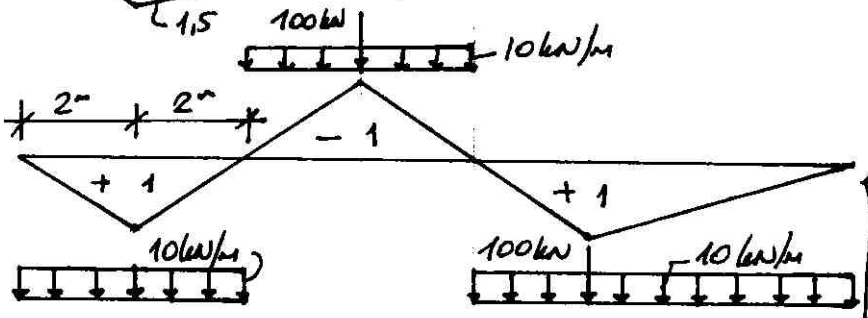
a) $?M_{dG_1}$, $?V_{eG_2}$ tesir çizgilerini çiziniz.

b) Verilen katar yükünden dolayı $\overset{max}{min}\, M_{dG_1}$ değerlerini bulunuz.









$$?M_{dG_1} = ?M_{oG_1} - 4 \cdot ?R_{AH}$$

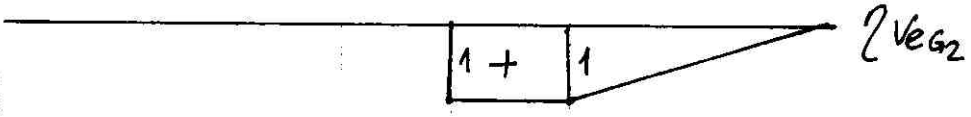
$$?M_{dG_1}{}_{d} = 1{,}5 - 4 \cdot 0{,}125 = 1$$

$$?M_{dG_1}{}_{G_1} = 0{,}5 - 4 \cdot 0{,}375 = -1$$

$$?M_{dG_1}{}_{G_2} = -0{,}5 - 4 \cdot (-0{,}375) = 1$$

$$\max M_{dG_1} = 10\left(\frac{1 \times 4}{2} + \frac{1 \times 6}{2}\right) + 100 \cdot 1 = 150\ kNm$$

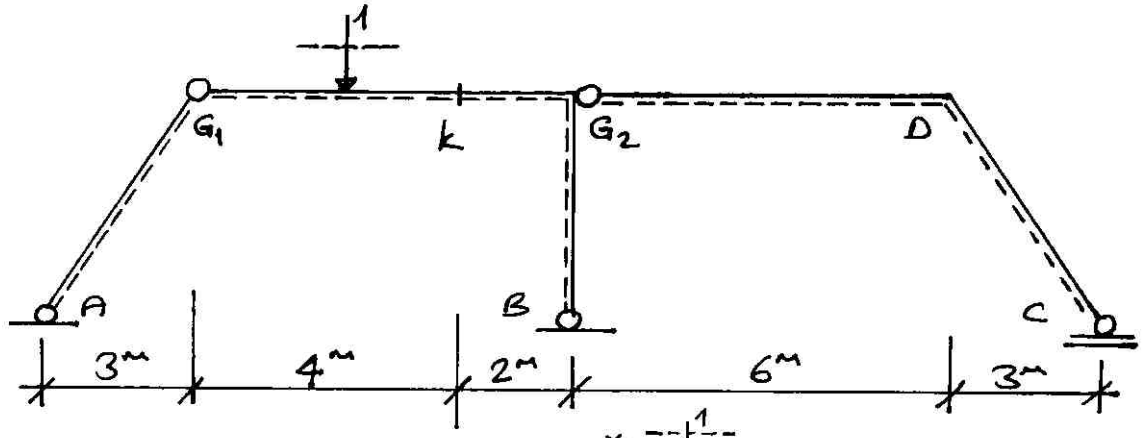
$$\min M_{dG_1} = -10\left(\frac{1 \times 4}{2}\right) - 100 \cdot 1 = -120\ kNm$$

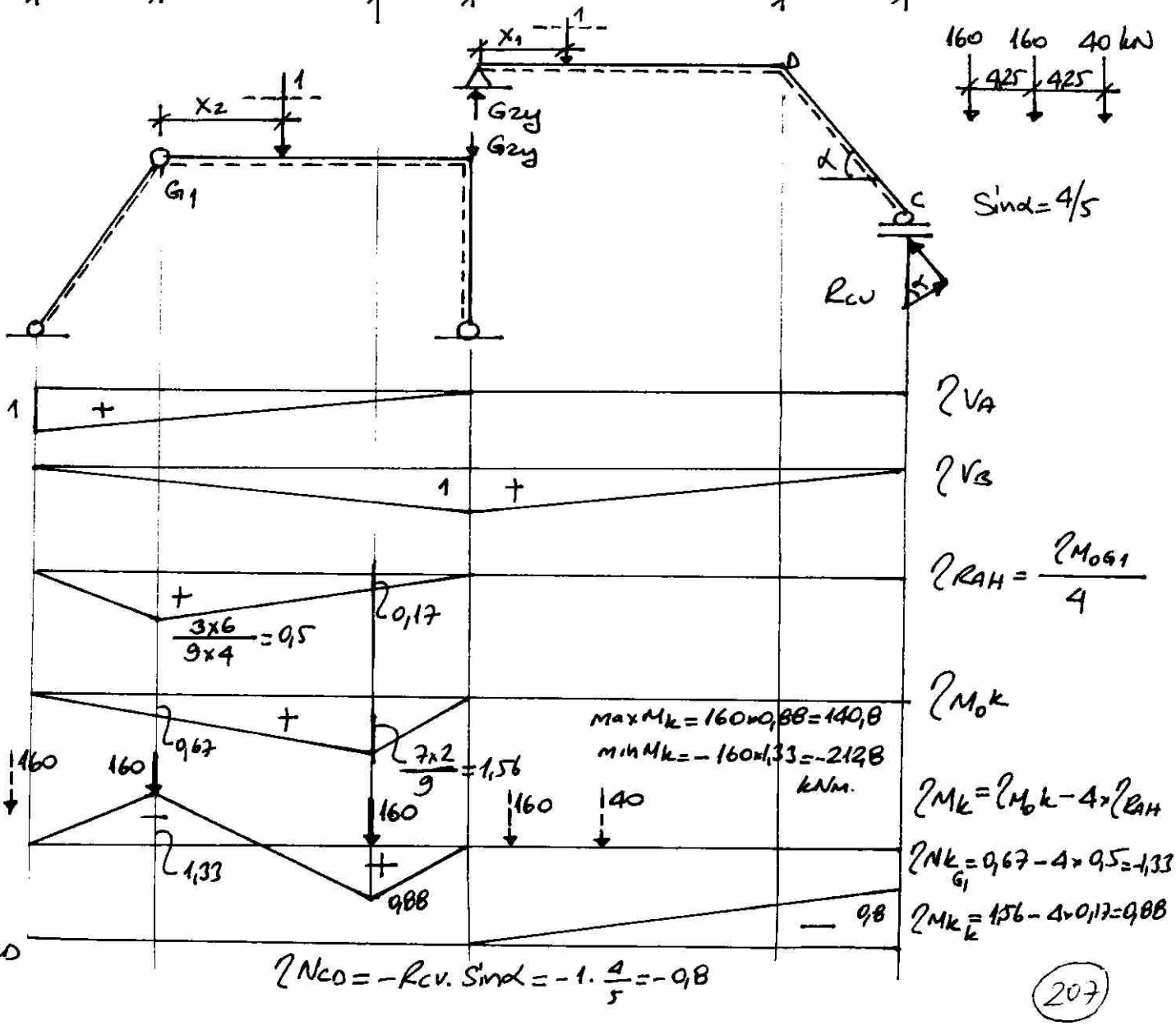


ÖRNEK:7) Şekilde görülen sistemde;

a) A ve B mesnetlerine ait yatay ve düşey mesnet tepkileri (R_{AV}, R_{BV}), k kesitine ait moment (M_k), C mesnedinin solundaki normal kuvvet (N_{CD}) tesir çizgilerini çiziniz.

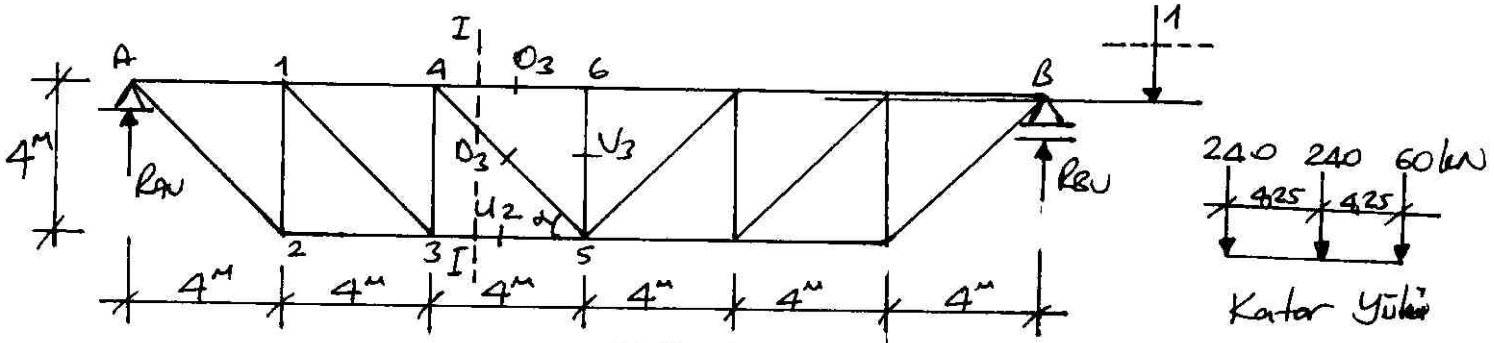
b) Verilen katar yüküne göre k kesitindeki momentin maksimum ve minimum değerlerini hesaplayınız.

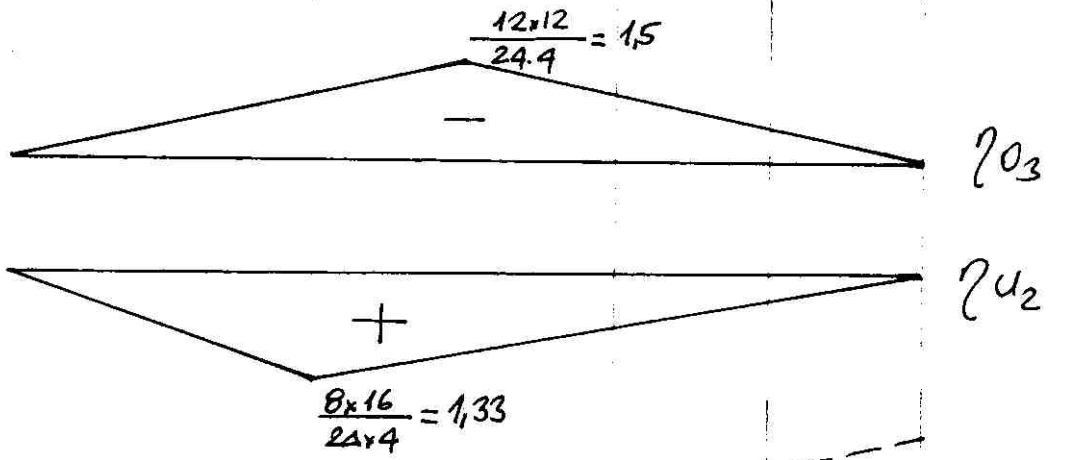


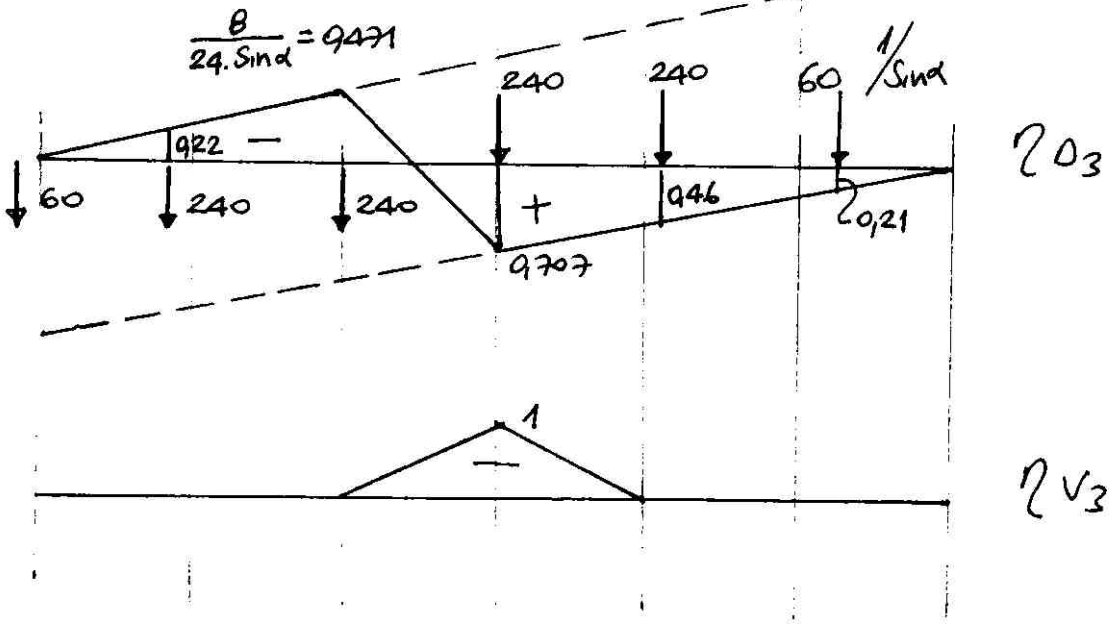


ÖRNEK : 8) a) Şekilde verilen kafes sistemin O_3, U_2, D_3 ve V_3 çubuklarına ait tesir çizgilerini çiziniz.

b) O_3 çubuğunun verilen katara göre max ve min değerlerini bulunuz







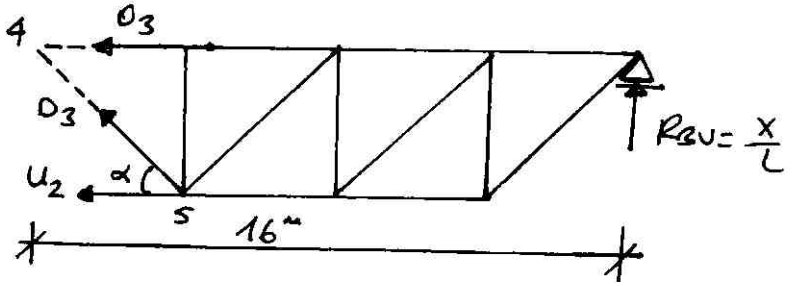
b) $\max D_3 = 240(0,707 + 0,46) + 60 \cdot 0,21 = 292,68$ kN

$\min D_3 = -240(0,471 + 922) = -165,84$ kN

I-I KESİMİ

η_{O_3} ve η_{U_2} çubukları

$0 < x < a_3, a_5$



$$\Sigma M_5 = 0 \rightarrow \eta_{O_3} = -\frac{x}{L} \cdot 12 \cdot \frac{1}{h}$$

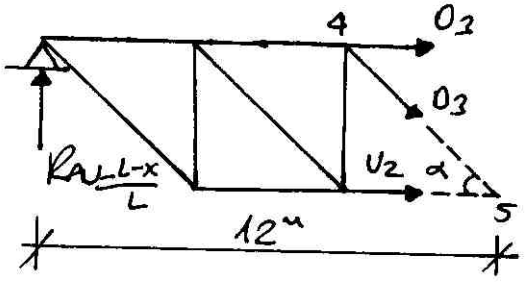
$$\Sigma M_4 = 0 \rightarrow \eta_{U_2} = \frac{x}{L} \cdot 16 \cdot \frac{1}{h}$$

$x = 0 \begin{cases} \eta_{O_3} = 0 \\ \eta_{U_2} = 0 \end{cases}$

$x = 8^m \rightarrow \eta_{U_2} = \frac{8 \cdot 16}{24 \cdot 4}$

$x = 12^m \rightarrow \eta_{O_3} = -\frac{12 \times 12}{24 \times 4}$

$a_5, a_3 < x < L$



$$\Sigma M_5 = 0 \rightarrow \eta_{O_3} = -\frac{L-x}{L} \cdot 12 \cdot \frac{1}{h}$$

$x = 12^m \rightarrow \eta_{O_3} = -\frac{24-12}{24} \cdot \frac{12}{4}$

$x = L \rightarrow \eta_{O_3} = 0$

$$\Sigma M_4 = 0 \rightarrow \eta_{U_2} = \frac{L-x}{L} \cdot 8 \cdot \frac{1}{h}$$

$x = 8^m \rightarrow \eta_{U_2} = \frac{24-8}{24} \cdot \frac{8}{4}$

O_3 çubuğu

$0 < x < a_4$

$$\Sigma Y = 0 \rightarrow O_3 \cdot \sin\alpha = -\frac{x}{L}$$

$x = 0 \rightarrow \eta_{O_3} = 0$

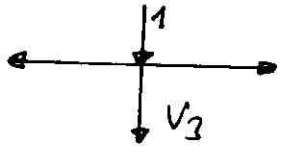
$x = 8^m \rightarrow \eta_{O_3} = -\frac{8}{24 \sin\alpha} = -0{,}471$

$a_6 < x < L$

$$\Sigma Y = 0 \rightarrow O_3 \cdot \sin\alpha = \frac{L-x}{L}$$

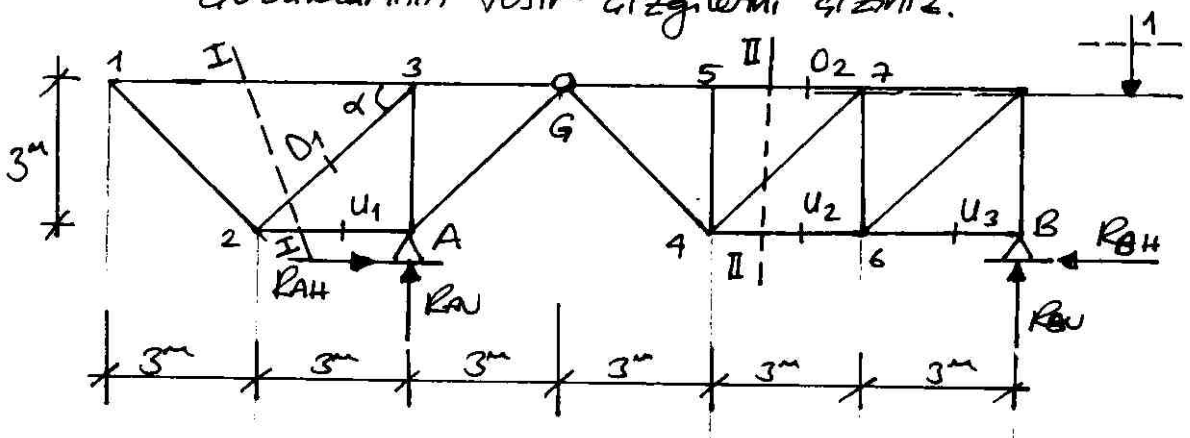
$x = 12^m \rightarrow \eta_{O_3} = \frac{12}{24 \sin\alpha} = 0{,}707$

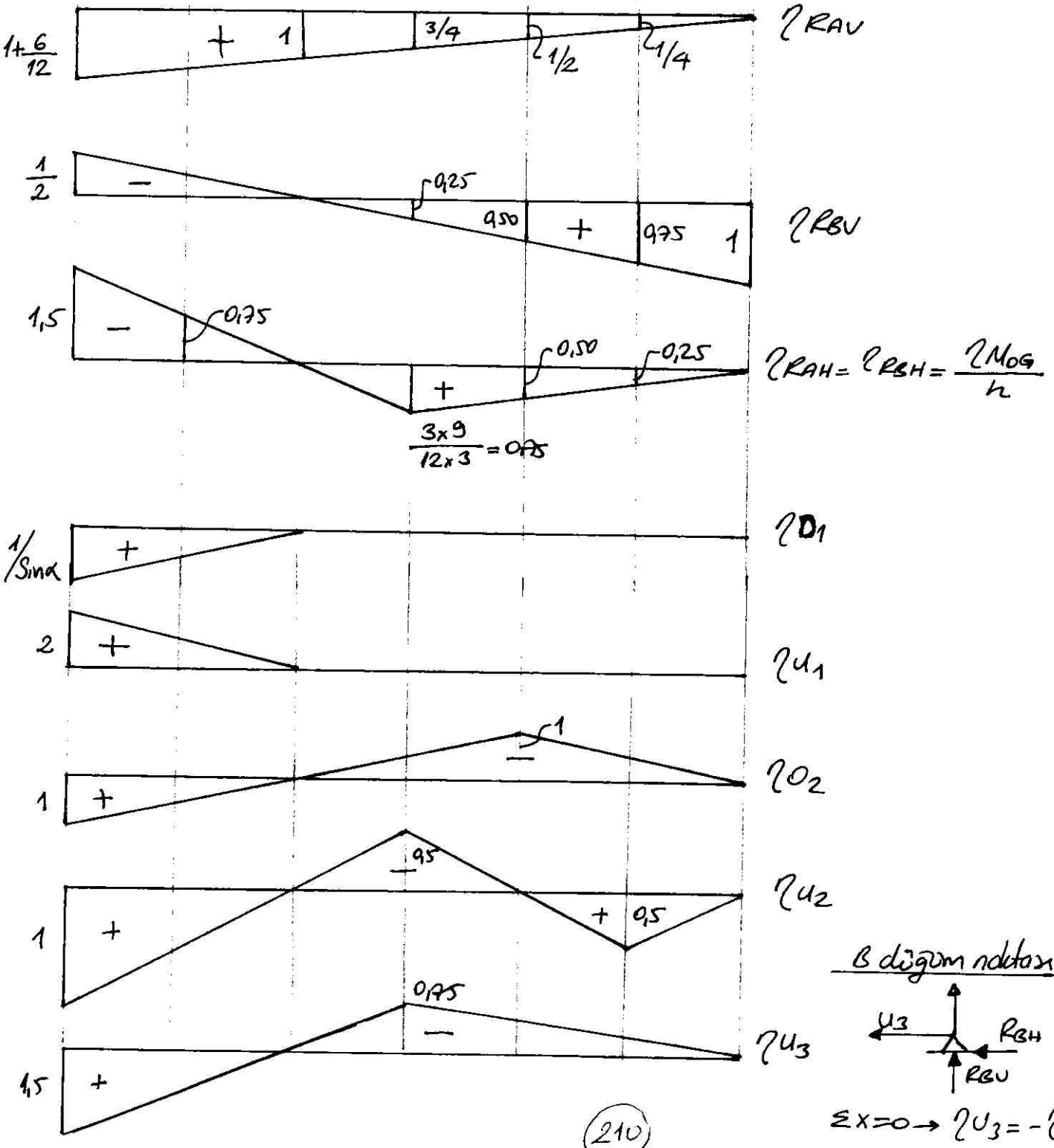
6 DÜĞÜM NOKTASI



$\eta_{V_3} = -1$

ÖRNEK:9) Şekilde verilen kafes sistemin R_{AV}, R_{AH}, R_{BV} ve (O_1, U_1, O_2, U_2 ve U_3) çubuklarının tesir çizgilerini çiziniz.



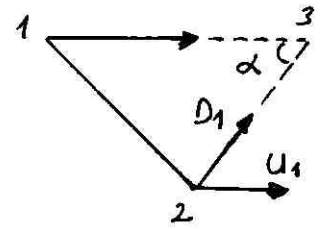


B düğüm noktası

$\Sigma X = 0 \rightarrow \eta U_3 = -\eta R_{BH}$

I-I KESİMİ

A<x<L



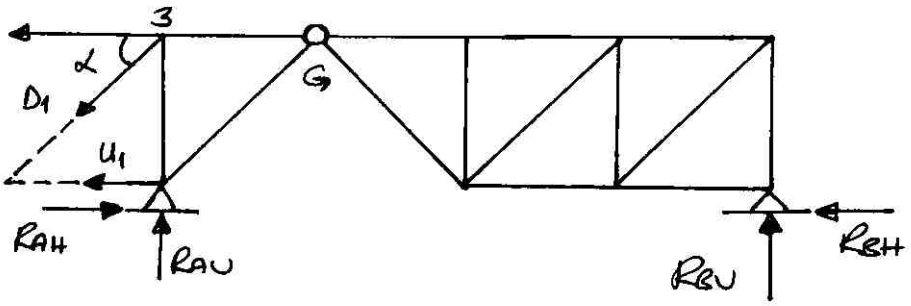
$\Sigma Y=0 \rightarrow D_1.\sin\alpha=0$

$\rightarrow D_1=0$

$\Sigma M_3=0 \rightarrow U_1.h=0$

$\rightarrow U_1=0$

1<x<A



$\Sigma Y=0 \rightarrow D_1.\sin\alpha = R_{AV}+R_{BV}=\frac{L-x}{L}+\frac{x}{L}$

$\rightarrow D_1 = 1/\sin\alpha$

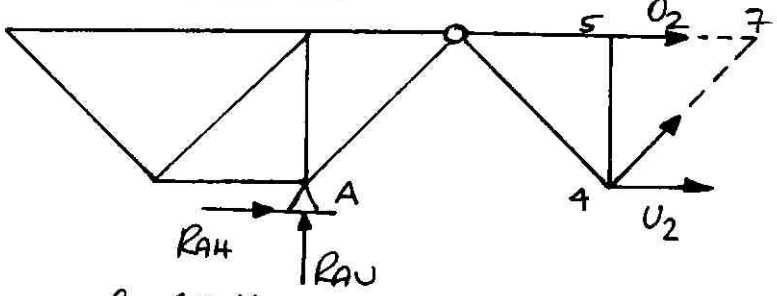
$\Sigma M_3=0 \rightarrow U_1.h = R_{BV}.L + R_{AH}.h - R_{BH}.h$

$\rightarrow U_1 = x/h \rightarrow x=6^m \rightarrow U_1 = 6/3 = 2$

$x=0 \rightarrow U_1=0$

II-II KESİMİ

a_4, a_6<x<L



a_4<x<L

$\Sigma M_4=0 \rightarrow O_2.h = -R_{AV}.6$

$\rightarrow O_2 = -\frac{R_{AV}.6}{3}$

$x=L \rightarrow O_2=0$

$x=a_4\,(6^m) \rightarrow O_2 = -\frac{1/2.6}{3} = -1$

a_6<x<L

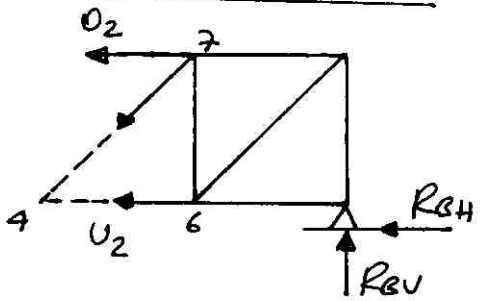
$\Sigma M_7=0 \rightarrow U_2.h = R_{AV}.9 - R_{AH}.3$

$\rightarrow U_2 = \frac{R_{AV}.9}{3} - R_{AH}.\frac{3}{3} = 3R_{AV} - R_{AH}$

$x=9^m \rightarrow U_2 = 3\times 1/4 - 0{,}25 = 0{,}50$

$x=L \rightarrow U_2=0$

0<x<a_4, a_6



0<x<a_4

$\Sigma M_4=0 \rightarrow O_2.h = -R_{BV}.6$

$\rightarrow O_2 = -\frac{R_{BV}.6}{3}$

$x=0 \rightarrow O_2=0$

$x=a_4\,(6^m) \rightarrow O_2 = -\frac{0{,}50.6}{3} = -1$

0<x<a_6

$\Sigma M_7=0 \rightarrow U_2.3 = R_{BV}.3 - R_{BH}.3$

$U_2 = R_{BV} - R_{BH}$

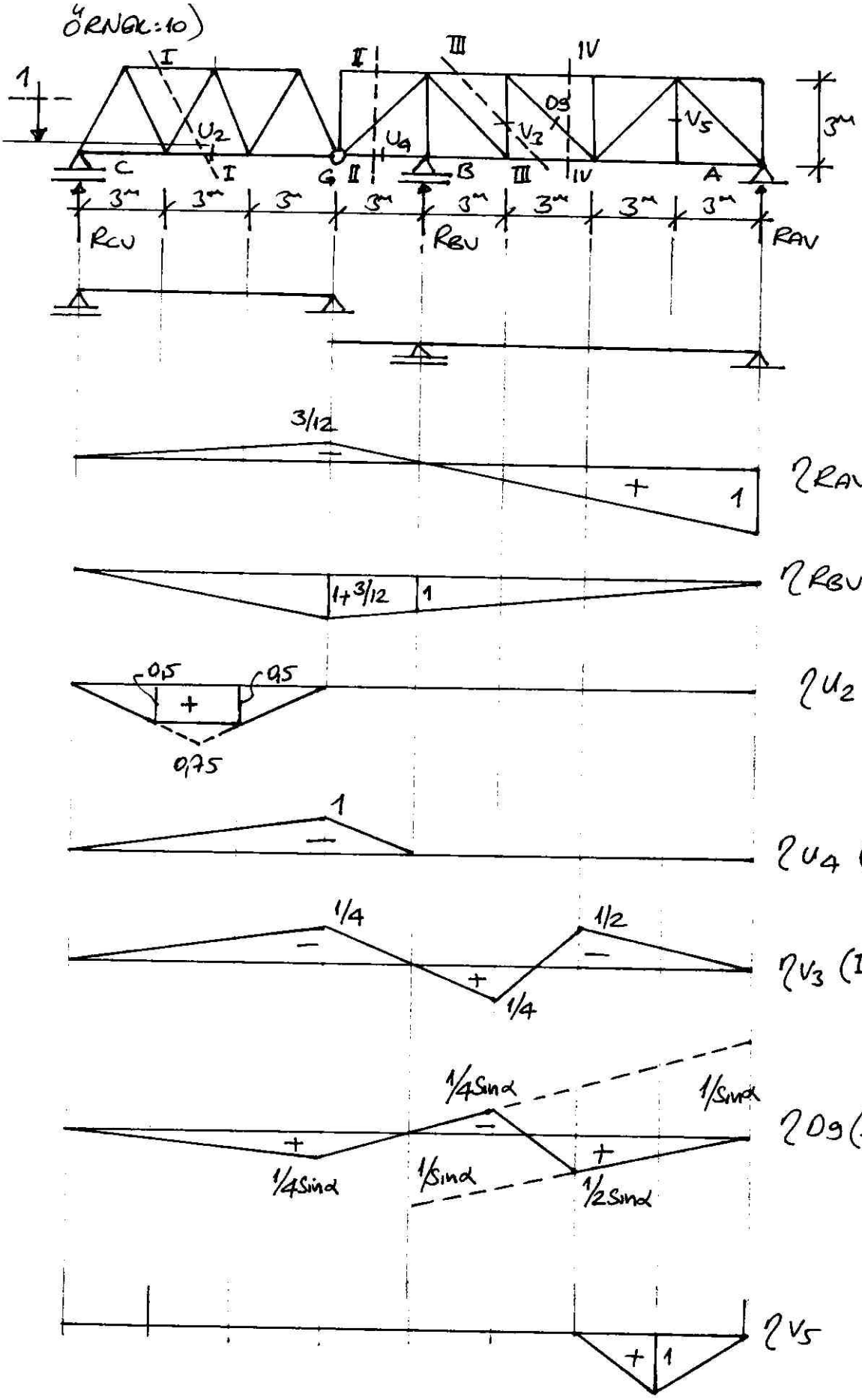
$x=0 \rightarrow U_2=0$

$x=3^m \rightarrow U_2 = 0{,}25 - 0{,}75 = -0{,}50$

$x=6^m \rightarrow U_2=0$

$x=9^m \rightarrow U_2 = 0{,}75 - 0{,}25 = 0{,}50$

ÖRNEK:10)


1
I
II
III
IV
U2
U4
V2
Dg
V5
3m
C
I
G
II
B
III
IV
A
3m 3m 3m 3m 3m 3m 3m 3m
RCV
RBV
RAV
3/12
+
1
ηRAV
1+3/12
1
ηRBV
0,5
0,5
+
0,75
ηU2 (I-I Kesimi)
1
ηU4 (II-II Kesimi)
1/4
1/2
ηV3 (III-III Kesimi)
1/4Sinα
1/Sinα
1/4Sinα
1/Sinα
1/2Sinα
ηDg (IV-IV Kesimi)
ηV5